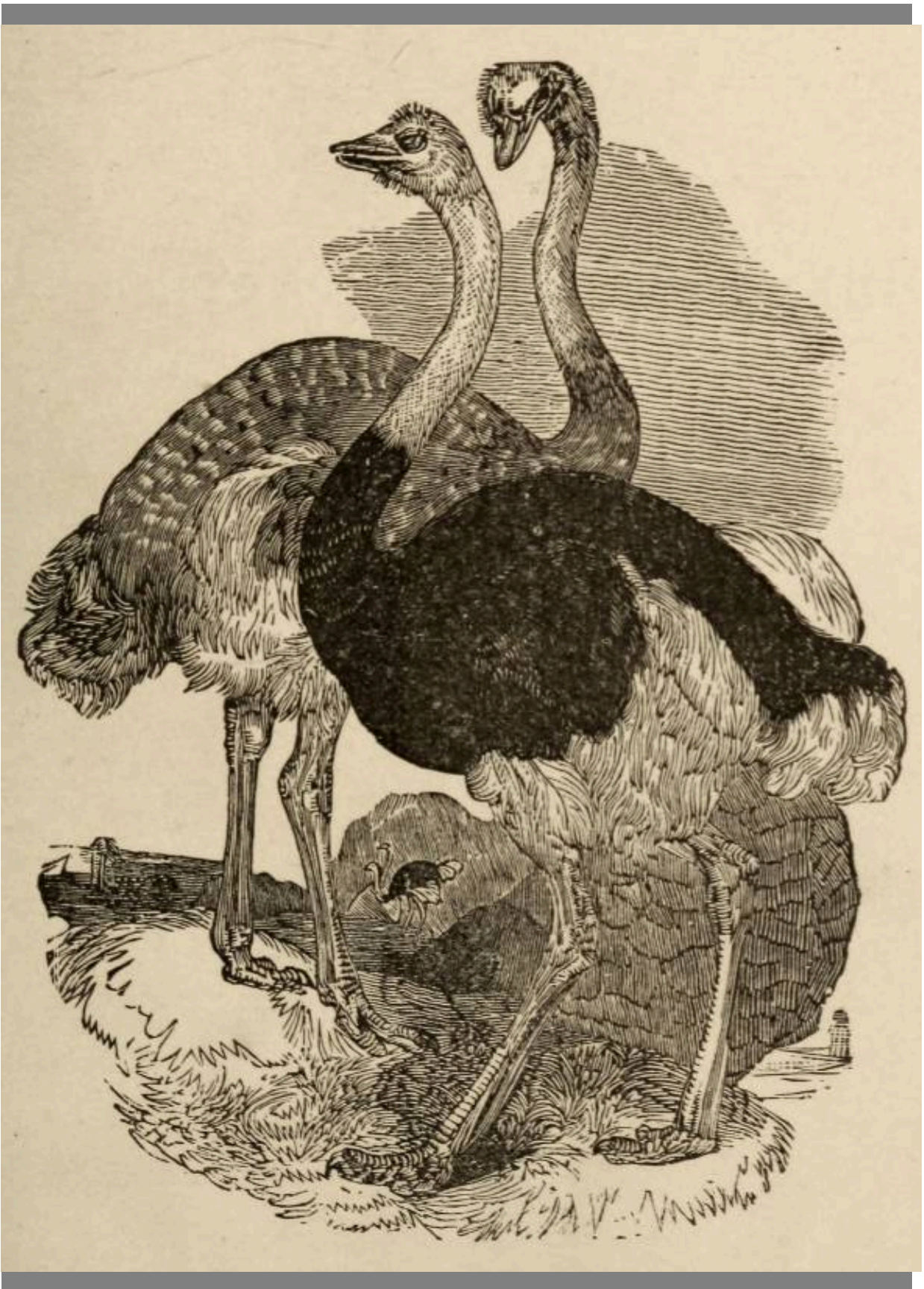






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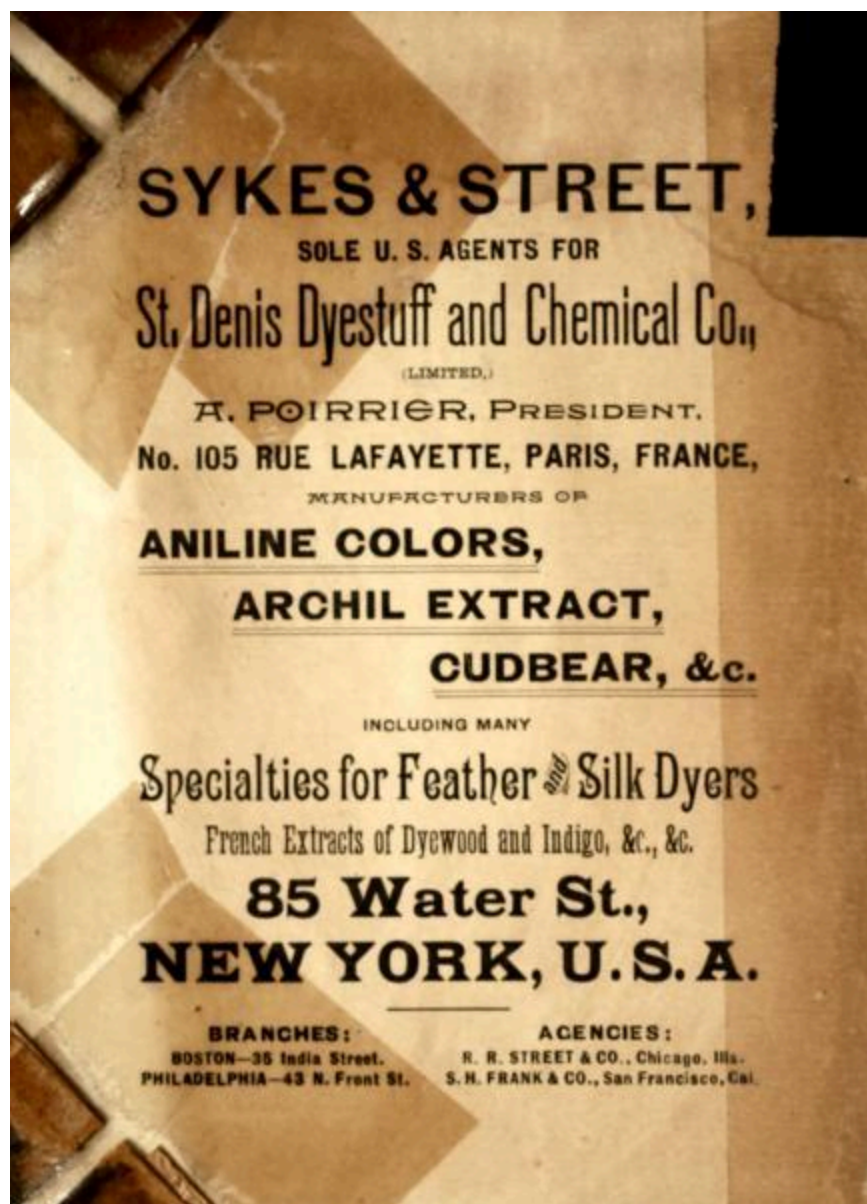
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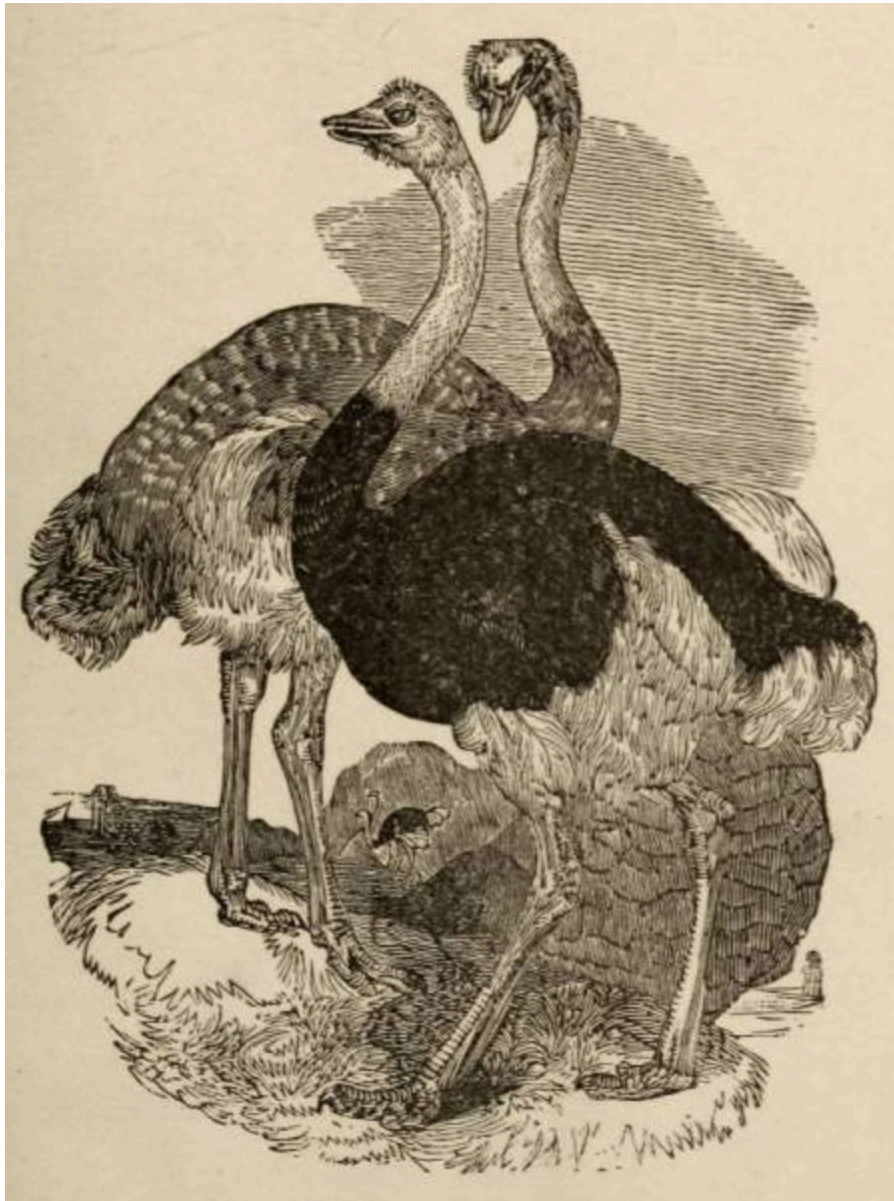
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GROUP OF OSTRICHES.

**THE
PRACTICAL
OSTRICH FEATHER DYER,**

BY
ALEXANDER PAUL.

REVISED AND CORRECTED
BY
DR. M. FRANK.

PUBLISHED BY
MRS. DR. M. FRANK,
"TEXTILE COLORIST,"
506 ARCH ST., PHILADELPHIA, PA., U. S. A. 1888.

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PREFACE.

In the preparation of this work it has been my aim to present Recipes, simple, yet complete in every detail, for dyeing every color and shade of color known. Reliability, practicability and rapidity I claim for this work, and would ask that it be judged not from a literary standpoint, but as a thorough and practical instructor in the art of Ostrich Feather Dyeing, as simplified and perfected by me during years of hard work and research. It is the first work of its kind ever put before the public in the English language, and will, in consequence, receive from those interested close scrutiny and criticism, which prompts the author to offer \$1000 to any person who will prove that the recipes herein contained, or any single one of them, will not produce the desired color or shade perfect and in the time mentioned. The old methodical orthodox dyers will find a decided advantage in being enabled to make colors in minutes, that heretofore required hours and days to complete. Technicalities and high-sounding phrases for the names of colors and terms of the dye-house have no place in this work. It is not necessary for a man to be a chemist to be a practical feather dyer, other authorities to the contrary notwithstanding. Good practical common sense and judgment and a knowledge of the nature of the goods you are handling, and throw theory to the winds.

ALEX PAUL.

TO THE
OSTRICH FEATHER
MANUFACTURERS, DYERS AND SCOURERS,
AND
INTERESTED PUBLIC OF AMERICA AND EUROPE,
THIS WORK IS RESPECTFULLY DEDICATED,
BY THE AUTHOR.

TESTIMONIALS.

The following are a few of the numerous testimonials received by Dr. M. Frank, Manager of "Textile Colorist," in evidence of our method:

CHICAGO, Feb. 23, 1885.

All I have to say regarding Mr. Alex. Paul's method for dyeing ostrich feathers are just as he represents, and after having taken a course I am perfectly satisfied.

I. F. SCHWARZ.

RICHMOND, VA., Jan. 20, 1885.

SIR,—After receiving a course of instruction of Mr. Alex. Paul, I think he is a thorough master of his art, and fully comes up his promises, and any one who wishes to learn the art could not do better than to engage his services.

JAS. F. THURSTON.

LOUISVILLE, KY., March 12, 1885.

This is to certify that I have this day received instructions from Mr. Paul, in the art of feather dyeing, and I can truly say that I am much pleased with his process, so simple, so quickly done, and produces such beautiful colors and shades. I paid \$150 to other parties for instruction in feather dyeing, and I can say that I knew but little about feather dyeing before to-day.

P. BARRISTER.

MILWAUKEE, Feb. 27, 1885.

We take pleasure in recommending the method of feather dyeing taught to us by Mr. Alex. Paul, for the sum of fifty dollars. We think it would be beneficial for any dyer to learn this art.

OTTO PIETSCH CO.

ROCHESTER, Feb. 4, 1885.

This is to certify that Mr. Paul has this day given me instructions in ostrich feather dyeing, for which I paid fifty dollars. I am perfectly satisfied that he has accomplished all that he undertook to do to my satisfaction, and think that it will prove to be money well invested.

WM. MAINS.

CANTON, O., Feb. 11, 1885.

I have taken this day a course of instruction in ostrich feather dyeing from Mr. A. Paul, for which I paid him fifty dollars. The same I consider the most simple and best method known; and is well worth ten times the amount.

C. PETER & SON.

UTICA, Jan. 31, 1885.

I have received a course of instructions from Alex. Paul, for which I paid him fifty dollars, and would state that I consider Mr. Paul a thorough master of the art of feather dyeing, and feel that five times the amount paid him would not be equivalent to the information received.

JOHN W. MCLEAN.

MILWAUKEE, Feb. 28, 1885.

Mr. Alex. Paul has given me instruction for dyeing and cleaning ostrich feathers. I feel satisfied to certify that his method cannot be excelled, and that the instruction is worth ten times the amount charged.

I. LEISER.

BALTIMORE, Jan. 14, 1885.

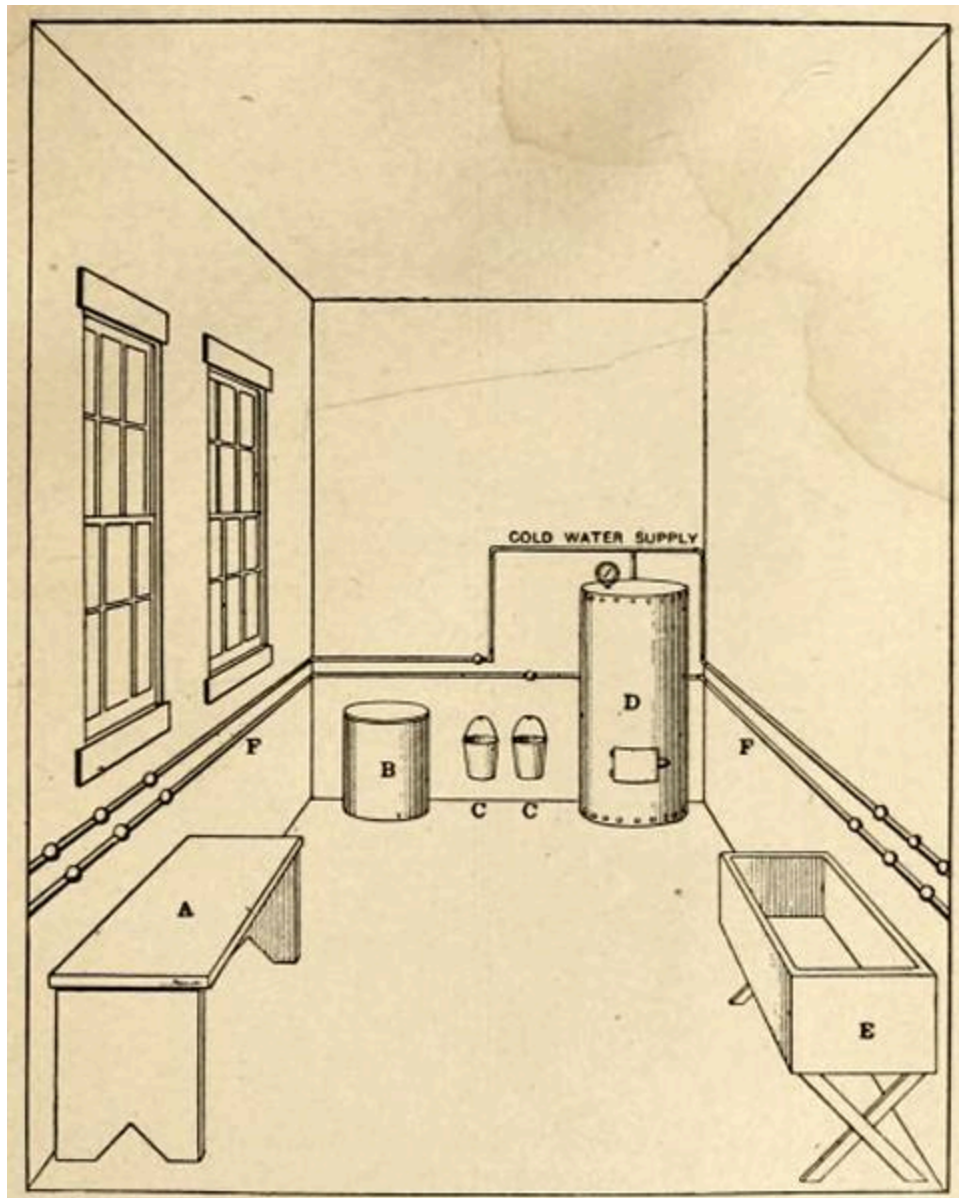
SIR,—I have received through Mr. Alex. Paul of your method of feather dyeing, and acknowledge that your method is far superior to my most vivid imagination of what can be executed in the art of feather dyeing. I would not sell the information I have obtained, nor would be without it for a great deal more than I paid for it.

E. BAUER.

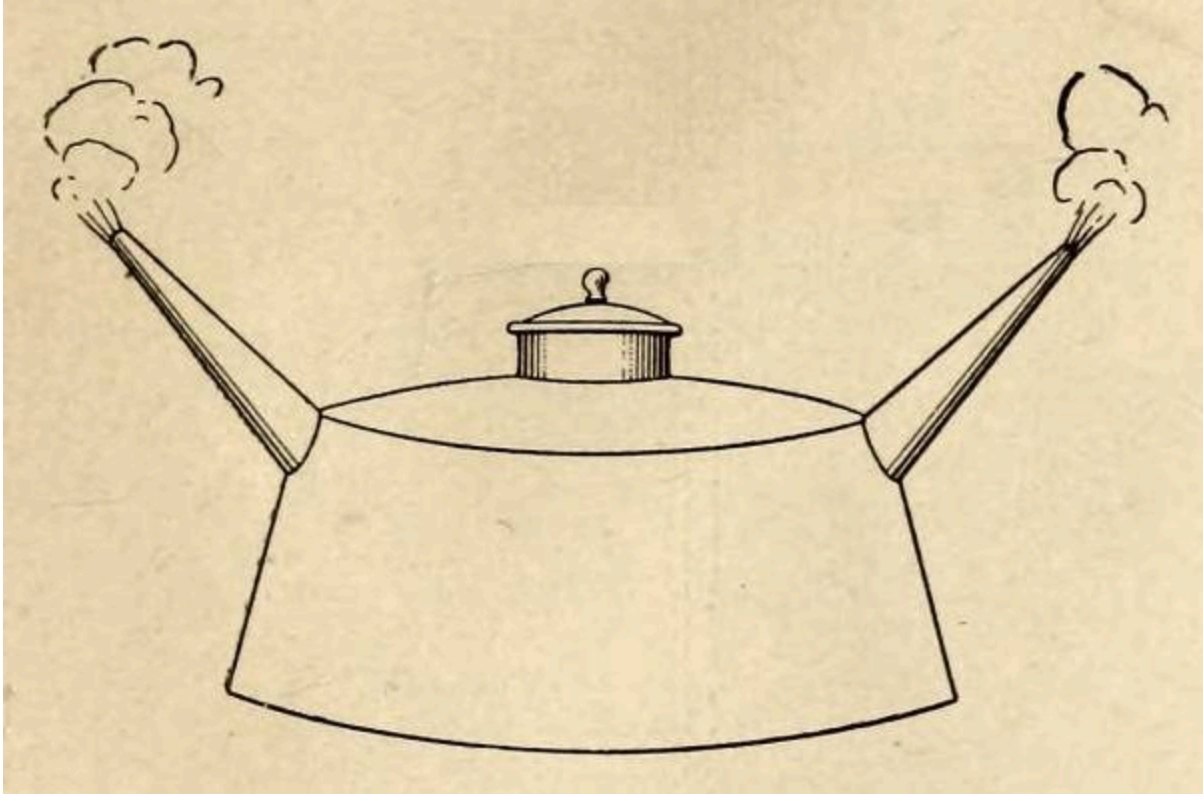
ALBANY, Jan. 28, 1885.

I am glad to have had the opportunity to learn the art of feather dyeing as taught by Mr. Alex. Paul, and will never regret it. It is the easiest, most economical and the best method known. I paid Mr. Paul fifty dollars for his instruction, but I would not be without it for five hundred. It is, without exception, the finest method extant.

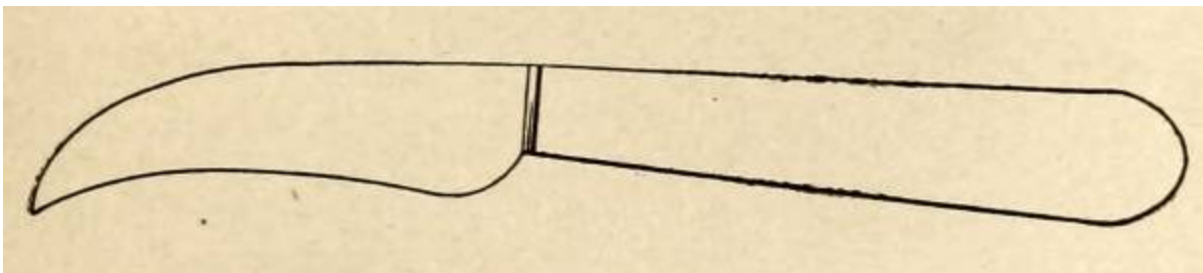
JOHN P. MAYER.



A, Work-bench. B, Hydro-extractor. C, C, Buckets. D, Boiler. E, Stationary Wash-tub. F, F, Hot-water Pipes.



Steaming Kettle



Curling Knife (Half Size)

OSTRICH FEATHER DYEING.

GROWTH OF THE OSTRICH FEATHER TRADE DURING THE PAST TWELVE YEARS IN THIS COUNTRY.

The manufacturers of America could have been counted on the fingers of one's hand a dozen years ago. At the present time New York alone can boast of between forty and fifty. Enterprising men in other cities and throughout the country are yearly becoming interested and endeavoring to take hold of this young and profitable business, and we can look to ostrich feather manufacturing at the present time as one of our staple industries. The greatest disadvantage manufacturers have had to contend with was a lack of knowledge of coloring. Our greatest chemists and aniline manufacturers have worked diligently, contributing largely to the progress of wool, cotton and silk dyeing, but the amount of dyestuffs used by the largest feather manufacturers was of such minor importance that it did not seem profitable for them to investigate; consequently the art of ostrich feather dyeing progressed very slowly. Feather dyers a dozen years ago were scarce, and the art (if in those days it could be called such), was controlled to a great extent by the French, who, judging by my experience with them, impressed me as being the most egotistical mortals, and decidedly orthodox in their methods, absolutely refusing to take hold of anything new that might prove beneficial to them, and so jealously did they guard their (as they considered them) secrets, that during working hours every one of them even their employers, were denied admittance to the dye-house.

Millions of dollars are at the present time invested in ostrich feathers in all conditions, in the cases of raw stock in the ware-houses and in the flourishing ostrich farms now in existence; and a milliner's window without its rich clusters of ostrich tips and plumes would to-day be a rare sight. They are used not only in the trimming of hats and bonnets, but fashion demands their use in trimming dresses, wraps, etc., and to a large extent they are being used in making handsome and very valuable fans. It is to be regretted that London and Paris markets are supplied with the choicest of the goods that come from the Cape, and America gets the leavings, although our market consumes equally as many, if not more. It is only a matter of time, however, when manufacturers will be importing raw stock direct.

THE BIRD, ITS PLUMAGE AND HABITS.

Years ago, before the trade had begun to assume its present proportions, the supply of feathers came chiefly from Egypt; the bird being hunted by the natives, and generally killed for its plumage, which was in quality far superior to the feathers which are to-day raised on farms at the Cape. The flues or fibres of the Egyptian were very close and compact and very strong in texture and of great durability, and having a great affinity for color, they were capable of standing a great amount of manipulation without receiving serious injury. A serious objection to them was that one-half, or more, were marked where the bird pecked them with his bill, giving them a moth-eaten appearance, and few of them could be used for white, as they were more or less stained on the ends, a dirty yellow, which soap would not remove and acid would only develop, there being at that time no known method of bleaching them, as the virtues of Peroxide of Hydrogen or Permanganate of Potash as bleaching agents were unknown to the dyers. Enterprising capitalists saw a profitable field for investment in the propagation of the bird, and, as a result, the supply has greatly increased, and the quality of the plumage is far superior in every respect to the wild Egyptian ostrich.

A full grown ostrich will weigh about three hundred pounds, and stands about seven to eight feet in height. In the breeding season they will travel in broods of from three to five in number, one of which is invariably a male. The hens lay their eggs in a pit scraped out with their feet, the sand forming a ridge around it. When they have accumulated a dozen eggs or so the male begins to brood, always taking his place on them at night, surrounded by the hens, while by day they will relieve one another. Again, at times the hatching has been left entirely to the sun. North African eggs present a smooth surface, while those of the South are pitted.

At the present time an ostrich farm is in progress in California; it is as yet a very young institution, and its success is being watched with interest, but, in my opinion, while the bird will live and thrive, the quality of the plumage will be very inferior to those in their native clime. So much has already been written concerning the bird's powerful digestive organs, and so exaggerated that we will not try to discredit or contradict it. It is hardly

necessary to remark that there is scarcely enough substance in ten-penny nails or doorknobs to fatten an ostrich on.



BRIEF SKETCH OF DYESTUFFS USED BY ME IN MY METHOD OF DYEING.

LOGWOOD.

Logwood is met with in commerce in the shape of large blocks, averaging about four hundred pounds each in weight. On the surface the wood is a dirty deep brown red, but within, where it has not come in contact with the atmosphere, its color is much brighter. The tree is a native of South America. It has been known and used ever since a short period after the discovery of America. During the twenty-third year of the reign of Queen Elizabeth an act of Parliament was passed, forbidding its use as a dyestuff, because it did not yield fast colors. This act was repealed, however, by an Order in Council of Charles II., which proceeds to set forth that great improvements have been made as regards the obtaining of fast colors from logwood. The following are the chief varieties of logwood, distinguished by names derived from the localities of exportation: Yucatan, Laguna, Domingo, Monte Christo, Fort Liberte, Jamaica, etc.

Logwood is to-day one of our most important dyewoods, as indeed it enters in feather dyeing into all of the dark or staple colors, such as black, navy blue, brown, green, garnet, etc. To extract the substance requires considerable boiling, especially if used in the form of chips; if it is used ground, which is by far preferable to chips in feather dyeing, it requires much less boiling to extract the substance. The dyer will often find logwood, that, although purchased under the name of a most excellent brand, will be far inferior to what he has been using, in which case it is well to look for an adulteration of some sort, which it is not at all easy to detect, only when it does not produce the desired result.

TURMERIC.

The substance known as turmeric is the under-ground stem of a plant which grows in a wild state in some parts of China and India. It emits a strong, but pleasant odor, and its taste is peppery, aromatic and bitter at the same time. The plant, however, is cultivated in Java and Bengal; the latter country

producing the better quality. Although turmeric is rich in coloring matter, its want of permanence is a hindrance to it. It is generally sold in powder, ground down very fine. It should be quite dry; if damp, it loses its color, turns a dull yellowish brown, and dyes flat shades. A good turmeric should show a beautiful lustre. It enters into a majority of the dark colors in feather dyeing, and, although used as a body for colors only, a great deal depends upon it as to the result.

BICHROMATE OF POTASH.

This dyestuff, known as red chrome and bichromate and often times simply as chrome, consists in one equivalent of potash, with two equivalents of chromic acid. It contains no water, and consequently cannot lose any weight by exposure to heat or dry air. It will not attract moisture from a damp atmosphere. It dissolves readily in ten times its weight of cold water, and is insoluble in alcohol. It forms bright red crystals, and the solution is of a deep orange yellow. Bichromate of potash is a most powerful oxidizing agent, and produces very complex and interesting changes in tinctorial bodies. It is an intense poison. Its most extensive application is now in the production of blacks, along with logwood; indeed, without its aid it would be next to an impossibility to produce a glossy and permanent black on ostrich feathers. In giving depth of shade to all dark colors it is used in preference to any thing else, and I have never found any to contain any adulteration that was perceptible, or that was a hindrance to its good qualities. It is used in ostrich feather dyeing always in a diluted form, in a very high temperature of water.

ARCHIL.

About the thirteenth century an Italian, Tederigi by name, during travels in the East observed the tinctorial powers of a certain class of plant of low organization, called lichens, and introduced the color into Europe under the name of archil. For this discovery he was amply rewarded by the government, besides amassing a large fortune, as the supply for years came from Florence. At first the weeds were collected on the shores of various islands in the Mediterranean; but on the discovery of the Canary Islands, in

1402, large quantities were obtained from there. Later on they were imported from Cape Verde; and now they are also obtained from Madagascar, Zanzibar, Angelo and Lima and various localities in South America.

The weed does not contain any coloring matter already formed, but under the influence of ammonia and the oxygen of the atmosphere gives rise to archil. The manufacture of archil was for centuries carried on in wooden troughs. Two hundred parts lichens were placed in the trough together with about two hundred and forty parts of decomposed urine, and the mixture well worked every three hours for forty-eight hours. Five parts of slaked lime, one part of arsenious acid and one and one-quarter parts of alum were then added, and the whole well stirred and allowed to ferment. The stirring was repeated, from time to time, for a month. The contents of the trough were then removed to casks, and left to stand, thus improving the color. Archil is also one of the most important dyestuffs used by the feather dyer, principally entering into the composition of garnet, plum, brown, etc. Contact with acid will destroy its coloring virtues by turning it a dull brown red.

SAFRANINE.

It is prepared by treating aniline oils successively with nitrous acid and arsenic acid, and one of an alkaline nitrate at about 212° Fah., for a short time. The product is extracted with boiling water, neutralized with an alkali filtered, and the color precipitated with common salt.

Pure hydrochloride forms thin reddish crystals, which are soluble in water and in alcohol, yielding a yellowish red solution. The most characteristic reaction of safranine is that when concentrated sulphuric acid is gradually added to its solution, the color changes to violet, then to blue, dark green and light green. Then, on diluting the solution with water, the same changes of color take place, only in the reverse order.

In feather dyeing safranine is used chiefly in making light colors of a pinkish hue; such as pink, terra cotta, and to give a tint to ecru, beige and such colors.

OXALIC ACID.

Oxalic acid, a most powerful acid, occurs combined chiefly with potash juices of plants of the genus *oxalis* and *rumex*. Artificially it was obtained by the action of nitric acid upon sugar and starch, but has been prepared latterly by treating spent dyestuffs with alkalies. Oxalic acid forms colorless transparent crystals, which are inodorous, intensely sour, and do not grow moist upon exposure. Should they become damp, some nitric or sulphuric acid, used in the preparation, has not been thoroughly removed. It is soluble in its own weight of boiling water, but requires about eight times its weight of water at 65° Fah. Oxalic is one of the largely used acids in feather dyeing, being used in a number of light colors for the purpose of developing the color. In developing blues it is invaluable. Other colors it will totally destroy, violet or safranine, for example; and it is used in place of sulphuric acid for the purpose of extracting color.

INDIGO BLUE.

Indigo is derived from several plants of warm climates. In the plant the color exists as a yellowish liquid; but when extracted and exposed to the action of the air it becomes insoluble, and takes an intensely blue color. The cultivation of the plant is carried on chiefly in India, Java, Egypt and Louisiana. Indigo comes in the market in lumps, which, if of good quality, presents a deep bluish purple color, and exhibits a fine reddish coppery lustre if rubbed with a hard, polished body. If very hard or heavy, or when the color is very dull, blackish, greenish or brownish, the quality is below the standard. It is, however, of very little consequence in ostrich feather dyeing, and its impurities would scarcely at any time be noticeable. It should, however, dilute thoroughly in boiling water, and if there remain a sediment of any proportion, the indigo is impure. Sulphuric acid is generally used to develop the color.

SULPHURIC ACID.

Sulphuric acid, commonly called oil of vitriol, a common, yet very important, acid. Although not used to any great extent in ostrich feather

dyeing, it occurs in commerce in various states and degrees of purity. It was at one time prepared by distilling dried copperas at a high temperature. It is now obtained in greater purity from the alkaline bisulphates. It is a clear colorless oily fluid, weighing about eighteen pounds to the gallon. If mixed with cold water, a great increase of temperature takes place. It rapidly destroys organic bodies, depriving them of their oxygen and hydrogen, and leaving the carbon behind, as a blackish mass. If any particle of organic matter falls into a carboy of acid, it is decomposed, and imparts a dark color to the liquid. It takes up water from the air rapidly, if left uncorked, and thus dilutes itself. Its use in feather dyeing is principally to extract colors that are too dark.

COPPERAS.

Copperas is generally prepared from the soft, white variety of iron pyrites, frequently found to a great extent in the coal measures. These, on exposure to air and moisture, decompose the latter, taking up oxygen, and are thus converted into sulphate of iron. Copperas forms pale greenish blue semi-transparent crystals, containing forty-five per cent. of water. If this be expelled, there remains a dull whitish powder. The crystals dissolve readily in one and one-half times their weight of cold water, and less than half their weight of boiling water. The direct uses of copperas have very much diminished in feather dyeing; as for dyeing black in conjunction with logwood it has been almost entirely superseded by bichromate of potash. In drabs and in saddening down light colors it is, to a certain extent, still used. It is used in quantities so small, however, that there is no serious results to be feared, as it must be used in quantity to injure the fibre.

BISMARCK BROWN.

Bismarck brown is a product that is used in feather dyeing to a considerable extent, chiefly in a diluted form. It dissolves readily in boiling water. It comes in the form of a powder, of a dirty black hue, and in liquid it is a heavy yellowish red. It makes a fast color, alkali having but little effect on it. Oxalic or sulphuric acid will brighten the color, and turning it more on the red order. In giving a brownish hue to such light colors as beige, ecru,

etc., it is invaluable. It is used by some in the topping of dark brown. It has such a great affinity for the fibre of feathers that it is very difficult to remove therefrom.

CONCENTRATED COTTON BLUE.

This blue appears to the consumer in the form of crystals or coarse powder of a purplish tint. It is not universally used among feather dyers, although it is the most reliable aniline blue in use. It is used in conjunction with oxalic acid to develop. It is fast to light, and possesses a great many advantages of value. It is soluble in water, either hot or cold, and is used in the production of the palest shades, as well as in the darkest navy blues.

ROCCELINE.

A patented product, that in feather dyeing is capable of taking the place of all other reds. It is the only dyestuff that satisfactorily takes the place of extract of safflower, producing, with the aid of a proportion of oxalic acid, the most beautiful shades of scarlet and cardinal. It is perfectly fast to light, dissolves readily in boiling water, and comes to us in the form of a dull red powder. Its adulterations, if it contains any, have never interfered with its success; in fact, to the feather dyer it contains virtues too manifold and valuable to enumerate.

DYEING RECIPES.

WHITE.

BLEACHING, OR WHAT IS COMMONLY CALLED CLEANING.

After stringing your feathers and marking your tickets, prepare luke warm soap-water and wash thoroughly between the hands to remove all dirt and grease. Rub the soap on the feathers, rinse thoroughly in luke warm water two or three times for the purpose of removing all particles of soap, which is very important; just as much so as removing the dirt. For one to one hundred feathers you can use a common porcelain wash bowl. Prepare bath by using one gallon of clear cold water, add to that a small handful of starch, powdered or lump starch will answer. Enter feathers, rubbing them thoroughly between the hands to expand the flues and get them in condition to receive the color, so as to insure an even shade; after which add about one-half teaspoonful of oxalic acid and a drop of diluted violet, just enough to give your bath a pale lavender tint. Enter feathers, and let remain in bath about one minute, keeping them under the surface and agitating by rubbing them between the hands; after which squeeze feathers out of bath and dry. The quickest method for a few feathers is to have a small quantity of clean, powdered starch, and rub them around in it. The starch will immediately absorb all moisture, and you have but to beat it out of the flues, as it dries either on a clean board or between the hands. It is but the work of a few seconds. This method of drying insures an unsoiled color, as the feathers are dry a few seconds after leaving the bath.

Great care should be used to bring your violet diluted thoroughly, so that no particles may enter the bath and spot your goods. In diluting your violet use boiling water, and shake well in bottle, and let it stand for a time, when all sediment will settle at the bottom, and will not again mix with your color.

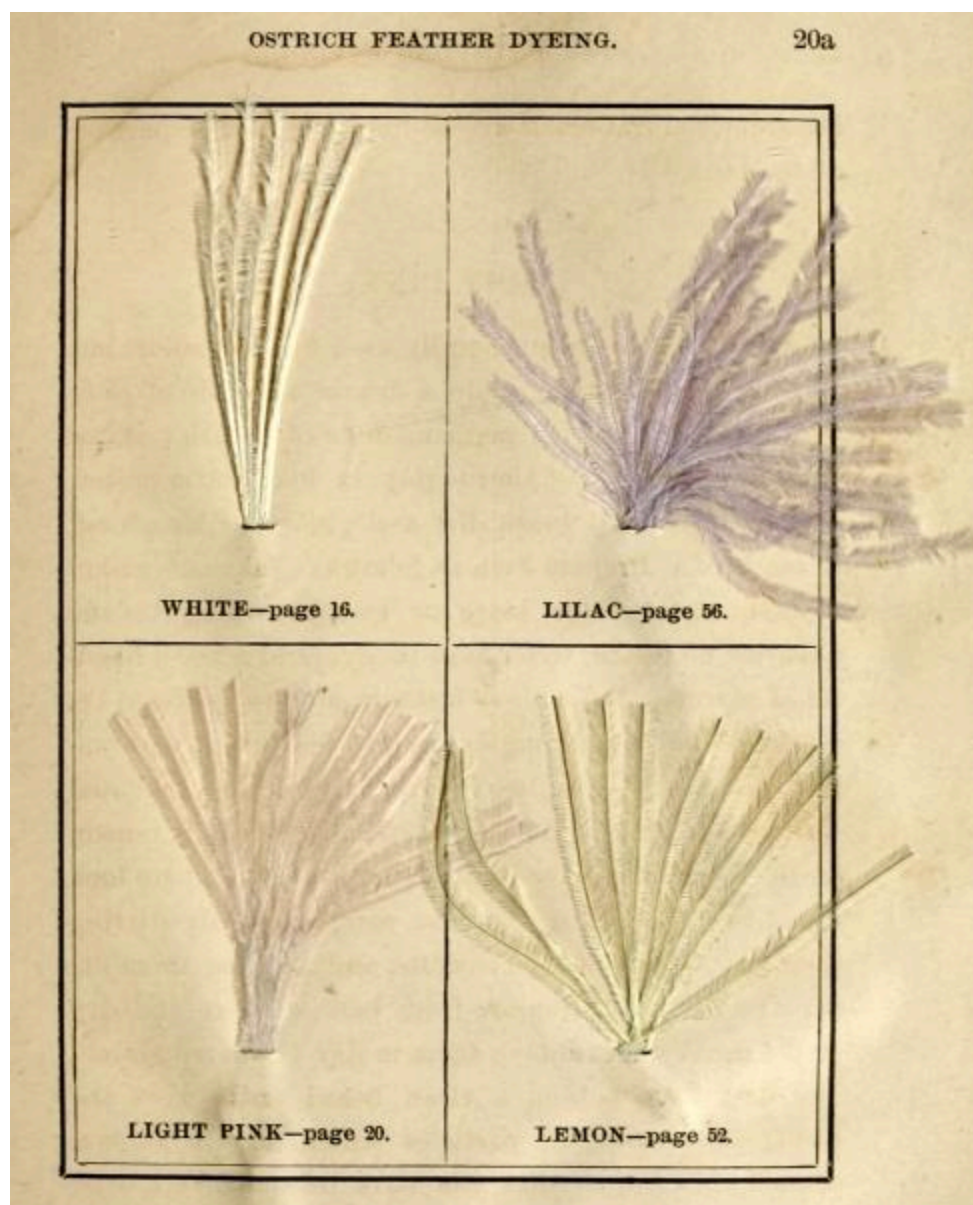
It is very important to use only the amount of oxalic acid mentioned in recipe, as a greater quantity would destroy your color by turning the violet a dirty blueish green, and much less than the quantity mentioned would have

a tendency to cast a lavender tint on your goods. Should you, by mistake or carelessness, spoil your white, proceed to rinse off all the starch in cold water first; then in luke warm water to remove all the acid from feathers, and then use soap and hot water, and wash well, and rinse. Mix a fresh white bath as directed in the recipe, and proceed this time with more care.

BLEACHING LIGHT COLORS WHITE.

Old faded light colors, such as blue, pink, ecru, corn, drab, etc., that you are desirous of bleaching white, can be accomplished in the following way. Wash feathers thoroughly in warm water, using soap. Add a small pinch of soda, after which rinse in about three warm waters to insure the removal of every particle of soap. Dilute in clean bowl or basin one-quarter ounce of permanganate of potash in one gallon of boiling water. The water must be as hot as steam or fire can make it. Enter feathers, and let remain in bath about one minute, a few seconds more or less will do no harm, nor will it make any material difference in the result; continually agitating in bath with clean stick, after which you will notice that the feathers have assumed a light, full brown color. Take out of the bath, but do not rinse them; let the loose color drain off for a few seconds, meantime empty bath and rinse your bowl thoroughly; then dilute half an ounce of oxalic acid or sulphurous acid in one gallon of boiling water. The water must be absolutely clean. Enter feathers, and let remain in until all the color has entirely disappeared, gently agitating while in bath. After the bath has become transparent and the feathers white, which will take about two minutes, empty out about two-thirds of the bath, and add cold water to reduce to hand heat; then add a small handful of starch and a drop of diluted violet, and enter your feathers, and let them remain in about one minute, squeeze out and dry in starch. Blue you will generally find the hardest of all light colors to remove for white, the soda and permanganate seeming apparently to decompose the color. The moment it enters the oxalic bath, it generally, to a more or less extent, develops the color again. Such being the case, after rinsing in luke warm water to remove acid, return to a weak soda bath for a minute, and then rinse and return to permanganate bath, rather weaker than the first one; in other words, repeat the first operation all through, only in weaker solutions.

This process can be used successfully in bleaching all light colors white. In bleaching natural blacks, however, it would not be practicable. A recipe for bleaching natural black will be found in another portion of the book.



WHITE—page [16](#). LILAC—page [56](#).
LIGHT PINK—page [20](#). LEMON—page [52](#).

LIGHT PINK.

White feathers are generally used for this color, but all light colors can be made a beautiful shade of pink by first bleaching with permanganate of potash. After washing and rinsing thoroughly in luke warm water, soap to remove all loose dirt and grease, or bleaching, if required. Prepare bath as follows: Take one gallon of luke warm water, more or less, according to the quantity of feathers you have to dye add a small handful of starch. Enter your feathers and rub around between the hands thoroughly to open the flues so as to insure an even shade; add a couple of drops of diluted safranine to bath. Enter feathers, and let them remain in the bath about one minute, or until feathers look about two shades darker than sample; gently stirring them around in bath meanwhile, and keeping them under the surface. Remove from bath, squeeze and dry in the usual way, rubbing them in dry powdered starch, and beat them out on a clean board or between the hands to remove all particles which might adhere. Should your sample that you have to match be a little on the yellowish order, a drop of diluted Bismarck brown added to bath will bring the desired shade; or if a very brilliant shade or rose pink, a drop of diluted violet added to the bath and increase temperature; a little judgment is always necessary; as, for example, should you require a dark shade, you would naturally let your goods remain longer in the bath than the time specified in recipe, or add a little more color, and if a very pale pink is wanted, less time and color should be used. Should you, at any time, find your color, after being dried, a couple of shades darker than your sample, rinse goods in luke warm water, and enter feathers, pass through for a minute, and dry.

LIGHT BLUE.

All other faded out light colors can be made into a delicate shade of sky blue by first bleaching with permanganate of potash process for the purpose of removing colors. White feathers that are only dirty and greasy must be thoroughly washed and rinsed in luke warm water, after which prepare bath as follows: For one gallon of luke warm water, more or less, according to the amount of feathers to be dyed, add a small handful of clean starch; enter your feather and rub them around in bath for a second between the hands to open the flues, to admit color evenly; add about one teaspoonful of oxalic acid, enter feathers and let remain in bath a few seconds longer; then remove feathers from bath, and add a couple of drops of concentrated

cotton blue diluted; re-enter feathers and let them remain in about half a minute; increase temperature of your bath a few degrees by adding some hot water; take feathers out of bath and add thereto a drop of diluted indigo blue; re-enter, and keep them well under the surface of bath to give them an even color, and allow to remain in about thirty seconds longer. Take them out of bath, squeeze out and dry, either in powdered starch or by beating on a clean board or table. Under no circumstances allow feathers to hang wet and motionless on line during process of drying without beating the starch out. The result of so doing would cause the feathers to look thin, shriveled, and injure the color and quality of goods. The same care should be observed not alone in this, but in all colors.

In light blues your bath should look about two shades darker than the sample to be matched. Where a darker shade is required, more color can be added; and, through carelessness or negligence, should you allow your color to become too dark, rinse off your feathers in cold water first to remove the starch, and then in luke warm water a couple of times to draw off all acid, and pass feathers a few seconds through a bath of luke warm water with a small pinch of soda in it, which will have the effect of drawing off all surplus color; after which rinse in luke warm water, and mix a fresh bath of luke warm water and starch and one-half a teaspoonful of oxalic acid; enter your feathers and carefully add color until you have obtained the desired shade.

ECRU.

All old colors, excepting dark brown, bottle green, navy blue, black, garnet, etc., can be dyed a good shade of ecru. Begin an old color by passing them through a solution of hot water, about one ounce of soda to a gallon of water, for about 30 seconds; after which take them out and rinse by passing them through clean boiling water, which will draw off more color than it would seem possible the feathers could contain. If all the old color, or enough of it, be not removed, put feathers through the permanganate of potash process. For dirty white feathers simply wash them thoroughly with soap and hot water, and rinse well; then prepare your bath as follows: One gallon of hand warm water, add a small handful of starch, and enter feathers, rubbing them around thoroughly, and getting the starch rubbed

into the flues; then add to bath a small quantity of copperas, about the size of a bean, and re-enter your feathers and let remain in bath about one minute or less; after which add a few drops of logwood liquor and about a teaspoonful of diluted aniline brown, first removing feathers from bath; enter feathers and let remain in bath about one minute, being careful to keep them moving in bath. If found a little too brown to match your sample, a small pinch of turmeric added to bath will reduce the shade. If they are found a little too yellow for sample, a drop of diluted violet will answer.

If the dyer, through his own carelessness, should allow his color to get too dark, proceed to extract color as follows: dilute in about one gallon of luke warm water one-half teaspoonful of oxalic acid. Enter feathers, first rinsing off starch in cold water; let them remain in about half a minute, and rinse off about three times in hot water to remove acid. The acid will turn the feathers a bright yellow, and after rinsing off well the yellow color will have entirely disappeared, and the feathers a light shade of dust. Prepare a fresh bath as per recipe, and, using more care, enter feathers and pass through until you have acquired the desired shade.

In the first bath, should a very dark shade be required, add a little more logwood and copperas than directed in the recipe, and if a very light color, a little less.

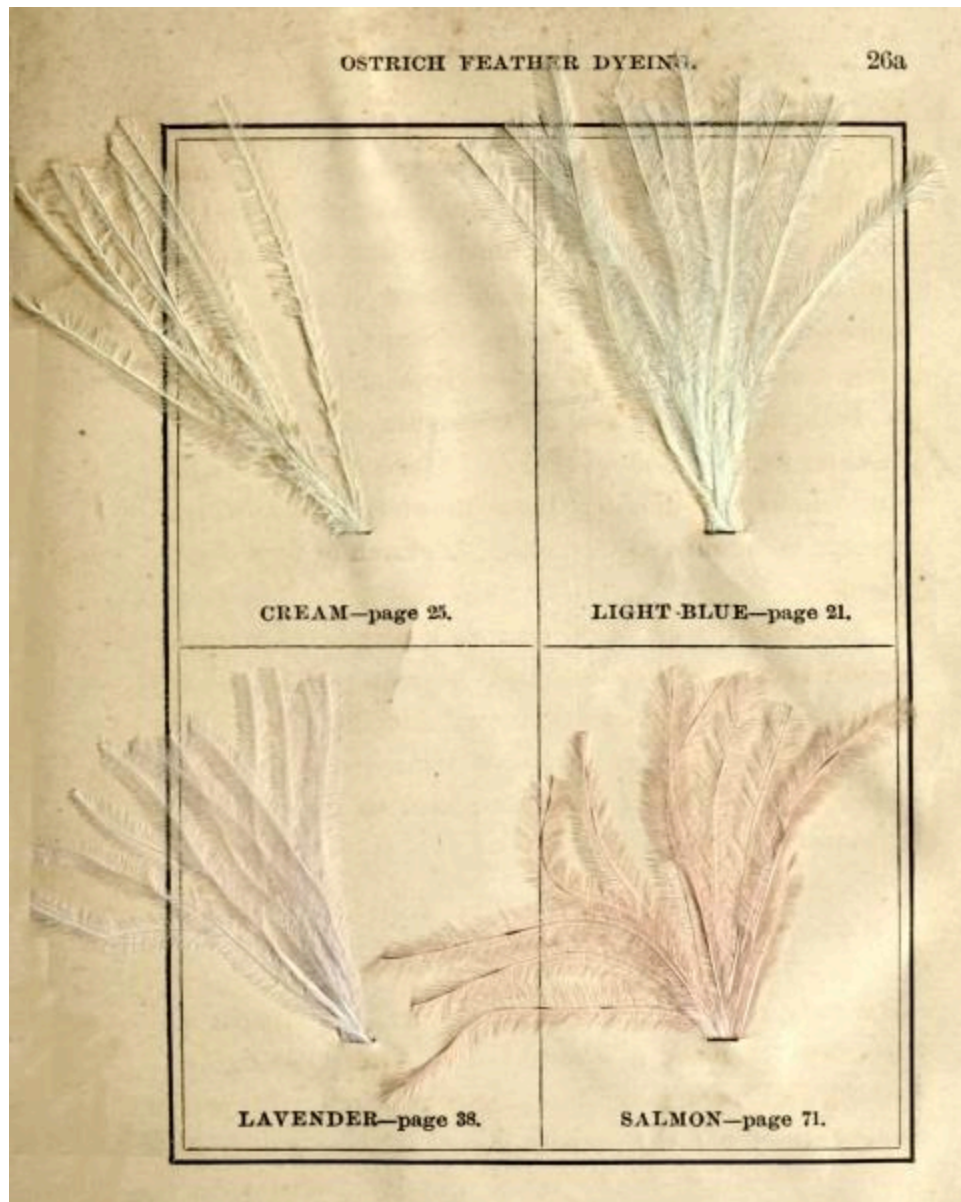
CREAM COLOR.

There are numerous methods of producing this most beautiful, yet simple, shade. Any yellow substance in conjunction with oxalic acid can be used with more or less fair success. A great many dyers use a few drops of diluted logwood, developed with the aid of oxalic acid. The color this produces is very satisfactory when finished, but no sooner is it exposed to strong light than the color becomes a dirty drab shade, caused by the acid leaving the feathers, the logwood becoming oxidized.

The best and most permanent shade of cream color is obtained in the following manner: Thoroughly wash and rinse your feathers to remove every particle of dirt, for it is as necessary to have the feathers clean as if they were for a white, and if they are very dirty or old faded out colors, put them through the permanganate of potash process, and then remove all color. Prepare bath of one gallon of luke warm water and a small handful of

starch; enter feathers and rub around in bath between the hands; meantime dilute in about one pint of boiling water a small five-cent package of essence of coffee (commonly called chicory), and boil for a few minutes; then add a few drops of the liquid to the bath, and add thereto a teaspoonful of oxalic acid. Re-enter feathers and let remain in bath about one minute, constantly moving them around; after which squeeze them out and dry, either in starch or on a clean board.

The result will be a rich and permanent cream. Should a pink or brownish tint be required to match sample, a drop of Bismarck brown added to bath will produce the desired result; or if wanted a little more yellow, a few grains of turmeric added to the bath will answer.



**CREAM—page [25](#). LIGHT BLUE—page [21](#).
LAVENDER—page [38](#). SALMON—page [71](#).**

SILVER GRAY.

A very delicate color, requiring feathers almost a pure white to make a clear shade. After thoroughly washing and rinsing, or bleaching if required, with permanganate of potash, prepare a bath of one gallon of luke warm water, and add a small handful of starch. Enter feathers and manipulate between the hands; then add to bath a small piece of copperas, about the size of a

pea, and a few drops of diluted logwood liquor; re-enter feathers and let remain in bath until in appearances they are two or three shades darker than sample; then add to bath a couple of drops of diluted violet, first removing feathers from bath; let them remain in a few seconds longer, and squeeze out and dry in the usual way. The violet gives your feathers the brilliant shade that is so desirable in silver grays.

Be careful in drying them not to use starch that has been previously used in drying feathers that have been dyed in acid baths, as it would be liable to spot your color. Should you, through carelessness or otherwise, allow your color to get darker than shade desired, rinse feathers off a couple of times in cold water to remove starch; then dilute half a teaspoonful of oxalic acid in a gallon of hot water, and pass feathers through it for a few seconds, and then rinse off twice in boiling water. After which prepare a bath same as per recipe, using more care, and pass feathers through until you have obtained the desired shade.

BISMARCK BROWN.

Wash and rinse your feathers, after which prepare a bath of one gallon of boiling water and about one ounce of turmeric and half an ounce of copperas; enter your feathers and let them remain in bath about two minutes, more or less, after which take out and rinse twice in cold water. Meantime have boiling a bath of half a pound of logwood to a gallon of water, and enter feathers at boiling temperature, letting them remain in about ten seconds or longer. Should a darker shade be desired, take out and rinse in cold water, after which dilute a half teaspoonful of aniline brown in a gallon of boiling water. Reduce temperature a little with cold water. Enter feathers and let them remain in about three minutes; then cool off a small portion of the bath, and add a small handful of starch, pass feathers through and dry.

If a lighter shade is wanted, add a drop of sulphuric acid to the starch bath and pass feathers through. If the sample to match be more on the yellow order, about twice the amount of turmeric in the first bath; and if desired more on the red, use no turmeric, only copperas, in the first bath. If a darker shade is wanted, let them remain a longer time than that specified in the logwood bath. Any light color can be used to make a Bismarck brown; but

if very dark colors are used, it is well to draw off some of the color, doing it in the usual way.



SEA-FOAM—page [70](#). SILVER GRAY—page [26](#).
ECRU—page [23](#). TRILEUL—page [58](#).

SEAL BROWN.

For seal brown it is not necessary to wash your feathers, nor to bleach off any color. Any old colors, excepting black, can be made a good shade of seal brown. Begin in bath by diluting about two ounces of turmeric in a gallon of boiling water (more or less matters not). Enter your feathers and keep them well under the surface of the bath about two or three minutes; after which take out and rinse in cold water twice. In the meantime boiling a bath of logwood about one pound to a gallon of water. If boiled on fire about fifteen minutes is necessary, and if boiled with steam a half hour is required. Enter feathers in logwood and let remain in about three minutes, keeping them well under the surface of bath, after which take out and rinse; if in cold water about twice, then dilute a half an ounce of bichromate of potash in a gallon of boiling water, and see that bichromate is thoroughly dissolved. Enter feathers and let them remain in about ten seconds, a longer time if a very dark shade is wanted; then take them out and rinse thoroughly in cold water; after which add to your logwood bath about one tablespoonful of extract of archil; bring bath to a boil and enter your feathers; cover up bath and let them remain in about four minutes; a little more or less time, in this bath is of no material difference in color, only to make the shade heavier or lighter. Take your feathers out of bath and rinse in cold water; mix a small handful of starch in about a quart of cold water, and pass feathers through and dry in the usual way.

If your color be darker than the shade you desire, add a drop or two of sulphuric acid to starch bath, and pass your feathers through for a few seconds. If found to be lighter than the shade you desire, rinse off the starch from your feathers in cold water; then dilute a quarter of an ounce of bichromate of potash in a gallon of boiling water, and pass your feathers through; after which rinse, starch and dry.

Another excellent method for quick seal brown is as follows: dilute two ounces of turmeric and half an ounce of copperas in one gallon of boiling water, and let them remain in about two minutes; take out and rinse, then enter in a strong bath of logwood at boiling, and keep under surface about three minutes; after which rinse; then mix a bath of a quarter to a half teaspoonful of aniline brown in a gallon of boiling water. Enter your feathers and let them remain in bath about three minutes; take out, rinse, starch and dry. If required darker, re-enter into logwood bath for a few

seconds. If wanted lighter, add a drop or two of sulphuric acid in your starch bath, squeeze out and dry in the usual way.

NAVY BLUE.

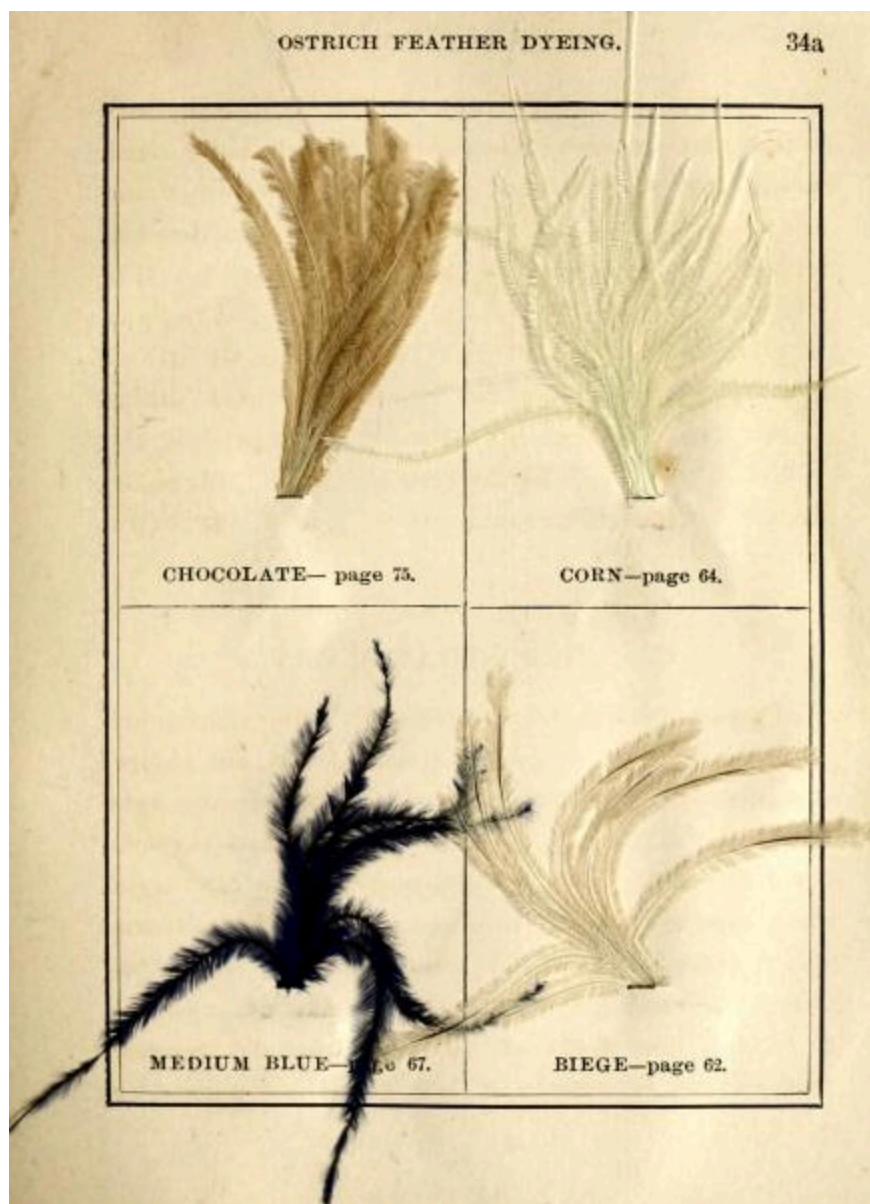
All light colored feathers can be used for navy blue without first either washing or bleaching out any of the color. But if your feathers be very dirty or greasy, especially the latter, wash them well in warm soap water and rinse. Prepare bath by diluting about one teaspoonful of concentrated cotton blue in one gallon of boiling water; add about a teaspoonful of oxalic acid. Stir around well to thoroughly dissolve aniline; then enter your feathers, and raise temperature of your bath to boiling. Let feathers remain in about three minutes; a minute more will not do any harm, only have a tendency to make your color a little richer; after which take feathers out of bath and rinse thoroughly in cold water for the purpose of removing all loose particles of color and the acid; having boiling meantime a bath of logwood of medium strength; enter feathers, letting them remain therein about one-half a minute; take out and rinse in cold water; dilute about half an ounce of bichromate of potash in a gallon of boiling water; enter feathers, let them remain in about half a minute, and stir them around well in bath; after which take them out and rinse in cold water and starch and dry. Should you desire a darker shade, rinse off starch, and return to logwood bath for a few seconds, rinse off and repeat bichromate of potash bath; then rinse, starch and dry. In this way, by repeating the logwood and bichromate of potash, you can darken your color down almost to a black.

Should you get your color darker than your sample to be matched, rinse off starch in clear cold water, and dilute a teaspoonful of oxalic acid in a gallon of hot water almost boiling and enter feathers, passing them through about a half minute; after which take out and pass through a basin of boiling water a few seconds. This will draw off the surplus of logwood and chrome, and then mix a starch bath luke warm; add thereto a half teaspoonful of oxalic acid for the purpose of bringing up the blue. This process will reduce your color three or four shades; then pass feathers and dry. This process of dyeing navy blue produces a rich, even shade that is perfectly fast to light and alkali, and with the smallest degree of judgment by the dyer it is impossible to have a failure.

CARDINAL.

Years ago the most successful shades of cardinal were produced by taking about equal parts of turmeric and oxalic acid and diluting in boiling water, entering feathers in same for a while; then adding thereto about half a cupful of extract of safflower and about the same amount of extract of archil, letting them remain in until the bath was cold. Not a bad recipe, but very expensive.

Prepare your feathers by washing and rinsing thoroughly, after which take about one gallon of boiling water, and add to it about one teaspoonful of oxalic acid, and enter feathers for a few seconds. Take out and add to bath a teaspoonful of rocceline powder, thoroughly dissolved, and re-enter feathers; raise temperature of bath to boiling, either with steam or fire, and let feathers remain in about four minutes. If quite a dark shade of cardinal be required, add to bath about a tablespoonful of extract of archil and let remain in a little longer, or a few drops of diluted violet in bath will answer instead. Then empty out all but a small quantity of your bath and cool off with cold water, and add a small handful of starch. Pass feathers through, squeeze out and dry. The result is a most beautiful shade of cardinal. This color is perfectly fast to light. If your shade to match should happen to be slightly on the yellowish order, a few drops of diluted aniline brown added to bath with rocceline will produce the yellowish tint. It is hardly possible to spoil this color, except by the extravagant use of one of the ingredients.



CHOCOLATE—page [75](#). CORN—page [64](#).
MEDIUM BLUE—page [67](#). BEIGE—page [62](#).

CRUSHED STRAWBERRY.

Prepare feathers by washing and rinsing thoroughly in luke warm water; or if old, dark, faded out colors, pass them through bleaching process of permanganate of potash; afterwards being careful to rinse all the acid out before entering bath. Prepare bath by diluting a small handful of starch in about a gallon of luke warm water, enter feathers and manipulate

thoroughly between the hands for a few seconds; take out, and add to bath a few drops of diluted safranine; re-enter feathers and let remain in bath about one minute, or until they have assumed a dark shade of pink; then add to bath a few drops of diluted aniline brown and a small pinch of copperas, and enter feathers, letting them remain a minute longer. Take feathers out, and dry in the usual way.

If a very dark shade is wanted, a few drops of diluted logwood added to bath at the time you add the copperas will have the desired effect; or a few drops of violet will answer in its stead. Should you find your color too much on the drab, a few drops of safranine added to bath will have the desired effect. Should you find that your color is entirely too dark for your sample, rinse off the starch in cold water; pass feathers through a solution of a half teaspoonful of oxalic acid in a gallon of hot water for a few seconds; then rinse in hot water twice to remove the acid, after which prepare a fresh bath as per recipe, using more care, and keep in until desired result is obtained.

PLUM.

Feathers that are any color excepting dark green or black can be dyed a beautiful shade of plum. Wash and rinse your goods, and prepare your bath as follows: one pound of logwood to a gallon or more of water, and boil fifteen minutes or longer, then add to bath about a quarter pound of extract of archil, and enter your feathers, letting them remain in bath about five minutes, after which take them out and rinse in cold water. Prepare a bath of half an ounce of bichromate of potash in a gallon of boiling water, more or less, and see that it is thoroughly dissolved; pass feathers through about ten seconds; then take them out and rinse twice in clear cold water; then dilute a small handful of starch in a half gallon of luke warm water, and add to it about half an ounce of soda; pass feathers through for about half a minute and dry.

Should color be found too light for sample, rinse off starch in cold water, and repeat bichromate of potash bath; rinse, starch and dry. An old logwood bath that has been used for other colors will answer for plum, and save boiling up a fresh bath.

OLIVE.

If your feathers to be dyed are very dark colors, such as brown, navy blue, green, garnet, etc., draw off some of the color by passing through a solution of boiling water and half an ounce of soda, and rinse in boiling water twice. Prepare bath by diluting two ounces of turmeric in about one gallon of water. Enter feathers and let them remain in about two minutes,—a longer time will not hurt; after which take them out and rinse in cold water twice. Have a medium strong bath of logwood boiling meantime, and enter your feathers, letting them remain in about two minutes; then take them out and rinse in cold water. Prepare a bath of one gallon of boiling water and half an ounce of bichromate of potash, and after it is thoroughly dissolved, enter your feathers and let them remain in about one minute, longer if a very dark shade be required. Take out and rinse, after which your feathers will have assumed a dark, dull olive, looking not unlike a faded out black. Next prepare a bath of two ounces of turmeric with about one gallon of boiling water, and add thereto a small pinch of green aniline, just enough to give your bath the appearance of being a couple of shades more on the green than the sample to be matched. Enter your feathers and let them remain in about three minutes; first, however, bringing your bath to a boil, after which take feathers out and rinse, starch and dry.

If feathers be found darker than sample to be matched, a few drops of diluted oxalic acid in your starch bath will bring the shade down; and if found lighter than sample, rinse the starch off thoroughly in cold water, and dilute a quarter ounce of bichromate of potash in a gallon of boiling water, and pass your feathers through for a few seconds. If wanted a very dark shade, they should, after having the starch rinsed off, be returned to the logwood bath, then rinsed and give the bichromate of potash bath as above. If found a little too much on the green for sample, a weak bath of turmeric, similar to the first bath of the operation will have the desired effect. There are also some shades of olive where it will not be found necessary to use any green at all; that is when the shade approaches the brown on the olive.

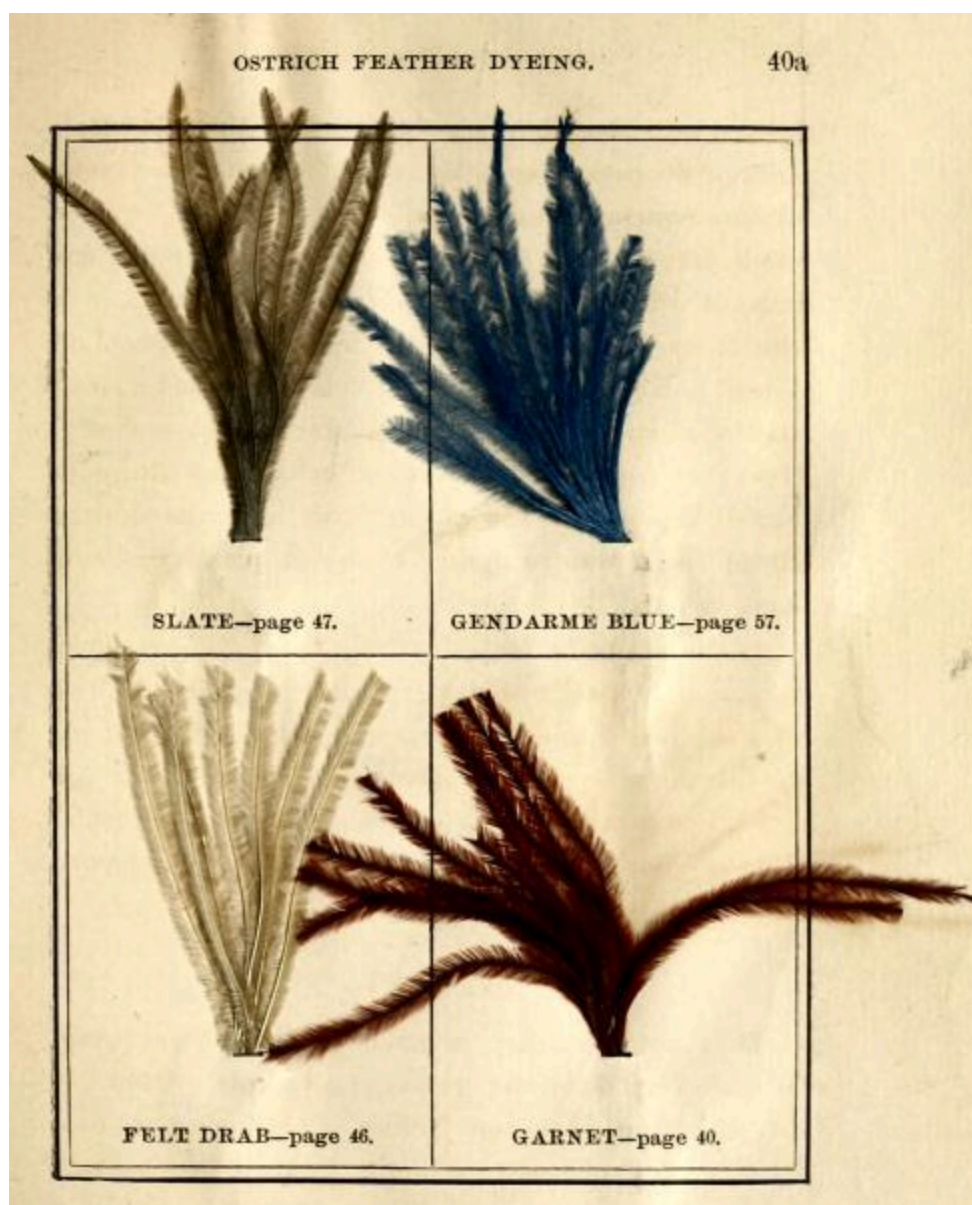
LAVENDER.

Feathers for lavender must be white, or nearly so, if you desire a good clear shade. All light colors can be used by first bleaching with permanganate of potash, or if only dirty white feathers, wash and rinse them thoroughly. Prepare bath of luke warm water and a small handful of starch, rub feathers around between the hands to expand the fibres; then add to bath a few drops of diluted violet. Enter your feathers and let remain about one minute in bath, keeping them meanwhile in motion; take out your feathers and add to bath a drop of diluted safranine; re-enter and raise temperature of bath a few degrees by addition of hot water; let your feathers remain about half a minute in bath; if wanted darker, add a few drops of diluted violet, and if lighter, less; after which take out your feathers and dry them in the usual way, being careful to use clean starch for drying. To use starch that had previously been used to dry light colors that contained acid, would most likely result in spotting your color, as the application of acid to any portion of the delicate color would turn it a greenish blue. If your color be found too dark for sample, you can either wash in a solution of soap water, or else pass feathers through a bath of a teaspoonful of oxalic acid to a gallon of luke warm water, after which rinse off well and put through fresh bath as per recipe.

OLD GOLD.

All light colors, such as light blues, pinks, drabs, yellows, etc., that you are desirous of making old gold need but to be washed with soap and hot water prior to entering in bath. Prepare your bath with two ounces of turmeric and one gallon of boiling water, more or less matters not. Enter your feathers, and let them remain in bath about two minutes, after which add a small pinch of copperas, about the size of a bean. Let your feathers remain in bath about one minute longer, after which take feathers from bath and add thereto a few drops of diluted Bismarck brown; let them remain in bath about one minute longer; take them out, cool off a small portion of the bath with cold water, add a small handful of starch, pass your feathers through and dry. If wanted a very dark shade of gold, a few drops of diluted logwood added to bath will have the desired effect; and if wanted lighter, a smaller quantity of copperas in bath.

If the shade be found entirely too dark for sample, a solution of oxalic acid in luke warm water will draw off a portion of the color and brighten what is left. If wanted a very yellowish shade of gold, use more turmeric, less copperas and no logwood, and be particular to have your bath at all times at a boiling temperature.



SLATE—page [47](#). GENDARME BLUE—page [57](#).
FELT DRAB—page [46](#). GARNET—page [40](#).

GARNET.

It is not necessary to wash your feathers, except they are very dirty and greasy. As a rule all old colors, excepting greens, navy blues or blacks, can be used for this color without bleaching. Prepare bath by boiling about one pound of logwood to a gallon of water or more about fifteen minutes; strain off liquor from wood; add about two tablespoonfuls of extract of archil, and bring again to a boil. Enter your feathers and let them remain in bath about four or five minutes, after which take feathers from bath, rinse twice in clean cold water, and dilute a small handful of starch in a little clear cold water; pass feathers through and dry in the usual way. Should your color be found too dark for sample to be matched, dilute a couple of drops of sulphuric acid in your starch bath, and pass feathers through for a few seconds; first, however, adding a little hot water to increase temperature.

If found lighter than the desired shade, rinse your feathers thoroughly in cold water and dilute half an ounce of bichromate of potash in about one gallon of boiling water; pass your feathers through for a few seconds, rinse thoroughly and dry. Great care is necessary in passing feathers through this chrome bath, as the color will oxidize very rapidly.

If your sample to match be more on the brown shade, a very little archil, not more than one-half the prescribed quantity must be used; and if more on the purple or plum, add more archil than the quantity specified.

In preparing bath, when you have added the archil, be careful in bringing it to boiling temperature that you do not allow it to boil any time, as that would have a tendency to dull your color. By keeping this bath clean it can be used several times, in fact, it improves with age; and, if kept in a crock, so that it will not come in contact with any metallic substance, and when needed just brought to boiling temperature; and if needed, a teaspoonful of archil added to it will produce very beautiful shades of garnet.

This bath can be used to make your plum colors; and if you have an old bath of logwood on hand it is not necessary to boil a fresh one, simply add the archil, and bring to a boil.

TERRA COTTA.

If white feathers, wash and rinse them thoroughly with hot water, and if faded out light colors, extract color by bleaching with permanganate of potash in the usual way; being careful to rinse well in hot water to remove all the acid used in bleaching before entering bath. Prepare bath as follows: about a gallon of luke warm water, and add a small handful of starch. Enter feathers, rub around in bath between the hands, take out and add a few drops of diluted safranine, and copperas about the size of a pea. Enter feathers and let remain in bath about one minute; take out and add about half a teaspoonful of diluted aniline brown; re-enter feathers and let them remain in about half a minute longer; after which dry in the usual way. If found too pink for sample, add a few drops more aniline brown, and return to bath for a few seconds. If found too yellow, add a few drops more of diluted safranine, and keep in bath a few seconds longer; if wanted darker, add a little more of each color, and keep in bath longer.

BOTTLE GREEN.

After washing and rinsing feathers thoroughly,—if dirty or greasy, extracting color if necessary,—prepare bath as follows: One ounce of turmeric diluted in one gallon of boiling water; enter your feathers and let remain in about one minute, after which take out and rinse thoroughly. Prepare a weak bath of logwood, about half a pound to the gallon of water, or about half the usual strength of an ordinary logwood bath for black; boil a few minutes, after which enter your feathers and let them remain in bath about one minute; then take out and rinse thoroughly in cold water; after which prepare a bath of half an ounce of bichromate of potash to one gallon of boiling water. Dissolve bichromate of potash, enter feathers and let them remain in about half a minute; a little longer if a very dark shade be required, and so much less time if a very light shade is required; after which take feathers out and rinse thoroughly in cold water. Dilute about one-half a teaspoonful of aniline green in a gallon of boiling water, and reduce temperature of bath a few degrees with cold water; then enter feathers and let them remain in bath about two or three minutes; remove feathers and cool off a small portion of the bath with cold water, and add to it a small handful of starch; pass your feathers through the bath, squeeze out and dry off in the usual way.

If found to be lighter than shade desired, rinse off starch thoroughly, and return for a few seconds to logwood bath without increasing temperature any; then rinse off in cold water, and pass through a weak solution of bichromate of potash, about one-quarter ounce to a gallon; after which rinse, starch and dry.

If found darker than shade desired, pass feathers through a solution of half a teaspoonful of oxalic acid in about one gallon of luke warm water for about thirty seconds; take them out of this and rinse twice through boiling water, and then give a weak bath of aniline green,—about half the strength of the first bath. If samples to be matched be more on the yellow or olive, use decidedly more turmeric in the first bath, and add a little, say about a teaspoonful, to the aniline green bath. If a green on the blue, it will be necessary to use only one-half the turmeric prescribed in the first bath.

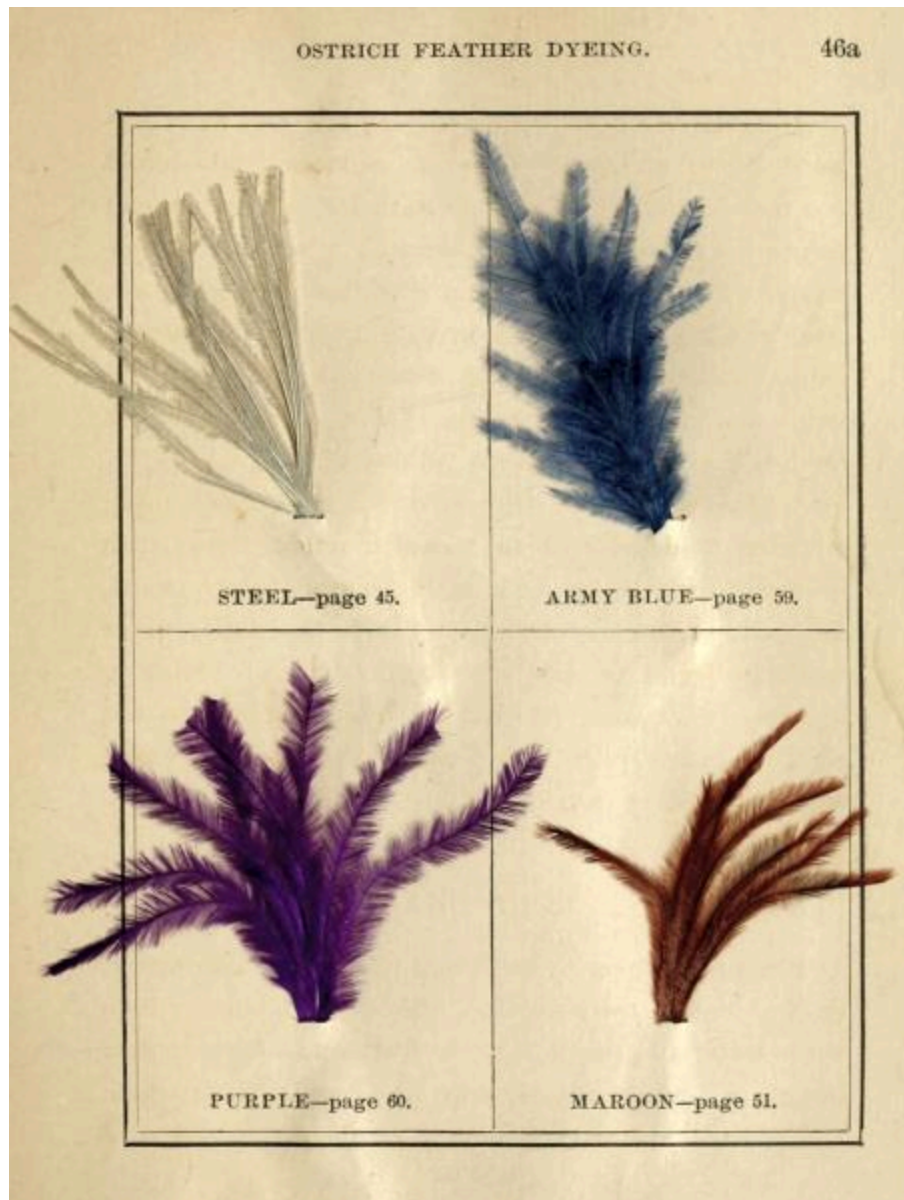
STEEL COLOR.

All light colors can be used to make a good shade of steel by first extracting colors by the usual process of bleaching with permanganate of potash; if white and dirty, wash thoroughly in hot water and soap and rinse. Prepare your bath as follows: To one gallon of luke warm water add a small handful of starch; enter your feathers, rub them around well in bath; after which add a small pinch of copperas and about a tablespoonful of logwood liquor, and let remain in about one minute; increase temperature of bath and add a few drops of diluted violet, first removing your feathers from bath; re-enter feathers and let remain about one minute, or until your feathers look about four shades darker than sample; after which take out and dry.

If found too light, return to bath and add more logwood liquor and a few drops more violet, and should you find them altogether too dark for sample, extract your color by passing them through a solution of one teaspoonful of oxalic acid in a gallon of hot water; after which rinse them off by passing them through a gallon of boiling water about twice, when you will find your color reduced four or five shades. The oxalic acid renders the feathers a bright yellow.

Boiling water will draw off the logwood and bring out your shade of drab in as much milder form; then proceed to mix a new drab bath the same as per recipe, only using more caution not to get it too dark; enter feathers, bring

to shade, using a drop of violet to brighten up color. Be careful in drying not to use starch that has previously been used on a color where acid was used to develop.



STEEL—page [45](#). ARMY BLUE—page [59](#).
PURPLE—page [60](#). MAROON—page [51](#).

FELT DRAB.

Prepare feathers by washing and rinsing thoroughly, or bleaching if needed; after which mix a bath of luke warm water and starch. Enter feathers and manipulate in bath a few seconds between the hands; after which add a small quantity of copperas, about the size of a pea. Enter feathers and let them remain in about half a minute; take out feathers and add a few drops of logwood liquor; re-enter feathers and let them remain in about half a minute; add to bath about a drop of diluted safranine, and if shade be wanted a little on the yellow, a drop of diluted Bismarck brown can be added. Allow feathers to remain in until they look about three shades darker than sample; then take out and dry as usual. If found either too dark or too light, treat precisely as preceding color (steel). Be careful not to use starch that has been used for an acid color.

SLATE COLOR.

To make this color all light colors can be used and some dark ones; only those, however, that do not contain much yellow, as, for example, blues, reds, etc. After preparing for bath by washing and rinsing, or by extracting color if necessary, mix a bath of logwood, about half the usual strength, and enter feathers. Bath must be at boiling temperature, and let them remain in about one minute; after which take out and rinse. Proceed to mix a bath of one quarter ounce of copperas and one gallon of boiling water; enter feathers and let them remain in bath about half a minute; take out and cool off a small portion of the bath, add starch and pass feathers through, squeeze out and dry.

If the color to be matched be very dark, repeat the bath of logwood and mix a bath of one-quarter ounce of bichromate of potash in a gallon of boiling water. Enter feathers and let remain in about half a minute; after which rinse off in cold water, and starch and dry. If a very brilliant shade be required, when you have rinsed feathers from bichromate of potash bath, wash thoroughly in soap-suds and rinse in luke warm water. Dilute a small quantity of starch in cold water, pass feathers through and dry. The above recipe produces a most beautiful shade of slate color, perfectly fast to light, and the depth of shade is regulated by the quantity of logwood. Should you find your color altogether too dark for sample, proceed to extract by passing through a solution of one teaspoonful of oxalic acid to one gallon of boiling

water for about half a minute, and then rinsing off twice or three times in boiling water; after which repeat in a milder form.

ORANGE COLOR.

Prepare feathers by washing and rinsing thoroughly. Prepare bath by diluting about two ounces of turmeric in a gallon of boiling water, and enter your feathers, letting them remain in bath about two minutes; then take them out and add a few drops of diluted Bismarck brown and about a teaspoonful of oxalic acid; re-enter your feathers and bring bath to a boil, and let remain in about three minutes; after which take out, and cool off a small quantity of bath, add a small handful of starch, pass feathers through and dry.

Should you desire a very full dark shade, use about twice the amount of turmeric, add a few drops more Bismarck brown; and if wanted much lighter, use less of each color. If wanted more yellow, use very small quantity of Bismarck brown; and if a very reddish shade of orange, a little more Bismarck brown than amount prescribed in recipe.

There are numerous orange anilines in the market that are used successfully in dyeing shades of orange, but it is almost necessary to have a different shade of aniline for every shade of color made. Should your sample to be matched be rather dull, use no oxalic acid in bath, as the oxalic acid is used in developing and brightening the shade. To remove the color, should it be too dark, the first method is to wash well in soap water, rinse and pass through a solution of oxalic acid in warm water, about half an ounce to the gallon.

SCARLET.

Wash and rinse your feathers thoroughly, and if required to remove a surplus of any old color, pass through a bath of permanganate of potash, as per recipe; after which prepare a bath of half a teaspoonful of oxalic acid to one gallon of boiling water and about a teaspoonful of turmeric; enter feathers and let them remain in bath about half a minute, after which take them out and add to bath about half a teaspoonful of rocceline; dissolve powder thoroughly, and return to bath; let them remain in about one minute

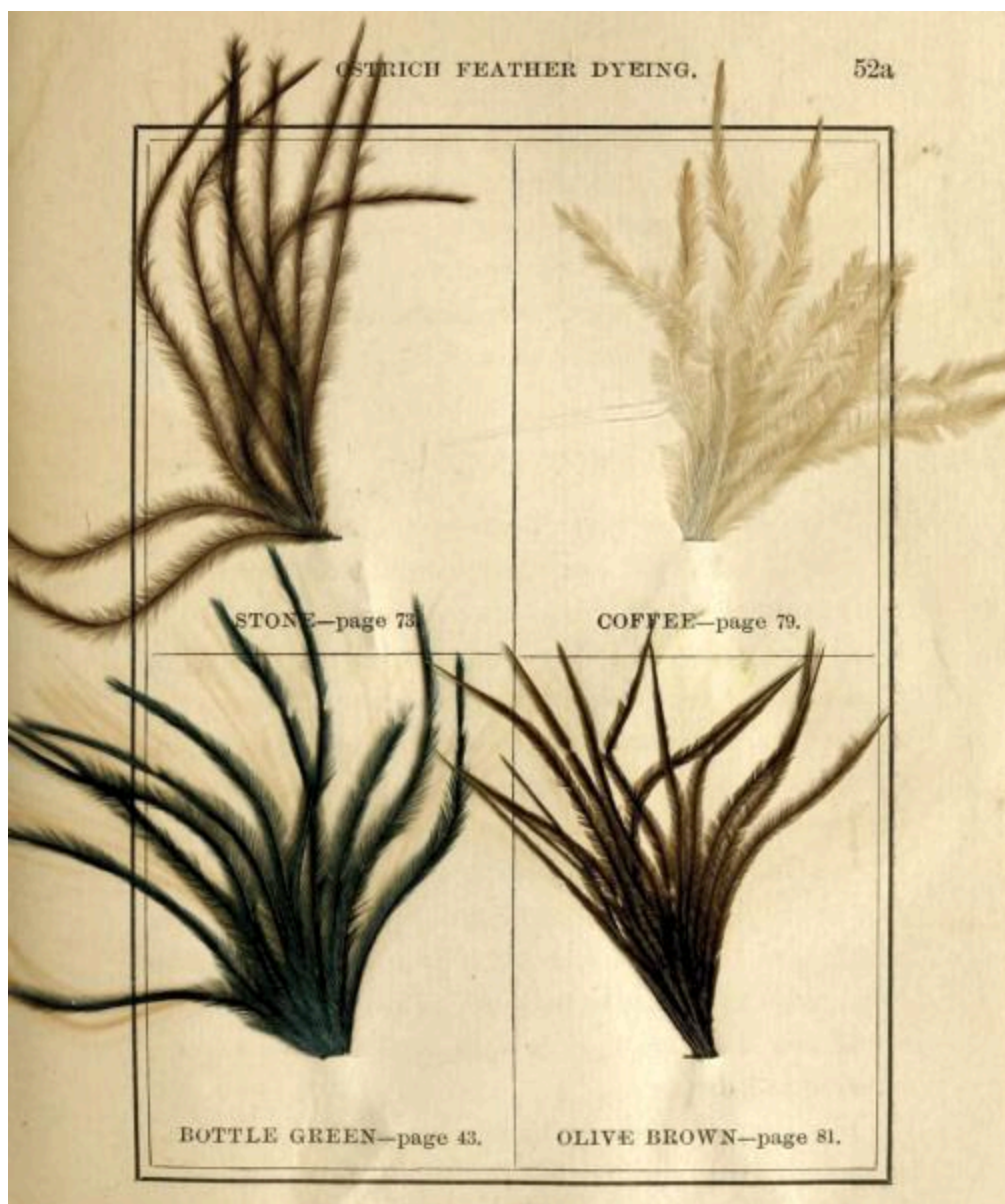
longer, then cool off a small quantity of the bath and add a small handful of starch; pass your feathers through, squeeze out and dry as usual.

If wanted a very dark shade, add a little more rocceline and let remain longer in bath. If shade be a little on the orange, use more turmeric and less rocceline; and if more on the cardinal, vice versa. Should you, through carelessness, get your color too dark, to remove color rinse off and wash thoroughly in a soap bath, and rinse off in boiling water about twice, which will have the effect of reducing the color several shades; mix a new bath as per recipe, and enter feathers, using more care and judgment and proceed to starch and dry as called for in recipe.

MAROON.

Almost any odd shades of color can be used without extracting colors, but if dirty or greasy, it is always best to wash thoroughly and rinse. Take your old logwood bath that has been used for black and other colors, or else boil a fresh bath of the same proportions, about a pound to the gallon. When at boiling temperature add thereto a half cupful of extract of archil, first removing the grounds of logwood from the bath; then enter your feathers and let them remain in the bath about four or five minutes; take them out and rinse thoroughly in cold water, and prepare a bath of one-half ounce of bichromate of potash to a gallon of boiling water, and thoroughly dissolve potash; after which pass your goods through for a few seconds only, and take out and rinse twice in cold water; dilute a small handful of starch in clean cold water, pass feathers through and dry.

Should a very dark shade be required, allow your feathers to remain in bichromate of potash bath a few seconds longer; take out and dry. Should you find your color too dark for sample, it is only necessary to add to your starch bath a few drops of sulphuric acid, and add a small quantity of hot water to increase temperature a few degrees, and pass feathers through. This bath, same as the garnet, can be used again, and improves with age if kept in a clean place. If you have an old garnet bath on hand, it will answer for maroon by bringing to a boil and adding about a teaspoonful more extract of archil to it.



STONE—page [73](#). COFFEE—page 79.
BOTTLE GREEN—page 43. OLIVE BROWN—page [81](#).

LEMON COLOR.

Wash and rinse your feathers thoroughly if dirty whites; if old faded out light colors, bleach with permanganate of potash; after which prepare bath as follows: One gallon of luke warm water and a handful of starch; enter your feathers and rub around between the hands for a few seconds; then add to bath a teaspoonful of oxalic acid, and dilute about a tablespoonful of

turmeric in a small quantity of water, and add a few drops of the liquor to the bath; re-enter your feathers and let them remain in about one minute or so; after which take them out and add a drop of diluted indigo blue; return feathers to bath and allow them to remain about one minute longer in bath, after which take out, squeeze and dry usual.

If a deep rich shade be desired, and you have no sample to match, use no indigo in the bath. Another excellent method of making lemon is to substitute an equal amount of picric acid for turmeric; and, should you find your color entirely too dark for your sample, rinse off your feathers in luke warm water, and proceed to wash with soap and hot water, and rinse thoroughly in boiling water; then prepare a fresh bath as per recipe, and enter your feathers, using much care. If found too light for your sample, add to bath a little more turmeric liquor, and return feathers to bath for a few seconds longer, and dry.

BLACK.

The most staple and important of all the colors. Some will argue that it is not a color; I, to the contrary, however, that it is not only a color, but a combination of colors, and it is the knowledge of how to properly combine them that results in the production of a very handsome and glossy black. Twelve years ago a bath of black that was commenced on Monday and was ready to go into the drying-room by Saturday was considered at that time a most expeditious piece of work; and, even up to the present time, some of our old orthodox dyers,—those old chronic, methodical dyers,—those who dye according to the most approved and advantageous methods of half a century ago,—still continue to occupy the greater part of a week in getting a black on what (by that time) is left of the feathers. Their object from the start is to produce a black, and they generally succeed.

Begin, if raw stock, by washing and rinsing thoroughly in order to remove all natural grease and dirt adhering to the fibre. If they are old colors to be redyed a black, it is not necessary to wash them nor to bleach them for the purpose of removing any of the color, as the black bath will overcome all the other colors; as, for example, a navy blue, a bottle green, garnet, etc., can be all entered at the same time, and put through precisely the same process, and they will all be the same shade of black when they are dried.

Prepare bath by diluting a quarter pound of turmeric in a gallon of boiling water and bring to a boil; after which enter your feathers, and let remain in bath about five minutes, keeping them well under the surface, and gently moving while in bath; after which take feathers out and rinse twice in clear cold water. Meantime dilute one pound of logwood in about one and a half gallons of boiling water, and boil for about fifteen minutes; after which enter your feathers and let them remain in bath about four minutes; then take out and rinse thoroughly in two waters. Dilute one ounce of bichromate of potash in one gallon, more or less, of boiling water, enough to completely cover up your feathers, dissolving bichromate of potash thoroughly. Enter your feathers, let them remain in bath about three minutes; after which take them out and rinse thoroughly. Meantime have logwood bath boiling, and return feathers to it. Cover up, and let them remain about eight minutes; take out and rinse twice as before. After rinsing, prepare a bath of about half an ounce of bichromate of potash and salts of tartar about the size of a pea in a gallon of boiling water; dissolve thoroughly. Let them remain in bath about three minutes; after which take out and rinse thoroughly in cold water. Then mix a bath of hot soap-suds, and enter feathers; wash well and rinse in luke warm water.

The washing and rinsing is not absolutely necessary, in fact, it can not much improve what is already a clean, glossy black. Washing, however, if productive of a change at all, must be beneficial. Then proceed to mix a small handful of starch in a small quantity of cold water; pass feathers through and dry. While your feathers are in the bichromate of potash bath, they must be kept moving in bath constantly and well under the surface. There is nothing to be added to make a successful result, except it be to caution you to adhere as strictly as possible to the recipe.

It often occurs that feathers are brought in to be dipped over that have faded out, or have grown rusty looking from exposure to light and long wear. The color can be restored by simply passing them through the last two baths for the same length of time that is allotted to the regular recipe. During the process of drying black be sure to have the starch beaten out as fast as it dries. It is best to dry them in the open air, and, if possible, allow them to hang in the sun for a while, as it improves the color. One especial advantage this black has over most others, is that it improves with age; and, instead of fading, the black will grow more intense.

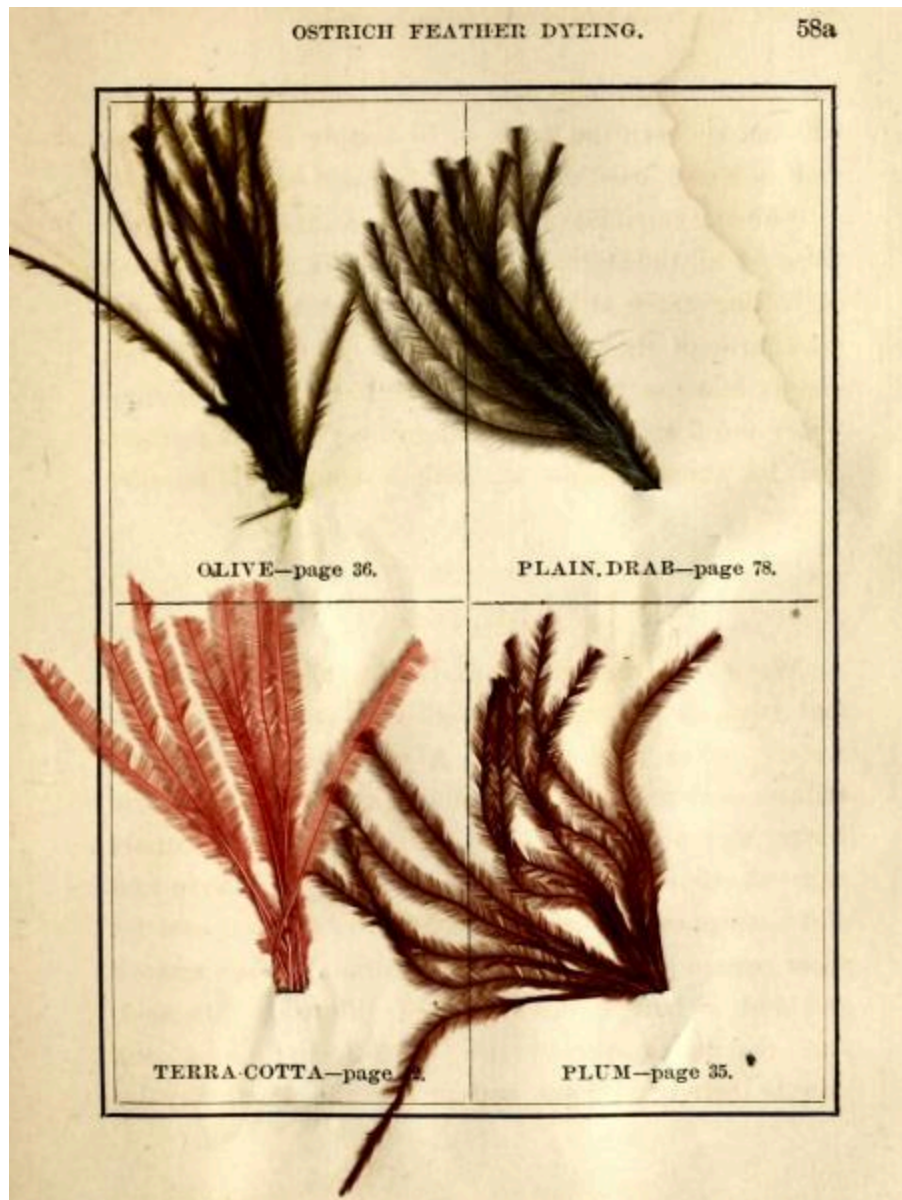
LILAC.

Wash and rinse thoroughly in hot soap water, and rinse in about four waters to remove any particle of soap that may adhere to the feathers; next prepare bath of one gallon of hand warm water, and add a handful of starch. Enter feathers and rub thoroughly between the hands; remove and add to bath a few drops of diluted violet, according to shade required; add about two drops of diluted saffranine, and re-enter feathers, let remain in bath about three minutes, squeeze out and dry in powdered starch in the usual way. Be sure your starch is clean and free from acid, and also that your board is in the same condition. Great care should be exercised to see that every particle of the violet is dissolved to avoid spots on the feathers. Should quite a bluish shade be desired, a drop of diluted aniline green added will produce the desired result.

GENDARME BLUE.

Prepare feathers by washing thoroughly, and rinse about four times in hot water to remove any particle of soap that may adhere to the feathers. Prepare a bath of a teaspoonful of indigotine powder to one gallon of boiling water. Mix thoroughly and enter feathers, and let remain in about one minute, after which remove and add about one teaspoonful of oxalic acid or same quantity of sulphuric acid, and re-enter feathers, letting them remain in bath about five minutes longer; then remove from bath and cool off. Reserve a small portion of bath, and cool off with cold water, adding a drop of sulphuric acid and a small handful of starch; pass feathers through and dry in powdered starch by rubbing between the hands or by simply beating out on a clean board, used only for drying acid colors.

Should you find your color too dark, thoroughly rinse off all the starch and pass feathers through a bath of boiling water and let remain about half a minute; pass through starch bath and dry. If found too light, simply increase temperature of bath by adding boiling water and few drops more indigotine; re-enter feathers and let them remain in bath a couple of minutes longer.



OLIVE—page [36](#). PLAIN, DRAB—page [78](#).
TERRA COTTA—page [42](#). PLUM—page [35](#).

TRILEUL.

Wash and rinse feathers thoroughly in hot water and soap, and rinse thoroughly in about four hot waters; then pass through a bath of plain boiling water; next prepare a bath of one gallon of luke warm water, and add a handful of starch. Enter feathers and rub thoroughly between the hands; remove and add a teaspoonful of oxalic acid; enter feathers and let

them remain in bath about two minutes; then remove and add to bath a few drops of diluted picric acid, and re-enter feathers; let remain in about one minute longer, take out and dry in the usual way by rubbing in powdered starch between the hands and beating out on a clean board until all the starch has been removed from the fibre. Should you find your color a shade too dark, mix a luke warm starch bath, and pass feathers through, keeping them under about half a minute, and dry as usual. Be careful that your picric acid is thoroughly dissolved, as otherwise it will be likely to spot your feathers, if the particles come in contact with the flues, and the spots are very hard to remove, as it would be necessary to put them through a bleaching process.

ARMY BLUE.

Prepare feathers by washing and rinsing thoroughly in hot water. Be careful about rinsing to remove every particle of soap that may adhere to the fibre, after which prepare bath as follows: One teaspoonful of indigotine powder, diluted in one gallon of boiling water, and add thereto about half a teaspoonful of oxalic acid, stirring around well to thoroughly dissolve every particle of color. Enter feathers and let them remain in bath about four minutes; after which take out and rinse in luke warm water to remove the acid in feathers; next prepare a bath of one gallon of hand warm water and add a small handful of starch; add thereto a cupful of boiled logwood liquor and a few grains of copperas, enter feathers, let remain in bath about three minutes; take out and dry by rubbing between the hands in powdered starch, and beat out on a clean board until all the starch has been removed. Should you find your color darker than shade required, prepare a bath of half a teaspoonful of oxalic acid in a gallon of hand warm water, and pass feathers through about half a minute; take out and pass through boiling water, after which pass through starch bath and dry. Should you find shade too light, add more logwood to bath, increase temperature, let remain in a couple of minutes longer and dry.

PURPLE.

Prepare feathers by washing in hot water and soap thoroughly, and afterwards rinse in about four hot waters to remove every particle of soap

and dirt; after which prepare bath as follows: Take one gallon of water about 200° Fah.; dilute therein half a teaspoonful of Violet 3 B., stirring it around thoroughly to dissolve every particle. Enter your feathers and let remain about five minutes; after which take out and pour out the bath, reserving some, and cooling it off with cold, clean water, add a small handful of starch and pass feathers through, first cooling them off by shaking them in the air; rub them between the hands in starch bath to aid the flue or fibre to expand; after which squeeze out and rub thoroughly between the hands, and beat out on a clean board until every particle of starch has been removed. Should you find the top or tips a darker shade than the bottom, or should they bronze or assume a metallic appearance, pass feathers through a bowl of boiling water with a small pinch of soda added, and rinse; after which pass through a new starch bath with a few drops of diluted violet added; take out and dry.

MEDIUM GREEN.

Prepare your feathers same as for bottle green. Prepare bath by diluting about one ounce of turmeric in a gallon of boiling water, and enter feathers, letting them remain in bath about two minutes; after which take out and rinse in cold water twice. Have boiling a medium strong bath of logwood, and pass feathers through for a few seconds, first cooling off temperature of logwood bath a few degrees with cold water; after which rinse off thoroughly, and prepare a bath of a quarter of an ounce of bichromate of potash in a gallon of boiling water, dissolve it thoroughly, and enter feathers; let them remain in this bath about ten seconds, and take them out and rinse thoroughly in cold water. Proceed to dilute one teaspoonful of turmeric and a half teaspoonful of aniline green in a gallon of boiling water, and reduce temperature a few degrees with cold water. Enter your feathers, and let them remain in bath about three minutes; then take them out and cool off a small portion of bath, and add a small handful of starch, and dry in the usual way.

If found to be too dark, add a few drops of diluted oxalic acid to starch bath, and pass your feathers through for a few seconds. If found too light, rinse off the starch in cold water and return to logwood bath for a few seconds,

without increasing the temperature any, and rinse off and give a weak bath of bichromate of potash, rinse off and dry.

BEIGE.

Prepare your feathers by washing and rinsing thoroughly, or if old light colors, bleach with permanganate of potash, being sure to rinse out in hot water to remove acid from feathers, before putting in bath. Dilute a small quantity of starch in a gallon of boiling water, and enter your feathers, rubbing them around in bath between the hands to expand the flues and admit the color evenly on feathers. After which add to bath a small pinch of copperas, about the size of a bean, and about a teaspoonful of turmeric, and enter your feathers, letting them remain in bath about one minute; take them out, and add about a teaspoonful of logwood liquor; re-enter your feathers, and let them remain in bath about one minute, first increasing the temperature by adding hot water; after which remove feathers from bath, and add thereto a few drops of diluted Bismarck brown. To bring the ecru tint desired, a few seconds before taking feathers from bath to dry, add a couple of drops of diluted violet, squeeze out and dry.

If a very dark shade of beige is wanted use a greater amount of logwood and Bismarck brown, and if lighter shade is desired, less color should be used. Should your color be found altogether too dark for sample, dilute about half a teaspoonful of oxalic acid in a gallon of hot water, more or less. Pass your feathers through for a few seconds, and rinse off twice in luke warm water and once in boiling water. Then mix a fresh bath of luke warm water and starch, and add thereto a small proportion of turmeric and diluted Bismarck brown, and copperas about the size of a pea. Enter your feathers, and, using care, bring to the desired shade.



NAVY BLUE—page [31](#). MAGENTA—page [69](#).
PEA GREEN—page [80](#). BRONZE—page [74](#).

CORN COLOR.

Prepare feathers by washing and rinsing thoroughly if dirty greasy whites, or bleach with permanganate of potash if faded out light colors. Prepare your bath as follows: Take one gallon of luke warm water and dilute therein a small handful of starch, and rub your feathers around between the hands. Add about a half teaspoonful of turmeric and dilute well in bath. Enter your

feathers and rub around well between the hands. Increase the temperature of your bath by adding hot water, and allow your feathers to remain in bath about one minute; then take them out and add a couple of drops of diluted aniline brown; re-enter feathers and let them remain in bath about one minute longer; then squeeze out and dry as usual.

If your shade to match be considerably on the yellow shade, use very little aniline brown, about one drop, and if more on the brown, use less turmeric. If your color be entirely too dark and dull looking, dilute half a teaspoonful of oxalic acid, and pass feathers through for a few seconds and rinse off in luke warm water. Prepare a fresh bath and enter your feathers, as per recipe; or, if wanted a very bright shade, wash off with soap and hot water, and rinse thoroughly in hot water. Then prepare a bath of one teaspoonful of turmeric, one teaspoonful of oxalic acid and one teaspoonful of diluted Bismarck brown in a gallon of luke warm water. Enter your feathers and keep in bath about two minutes, add a little starch to bath, and pass feathers through for a few seconds longer, squeeze out and dry in the usual way.

ELECTRIC BLUE.

Feathers must be white, or nearly so, to make a good clear shade of electric blue. Prepare your feathers by washing with soap and hot water if dirty whites, and if old, faded light colors bleach with permanganate of potash. Prepare your bath as follows: Take half a teaspoonful of cotton blue and a half teaspoonful of oxalic acid,—a little more or less matters not,—in a gallon of boiling water. Enter your feathers, and let them remain in bath about five minutes; after which take out and rinse twice in cold water and once in hot water to remove all acid and loose color. Prepare a bath of about one cupful of logwood liquor and a small pinch of copperas in a gallon of hot water, not quite boiling, however, and pass feathers through for a couple of minutes. Cool off a little of your bath, and add a small handful of starch and a few drops of violet, pass feathers through and dry.

MEDIUM BROWN.

All light colors can be made a handsome shade of medium brown without removing the color by bleaching or without washing, unless very dirty and

greasy. Prepare your bath by diluting about two ounces of turmeric and a half ounce of copperas in one gallon, more or less, of boiling water. Enter your feathers, keep them well under the surface of bath, and let them remain therein about two minutes; after which take out, rinse twice in cold water. Have boiling meantime a medium strong bath of logwood, about the same proportion as for black; boil about fifteen minutes, and enter your feathers, allowing them to remain in about one minute; after which take out and rinse off twice in cold water; then dilute about a half teaspoonful of aniline brown in a gallon of boiling water, and after dissolving well, enter your feathers, and let them remain in bath about two minutes; take out and rinse in cold water; after which dilute a small handful of starch in a small quantity of luke warm water, and add to that a couple of drops of sulphuric acid; pass feathers through for a few seconds, squeeze out and dry.

Should your color be too dark to match sample, return to starch bath, add a few drops of sulphuric acid, let feathers remain in about half a minute, and dry. If a darker shade is wanted, it is necessary to rinse off starch in cold water, and return your feathers to logwood bath for a few seconds, rinse off and repeat Bismarck brown bath as before. By this process, with a little judgment, all shades of brown can be produced in the most satisfactory manner.

MEDIUM BLUE.

Prepare your feathers by washing and rinsing thoroughly in hot water; light faded out colors need not be bleached, but thoroughly washed in hot soap suds instead. Prepare your bath as follows: Take one teaspoonful of concentrated cotton blue and one teaspoonful of oxalic acid, dilute it in one gallon of boiling water. Be careful to see that the blue crystals are well dissolved. Enter your feathers, and let them remain in bath about four minutes, keeping them well under the surface. Meantime keep them gently agitated to insure an even color; after which take out, rinse, starch and dry.

If your feathers be found too dark for sample, or too much on the purple, rinse off, starch in cold water thoroughly, and pass through a bowl of boiling water, starch and dry, using a few grains of oxalic acid diluted in starch bath.

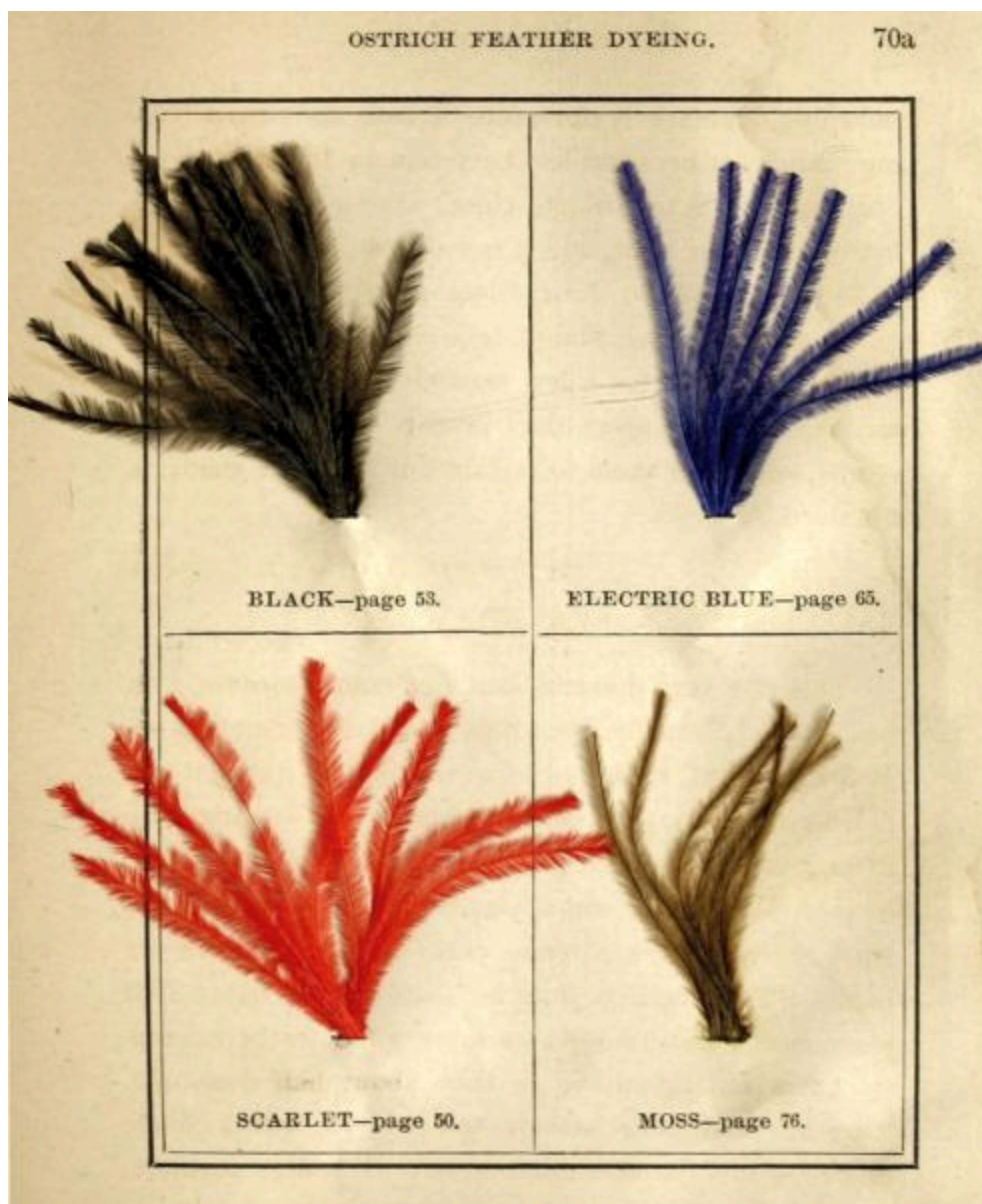
If a very light shade be desired, use but half the quantity of cotton blue, and do not allow them to remain in bath quite so long a time. If a much darker shade be required than the foregoing recipe will produce, then rinse off your feathers thoroughly in cold water, to remove all starch, and pass feathers through a medium strong bath of logwood at boiling temperature for a few seconds, and rinse off twice in cold water; dilute a half ounce of bichromate of potash in a gallon of boiling water, and pass your feathers through for a few seconds only; rinse, starch and dry. Should you get your color too dark by this process, pass your feathers through a solution of half a teaspoonful of oxalic acid in a gallon of boiling water, and rinse off in boiling water twice; then dilute a small quantity of starch in luke warm water, add a few grains of oxalic acid to it, pass feathers through and dry as usual.

MAGENTA.

Prepare your feathers, whether dirty whites or faded out light colors, by washing thoroughly in hot soap suds and rinsing well in hot water. Prepare your bath as follows: Take about a half teaspoonful of safranine and dilute in one gallon, more or less, of boiling water, and add thereto a half tablespoonful of extract of archil. Enter your feathers and let them remain in bath about two minutes; after which take out and add to bath a few drops of diluted violet, and re-enter your feathers, letting them remain in bath about one minute longer. Then take out and rinse in cold water, and dilute a small handful of starch in bowl of luke warm water; pass feathers through and dry.

If found too red for sample, rinse off and add to bath a tablespoonful of extract of archil; return feathers to bath for about one minute, first, however, increasing temperature; next rinse, starch and dry.

If found to be too much on the plum for sample, rinse off and add to bath about a quarter teaspoonful of safranine, increase temperature of bath to almost boiling; enter feathers and let them remain in bath about one minute; after which rinse, starch and dry. If found to be too light, add a few drops of diluted violet to bath; and, if too dark, dilute a half teaspoonful of oxalic acid in one gallon of luke warm water, and pass feathers through for a few seconds, rinse off twice or more in boiling water; then prepare bath same as per recipe, and allow them to remain until desired shade is obtained.



**BLACK—page [53](#). ELECTRIC BLUE—page [65](#).
SCARLET—page [50](#). MOSS—page [76](#).**

SEA FOAM.

This is a very delicate shade of color bordering on pea green. Your feathers must be white, or nearly so. If dirty whites, wash and rinse thoroughly; and, if old faded out colors, pass through bleach of permanganate of potash; after which prepare your bath of one gallon of luke warm water and a small handful of starch, and enter your feathers, rubbing them around between the

hands. Take feathers from bath and add about a half teaspoonful of turmeric; re-enter your feathers, keeping them moving around in bath about half a minute. Then take out your feathers and add to bath a couple of drops of diluted aniline green. Re-enter feathers, first increasing the temperature of your bath a few degrees by adding hot water, let them remain in bath about two minutes longer, squeeze out and dry in the usual way.

Should your sample be more on the green, you will simply add a few drops more diluted aniline green; and if more on the yellow, you can use less. If the shade to be matched be darker than your feathers, add more of each color in the preparation of first bath. If a rather dull shade be desired, which in this color is quite frequently the case, a small pinch of copperas about the size of a pea will have the desired effect.

Should you find your color entirely too dark for your sample, wash off thoroughly in soap suds, and rinse in hot water; after which dilute a half teaspoonful of oxalic acid in a gallon of luke warm water, pass feathers through for a few seconds and rinse off in luke warm water. Then prepare your bath as per recipe, using a little more care and judgment in your second attempt.

SALMON.

Have your feathers white, or nearly so, by washing if dirty, or bleaching with permanganate if needed, being careful to rinse thoroughly for the purpose of removing any acid or soap; after which prepare your bath as follows: Take one gallon of luke warm water and a small handful of starch. Enter your feathers and rub around between the hands for a few seconds; then add to bath a few drops of diluted safranine and copperas about the size of a pea. Let your feathers remain in bath about one minute; after which take out and add to bath about one teaspoonful of diluted Bismarck brown, first increasing temperature of bath a few degrees with hot water; re-enter your feathers and allow them to remain in bath about a minute; after which squeeze out and dry in the usual way.

If your sample to be matched be more on the pink, use less aniline brown; and if more on the yellow, use less safranine and more aniline brown. Should you desire a much darker shade, use more of each color than laid down in recipe, and add a few drops of logwood liquor. If your feathers be

found altogether too dark for sample, rinse off starch in cold water and dilute a half teaspoonful of oxalic acid in luke warm water, and pass your feathers through for a few seconds, take out and rinse a couple of times in hot water (not boiling). Prepare bath again as per recipe, using greater care. This shade of color is on the order of the terra cotta and crushed strawberry, and can be made in the same bath by adding color or diluting. Be careful in drying to use only clean starch and a clean board that has not been used with any acid colors.

STONE COLOR.

Stone color is a shade varying very slightly from slate and smoke color. All light shades can be used for this color; first preparing them by washing and rinsing them thoroughly. Prepare a medium strong bath of logwood by boiling for about fifteen minutes; after which enter your feathers, and let them remain in bath about two or three minutes, longer if a very dark shade be required; then take them out and rinse in cold water twice. Prepare a bath of half ounce of bichromate of potash in one gallon of boiling water, and dissolve thoroughly. Enter your feathers, and let them remain in bath about two minutes, keeping them well under the surface of bath and moving at the same time, to assist in producing an even color; after which take out and rinse off about three times in cold water, and prepare a bath of hot soap water. Enter your feathers, and wash thoroughly, adding to bath a small pinch of soda; after which rinse carefully in hot water; dissolve a small handful of starch in cold water, pass your feathers through, squeeze out and dry in the usual way.

If your feathers be found much too light for your sample to be matched, rinse off starch in cold water, and return your feathers to logwood bath for a few seconds; dissolve a small pinch of copperas in a gallon of boiling water, reduce temperature a little and enter your feathers, letting them remain in bath a few seconds. Take out and pass through starch and dry. If found to be altogether too dark, dilute a teaspoonful of oxalic acid in a gallon of hot water; pass feathers through a few seconds and rinse off in boiling water twice; wash, starch and dry.

BRONZE.

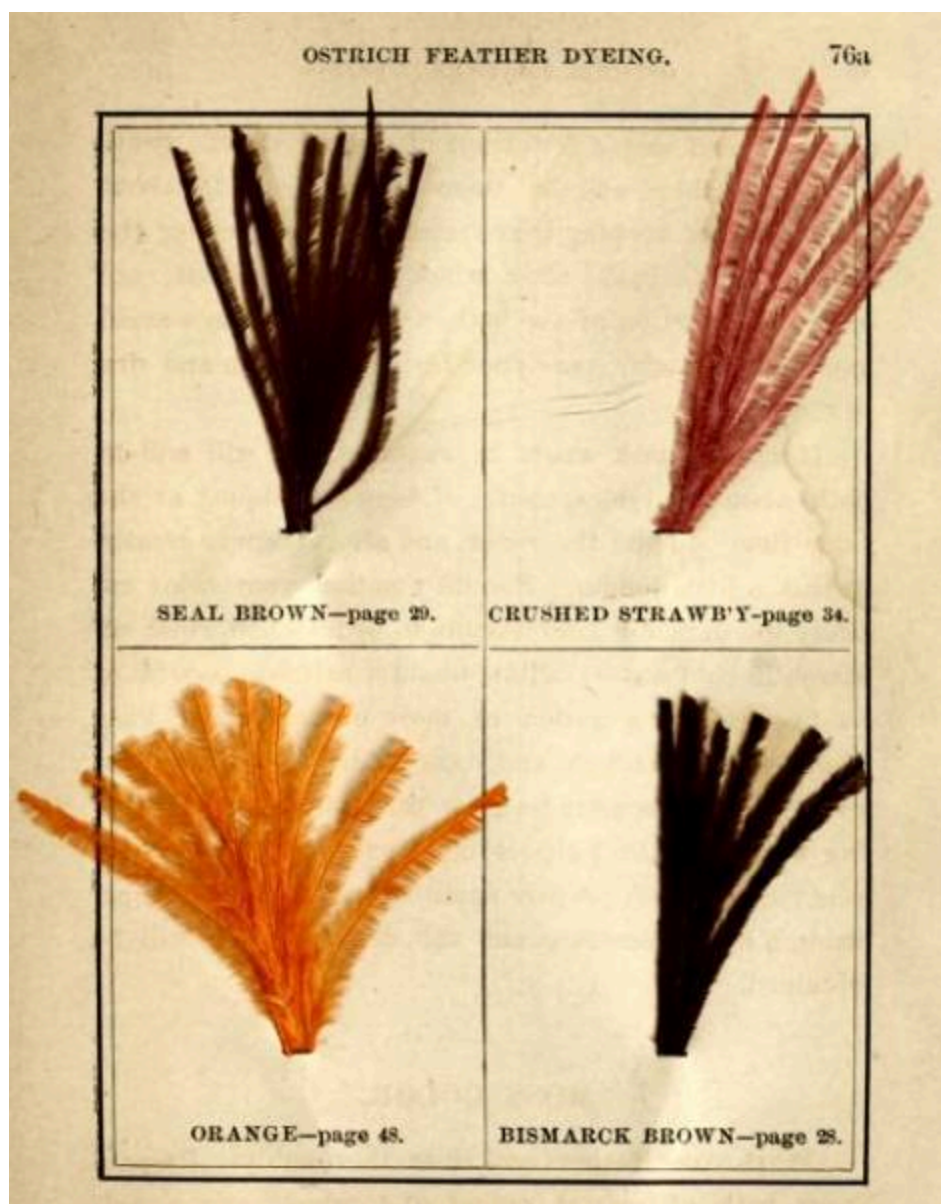
Wash and rinse thoroughly, using soap for washing, and rinse out in hot water about four times; after which prepare a bath of one quarter pound of turmeric to one gallon of boiling water. Enter feathers and let remain in bath about three minutes; take out and rinse. Boil a bath of half pound of logwood to one gallon of water about ten minutes; enter feathers and let remain in bath about four minutes; take out and rinse. Then prepare a bath of half an ounce of bichromate of potash and one gallon of boiling water, and let feathers remain in bath about two minutes, take out and rinse. Next prepare a bath of one quarter pound of turmeric and one-quarter teaspoonful of Victoria green crystals, and add one gallon of boiling water. Enter feathers and let remain in bath about four minutes; take out, cool off a small portion of the bath and add a small handful of starch. Pass feathers through and dry in powdered starch by pressing between the hands; then beat on a board or table until all the starch is removed from the feather.

CHOCOLATE.

Prepare your feathers by washing and rinsing thoroughly; and, if necessary, bleach with permanganate of potash. After doing this, rinse thoroughly in hot water for the purpose of removing all acid from the fibre. Prepare your bath of one gallon of water at boiling temperature; add thereto a teaspoonful of turmeric and a small pinch of copperas about the size of a bean. Enter your feathers and allow them to remain in bath about one minute or longer. Take out your feathers, and add to bath about one tablespoonful of diluted Bismarck Brown and a few drops of diluted violet; re-enter your feathers, and let them remain in bath about three minutes, keeping them meanwhile well under the surface of the bath; after which take them out, cool off a small portion of the bath, and add thereto a small handful of starch; pass your feathers through and dry in the usual way.

If a very dark shade be required, you will add to bath about a tablespoonful of logwood liquor at the same time you add the violet, and allow them to remain in bath a little longer. Should you find your color entirely too dark for your sample to be matched, rinse off starch in cold water; dilute about a half teaspoonful of oxalic acid in a gallon or more of hot water. Pass your feathers through, and rinse off in luke warm water twice; then pass your feathers through a bath of boiling water, for the purpose of effectively

removing the acid; after which prepare again as called for in recipe, using a little more care, and the desired result will be obtained.



SEAL BROWN—page [29](#). CRUSHED STRAWB'Y—
page [34](#).
ORANGE—page [48](#). BISMARCK BROWN—page [28](#).

MOSS COLOR.

Wash your feathers and rinse thoroughly. Prepare your bath of quarter pound of turmeric and a half ounce of copperas diluted in a gallon or more of boiling water. Enter your feathers and let them remain in bath about two minutes; after which take out and rinse twice in cold water. Meantime have a medium strong bath of logwood boiling, and enter your feathers, letting them remain in about one minute, take out and rinse. Then prepare a bath of about two ounces of turmeric and a small pinch of aniline green in a gallon of boiling water. Enter your feathers and allow them to remain in bath about three minutes or longer. Take out and cool off a small quantity of bath with cold water; add a small handful of starch, pass your feathers through and dry.

If your color be found too much on the green for your sample to be matched, add to starch bath a few drops of sulphuric acid; or, instead, rinse off starch and mix a bath of two ounces of turmeric in a gallon of boiling water; pass your feathers through for a minute or so, starch and dry.

If found to be too much on the yellow or olive, add to your bath a few grains of aniline green, and return them to the same for a few seconds, first rinsing off starch in cold water. If found too light, pass for a few seconds through a weak bath of bichromate of potash; and if too dark, dilute a few grains of oxalic acid in hot water, and add to your starch bath a few drops. Pass your feathers through for a few seconds and dry in the usual way.

PLAIN DRAB.

If your feathers are old, dirty whites, wash and rinse them thoroughly. If light colors, remove the same by passing through permanganate of potash process, and use great care in rinsing to remove all the acid before entering in bath. Prepare your bath with one gallon of luke warm water and a small handful of starch; enter your feathers and rub them around well in bath between the hands to expand the fibres. Take out your feathers, and add to bath a small piece of copperas about the size of a bean and about a quarter cupful of logwood liquor; re-enter your feathers, and let them remain in bath a few minutes, meantime adding a small quantity of hot water to increase temperature of bath; then add a couple of drops of diluted safranine to bath, let remain in bath one minute longer, squeeze out and dry as usual.

If wanted more on the shade of felt drab, use, instead of safranine, a few drops of Bismarck brown; and if wanted more on the steel, use a few drops of diluted violet in bath. If a darker shade should be desired, use only a little more logwood liquor, and allow them to remain a short time in bath.

Should you find your color to be altogether too dark for sample to be matched, rinse off starch, and dilute a half teaspoonful of oxalic acid in hot water; pass your feathers through, rinse off a couple of times in luke warm water and lastly through boiling water, for the purpose of removing all acid. Then prepare a fresh bath according to recipe, and pass through until you have obtained the desired shade.

COFFEE COLOR.

Old faded out light colors need only to be thoroughly washed and rinsed to prepare them for this color; and darker colors can be prepared by bleaching with permanganate of potash, taking care to rinse thoroughly in hot water for the purpose of removing all the acid. Prepare your bath of about one teaspoonful of turmeric and copperas about the size of a bean in a gallon of boiling water. Enter your feathers and let remain in bath about two minutes; remove feathers from bath and add a half cupful of logwood liquor and return feathers to bath, letting them remain in about one minute; after which remove feathers and add to your bath about two tablespoonfuls of diluted Bismarck brown and hot water to increase temperature of bath; re-enter feathers and allow them to remain in about two minutes; after which cool off a small quantity of the bath and add a small handful of starch; pass feathers through and dry.

If found to be too light, return to bath, first adding more logwood liquor and Bismarck brown, and let them remain in bath about one minute. If too dark for your sample to be matched, dilute a few grains of oxalic acid in luke warm water; pass feathers through for a few seconds and rinse off three times in luke warm water. Then prepare bath as per recipe, using more care in the preparation.

If found too much on the yellow, a few drops of diluted safranine added to your bath will produce the desired effect. Use clean starch in drying; if a table or board is used, see that it is perfectly clean and free from acid.

PEA GREEN.

Prepare your feathers by washing thoroughly in hot water, and rinse thoroughly to remove any soap that may adhere to the feathers. Then prepare a bath by diluting a handful of starch in a half gallon of hand warm water, and rub feathers around between the hands. Remove feathers and add a few drops of diluted Victoria green and a couple of drops of diluted picric acid. Enter feathers, letting them remain in bath about two minutes, keeping them well under the surface to insure an even color.

If wanted a shade more on the yellow, add a drop more of picric acid; and if more on the blue, leave the picric acid out entirely. Take out and dry in starch, being careful to beat out on a clean board in the usual way.

OLIVE BROWN.

Wash feathers thoroughly in hot water and soap, and rinse about four times in hot water; after which prepare a bath of half a pound of logwood; first enter feathers in one-quarter pound of turmeric and one gallon of boiling water; let them remain in bath about four minutes. When logwood bath has boiled sufficiently, say ten minutes, rinse feathers out of turmeric in cold water; and enter in logwood, letting them remain in bath about six minutes; take out and rinse. Prepare a bath of half an ounce of bichromate of potash and one gallon of boiling water; enter feathers and let remain in bath about one minute; take out and rinse thoroughly in cold water. Mix a bath of one ounce of turmeric to one ounce of archil and half the old logwood bath; bring to a boil and enter feathers, letting them remain in bath about six minutes; take out and rinse. Then mix a bath of luke warm water and starch, add a couple of drops of sulphuric acid and a couple of drops of picric acid diluted, pass feathers through, squeeze out thoroughly and dry by rubbing in powdered starch between the hands; beat out on a clean board until all the starch is removed from the feathers.



MEDIUM BROWN—page [66](#). **OLD-GOLD**—page [39](#).
CARDINAL—page [33](#). **MEDIUM GREEN**—page [61](#).

PROCESS OF DEGRADING OR BLEACHING NATURAL GRAY OR BLACK WHITE.

Begin by washing and rinsing your feathers thoroughly; after which soak in a bath of compound of one gallon of ammonia to eight gallons of water for about eight hours; take feathers out and squeeze out the excess of ammonia which is in the flues. Put your feathers in the peroxide of hydrogen with an

addition of twelve to sixteen ounces of ammonia to one five gallon can or demijohn, and let it work slowly, stirring feathers from time to time for about six hours; after which lay your feathers on one side of the tub and add to the peroxide of hydrogen bath about four ounces more of ammonia; stir the bath well to insure a thorough mixture of the peroxide of hydrogen with the ammonia.

The peroxide of hydrogen will continue to work for about twelve hours more, until it becomes thoroughly exhausted; after which take out your feathers and rinse a few times in luke warm water. Then proceed to put them in a second bath of peroxide of hydrogen to be prepared as follows: To a half gallon demijohn of peroxide of hydrogen add two and a half gallons of water, and add thereto about eight ounces of ammonia. Then enter your feathers, and allow the bath to work a few hours; again add about two ounces of ammonia by the same process as before, and then let it work a few hours longer, or until the bath becomes exhausted. To ascertain whether total exhaustion has taken place, take a small portion of the bath in a glass and dilute therein a few grains of permanganate of potash; if it be not totally exhausted, bubbles will appear on the surface; if exhausted, none will be noticeable.

After your feathers have been removed from the bath they must be carefully rinsed off in three or four waters, a few degrees more than luke warm. Then prepare a warm soap bath, and allow your feathers to remain in a few minutes; after which rinse off thoroughly in luke warm water; dilute a small handful of starch in a quantity of cold water, pass your feathers through and dry.

All natural color will have entirely disappeared. Whatever portion of the amount of feathers you have just bleached are for whites, before drying them up, prepare a bath as per recipe for white, pass through and dry in the usual way. This process of bleaching is used only when it is desirable to make light colors from gray or natural black feathers, but feathers for navy blue, seal brown, bottle green, etc., will not be improved by bleaching. The shade of color can be evened off in the bath.

HINTS ABOUT THE DYEHOUSE.

In dyehouses where steam is used, it is necessary to boil your bath a longer time than where the bath comes in direct contact with the fire. The accommodations of a dyehouse for the re-dying of ostrich feathers need be very simple and inexpensive; in fact, I have seen a dyehouse where old re-dyed transient work to the amount of fifty dollars per day was accomplished with a small cooking stove, a wash-boiler, a wash-bowl and a tin dipper; costing in all less than six dollars. Of course, in the manufacture of raw stock it is necessary to have larger vessels and much better facilities; for instance, instead of from ten to fifty, or even a hundred feathers, you will of necessity be compelled to dye lots of from five to ten pounds of goods at one time. Two stationary tubs or vats, one for use in washing white and bleaching, and the other for black, with water pipes and steam pipes and connections; a few large porcelain lined or copper basins for dark colors are essential; it is also well to have an outer room or inclosed closet to keep your dyestuffs in, as it is important that they be kept clean. When cans of color are opened for the purpose of diluting a portion or making a color, have the cover replaced and returned to closet when through with it.

Have bench or table whereon rests your basins, while you match shades in making colors, if possible, where a north light will strike it; and if cold weather and the windows closed, keep the glass clean. You will often get various reflections in the dyehouse that cause a great deal of trouble to the dyer; as, for example, if the sun should be shining on a red brick wall and the reflection beating into the dyehouse, it will often lead the dyer astray, and while he thinks he has a perfect match, when the color goes into the office there is a decided difference.

The great majority who are expected to be benefitted by this work are not ostrich feather manufacturers, but the job dyer; and it is my object to simplify the dyehouse as well as the methods of dyeing. A small corner of the dyehouse can be used, and a couple of ordinary wash-bowls, a common wash-boiler and a tin dipper are really all the utensils that are practically necessary to complete the dyehouse for the renovator. A couple of hours in

the morning devoted to feather dyeing, and a good practical man can turn out fifty dollars worth at a cost of only his two hours labor, and perhaps fifty cents worth of color. Feathers can be dried in an ordinary hot room or, if warm weather, out in the open air. The dry room where large quantities of feathers are dried should never be too warm, as the feathers are apt to dry up quicker than the boys can beat the starch out of them; and, as a consequence, the flues or fibres are not expanded as they should be, and the feathers are much harder to curl. The board or table used to beat the feathers on must be perfectly smooth, as there is otherwise danger of tearing out the flues.

The drying of feathers is quite an important operation, and if not understood, can result in ruining a great many by drying them improperly, allowing the starch to dry up on the flues without beating it out, and by breaking the quills. The dry room is only used when the weather is too inclement to dry in the open air, or when you have not got outside accommodations. The yard or roof is far preferable to the dry room, and especially so for white and black feathers. After having been washed and the starch thoroughly removed, it will improve them greatly to expose them to the sun for an hour or two. Colors, especially delicate shades, should not be allowed to hang in the sun only during the actual time required for drying a black made by our process; it greatly improves upon exposure to the sunlight, giving it an advantage over all others. Baths of logwood or old garnet baths that you are desirous of saving for future use, it will be well to remove them from the copper or tin basins or pans to wooden buckets or crockery jars, and cover them up for the purpose of excluding all foreign matter.

MISCELLANEOUS INFORMATION.

In the re-dying of old feathers the first thing necessary is to enter them in book by whatever system you may think best; after which they are assorted as to color, the blacks, browns, greens, blues, etc. Put in separate lots and then string them and mark your tickets. You will often find when you have selected your colors a number of different shades to be dyed one color; as, for example, when you come to string your browns, you will find a blue, a green, a garnet, a drab, and perhaps a dozen different shades of colors;

string them all on at once and enter together, and dye a good medium shade of seal brown and dry; after which you proceed to take them off the string, and place them with their respective tickets. You will now find, perhaps, one a shade too dark for your sample, another perhaps a shade too light. The former you would pass through a weak solution of sulphuric acid in starch bath, and the latter through a weak solution of bichromate of potash. Another one you may find a little too red for sample, or too yellow. These, in turn, you can bring to match your sample as per recipe for brown.

In beginning the days work, it is well to do all your bleaching and cleaning first, while your hand basins and dyehouse are in a clean condition; after which the blacks, as they require logwood good and pure, and the same logwood used for them can be used for all other colors where logwood enters into their composition. Consequently one bath of logwood boiled in the morning will do all the work for the day.

In Chicago I remember, while giving instruction to a gentleman, who had come down from St. Paul, Minn., for the purpose of learning the art, that in one afternoon I taught him how to make every color and shade of color known, and my logwood bath that was used during the whole day's work was boiled in a small sauce-pan that held about two quarts. It had been used in making black, browns, greens and navy blues of all shades, and was still in good enough condition to make any color, excepting perhaps black.

Keep your bath of logwood covered at all times when not in actual use, and, indeed, then, if convenient, to prevent any foreign substance from entering it. It is the custom of a great many ostrich feather dyers to keep a quantity of starch in the dyehouse for the purpose of dipping their feathers into it and partially beating them out prior to removing them from the bath for the purpose of drying the ends up to see if they match sample. This is a very bad practice, for the loose starch flying through the dyehouse will settle on the uncovered colors and cause not a little annoyance and trouble. Keep the starch out of the dyehouse; keep it in the drying-room where it belongs. In drying your feathers out of the baths in starch it is well to have two boxes,—one to be used for colors that contain acid; as, for example, light blues, lemon, etc.,—the other for those colors that contain none; such as drabs, pinks, etc. In dissolving colors use ordinary bottles, and be sure to always use boiling water for the purpose of diluting. Let the proportions be about

one teaspoonful of color to one pint of boiling water. Shake gently to thoroughly dilute aniline, and cork or cover bottles to keep out dirt.

Colors that are used in making very delicate shades, such as pinks or light blues, it is well to tie around the top of the bottle in place of a cork a small piece of muslin. It will act as a strainer, and prevent particles of color that may not have been thoroughly dissolved from passing into the bath and spotting your goods. Do not be too careful of the hands and afraid of getting them covered with dyestuffs; use them in the bath instead of sticks at all times, excepting where the liquid is too hot to permit it. The best method of cleaning the hands, no matter how dirty, is to pass them through a solution of soda, about one-quarter ounce in a small quantity of hot water; rinse off in cold water, and take about a teaspoonful of chloride of lime, moisten with water and rub the hands gently with it until all color has entirely disappeared; then wash with soap and hot water.

WASHING RAW STOCK.

First string your feathers, being careful to place the string on the end of quill so as not to get any of the flues under the loop; then slice down according to quantity of feathers to be washed, from one to more pounds of soap in boiling water, and boil down to a liquor; after which fill a clean tub half full of luke warm water, and pour soap into it; then enter your feathers and give them a slight rubbing. Then push them well under the surface of the water, cover them up and allow them to remain over night. In the morning run off dirty water and squeeze out your feathers; enter your feathers in a tub of clean luke warm water and use an ordinary wash board and a soft scrubbing brush. Rub bar soap on feathers, and brush gently, being very careful not to tear out the flues. Soap and brush one string at a time, manipulate them much after the manner of a woman handling a large wash. Be careful to give minute attention to the bottom portion of the feathers, as the flues are always more closely stuck together with the natural grease of the bird, and it often requires an amount of hard labor to remove. Repeat the washing operation and rinse off in about three luke warm waters, starch and dry.

In starching rub the feathers around well between the hands for the purpose of getting all the flues thoroughly expanded, squeeze out of bath and hang

on lines to dry. Put no more out at once than the dyers can comfortably handle, as it is well to have them beat out on board at regular intervals of a minute or so; thereby expanding the flues to their utmost. The process of selecting the different grades or qualities follow, and it is necessary for the person performing this work to be familiar with the application of dyestuffs to feathers, to insure the dyer less trouble; as the different qualities all put in the bath together, and going through exactly the same process will come out different shades of color, will cause the dyer a great deal of trouble and labor getting them all an even color. When a batch of feathers are intended for white it will not be necessary to dry them first; simply wash and rinse, and prepare your white bath as per recipe, and pass them through it. It is scarcely necessary to remark here that natural black and gray feathers must not be washed at the same time with whites, as the latter would not be improved.

Strings should not contain more than fifty plumes, for, if they are made much longer, it would be awkward to handle them. Tips, however, are often strung three or four in a bunch, according to size, and an ordinary string may contain two or three hundred. In washing natural black tips it is advisable to use a brush on them during the first rinsing to remove all particles of soap therefrom.

SHADING.

Shading from dark to light colors is the result of submerging one portion of the feather in the bath and withholding the balance. Great care and not a little skill is needed to produce a satisfactory result. There are various ways of handling the goods, covering up the portions to remain the light shade or holding them out with the hands. Spotted or speckled feathers are produced by first dyeing the light shade that you desire to be spotted, and then wrapping around a round stick with cord, according to the size you desire to have the spots, you will regulate the weight of cord used. After having bound the cord tightly around the feather and stick, which must then be tied firmly to keep from slipping, pass through boiling water for a few seconds for the purpose of expanding the wood and contracting the cord, thereby making the cord much tighter. After you have made them whatever dark color you desire, take out, starch and pass through dry starch; then remove

cord and dry your feathers, when you will find that the portion covered by the cord will be the light shade, and the feathers have the appearance of being dotted all over.

Natural blacks or grays can be speckled as follows: Go through the same preparations of binding around stick with cord and degrading or bleaching them white. The result will be that the portion covered with cord will be same as before entering the bath, a black or dark brown, and the body of the feathers will be white. Should you desire the feathers dyed any light color to contrast with the dark spots; before removing the cord, mix your bath and dye as per recipe, dry as before directed, and the result is very beautiful. Some very nice effects are produced in shading by taking natural grays or bioucs, that is, feathers that are one portion white and the balance in spots, black.

PARING, STEAMING AND CURLING.

Feathers that have just come out of the dyehouse for the first time require paring, which consists in removing the quill from the inner portion of the feather, thereby making the feathers more elastic. The feathers must first be thoroughly dried; they are then taken, one at a time, held between the thumb and two fore fingers of the left hand, while, with a knife held in the right, the inner quill is rapidly removed close to the flues or fibres. This branch of the business is in itself a trade, and requires a great amount of skill and caution to prevent cutting through the quill. The feather can be made still more limber by scraping the quill with a piece of glass. Of course, this process of paring the quill is only used in new work. In re-dying old feathers it is never needed; in old work it is only necessary to dry up thoroughly, steam and curl. A great many have no knowledge of what relation steaming has to the finishing of feathers. It has the effect of making all the flues lie perfectly straight beside each other, and also dampens the feathers just enough to assist the curler in her work.

It is necessary to have a steamer made as follows: get a kettle that will hold about one gallon or more of water, made out of plain tin, with a spout commencing at the base about two inches in width and tapering up to a half inch in width at top. The spout should be about eighteen inches in length; the total cost should not be more than one dollar. Never have it more than

half full of water, and you can boil it on either an ordinary stove or common gas or oil stove.

You may ask why steam from the boiler, or out of an ordinary tea-kettle would not answer? It is too wet. Instead of having the desired effect it wets the flues, while the other dampens it just enough. The steam emitted from the steam kettle is drier than any other.

When the steam is passing through the tube take hold of the feathers by either end and pass backward and forward for a few seconds about two inches above the top of pipe, and lay down perfectly flat, one on top of the other. Curling is a trade that can only be thoroughly mastered by practice; the principles can be taught, but only practice will make perfect. It does not, however, require a great while. I have known persons that within three months had become first-class curlers, practicing a short time each day.

The feather is held between the first and second finger and thumb of the left hand and a few flues taken up at a time with the knife held in the right hand, and gently drawn along the round dull edge of the knife, and allowed to drop in a half circle; begin at the bottom of the right hand side of the feather, work up to the top and around and down the other side; and in laying up take up about three flues at a time, skipping about six. Feminine fingers are generally better adapted to this work than others, and, in fact, it is more of a woman's work than a man's.

Tips are generally bent and branched. You can give the feathers a nice droop by taking the quill between the thumb and fore-finger, and with the thumb pressing the quill through between the first and second finger. Begin about the middle of the feather, and, shifting about a quarter inch at a time, pass swiftly up towards the top, when the feathers will have a very beautiful droop. Plain wire stems can be used. Take thin wire, cut about five inches in length, and twist one end of it on stem or quill of your feathers so as to hold; then take tissue paper, cut in strips about a half inch wide, and in color corresponding with the shade of feathers; wrap it around wire to entirely cover it up, and then branch tips, two or three in a bunch, as suits your fancy.

NOTE OF THE PUBLISHER.

The old maxim, that "seeing is believing," applies perhaps nowhere more than in dyeing. All those who have availed themselves of the opportunity to see the method of dyeing ostrich feathers practically executed before their eyes by the author, as described in the foregoing pages, are satisfied of and willingly testify to its superiority over any of the methods heretofore known, practiced and often acquired at the cost of much money, time and trouble, and which, in many cases, when put to the practical test, failed to give the desired results. Yet there are probably many more disbelievers than believers in any new method, however freely and truthfully certified to, who mistrust the quick work of our new processes of ostrich feather dyeing, and who would rather prefer to operate after a somewhat slow but (in their opinion) therefore surer, older method. They shall not be disappointed by perusing our book and in looking up something which they would want to try in practice, and for them especially we supplement our book with the following Appendix, containing a number of practically tested recipes for dyeing ostrich feathers.

APPENDIX.

GENERAL REMARKS.

The cultivated taste of the present age, requiring a large variety of natural and artificially produced or embellished material for adornment, employs almost any kind of bird's feathers, either in their natural coloring or dyed. None of them, however, are used in the condition as they are plucked from the body of the live or dead bird, but all must undergo a cleaning process, which not only serves to improve their appearance, but is an exceedingly essential requisite for the preservation of the material from decay and the attacks of moth and other insects, and is, above all, the first condition and indispensable preparatory operation for dyeing feathers, whether the costly feather of the ostrich or the common feather of our domestic chicken or pigeon. The cleaning or washing process is the same for all kinds of feathers; the ostrich feather, however, requires drying after every treatment in a bath, and a special operation for the purpose of opening the fine flues, which gives the plumage of the ostrich its characteristic and distinguishing beauty and rich, downy appearance of luxurious softness.

The feathers of the ostrich, which are used for dress-feathers, are taken from the wings and tail of the bird, whose spurred wings, by their peculiar construction render it entirely unfit for flight. The wings seem rather only fit to serve for the purpose of holding the body of the bird in equilibrium while running, and of preventing it from sinking to any depth into the loose sand of the deserts, which are the home of the ostrich. The natural colors of the ostrich feathers are white, black and gray, or rather a dark drab. They are, therefore, sorted according to their natural color, to be bleached white, or dyed in light colors, or to be used for dark shades. Practical men in the general dyeing business, and in garment dyeing or re-dyeing, hold that it is unnecessary to bleach, respectively strip, the material for dyeing dark colors, and garment dyers strip their material only to a certain extent, so as to leave upon it a bottom of color which they can advantageously use for their new dye. This method appears correct, if as "practical" as all that is designated, which results in a saving of expense or labor; but it is evident that a clear color of the highest possible beauty can never be obtained upon a bottom of a different hue; the bottom color will always, more or less,

show and impair the purity of the topping color; but compound or mixed colors can be thus produced in an advantageous manner and to good effect, if the color of the bottom enters into their composition. The same is unquestionably the case with ostrich feathers, and the dyer is often compelled and must be prepared to bleach the gray or black and white feathers in order to dye them any light shade. The bleaching of naturally purely black feathers is probably but seldom required, as these are ordinarily left as nature has made them, but merely cleaned to heighten their beauty and gloss.

Owing to the delicate nature of the material, the dyeing of ostrich feathers bears much similarity to that of silk; both being high in price, carelessness and negligence in their treatment is apt to entail heavy losses. The utmost cleanliness of all utensils is an absolute requirement; dyestuffs, drugs and chemicals must never be added to baths in substance, but always in solution, and never while the material is in the bath; but the material must be taken up while the dyestuff or salt, etc., solution is being added to the bath, and only re-entered after stirring well. Solutions, as well as decoctions, must always be filtered, respectively strained, before adding them to the dye bath, even if they have been prepared beforehand, because any undissolved or solid particle of substance deposited upon the feathers would necessarily produce a spot or mark of a darker or lighter shade, as the case may be, according to the character of the undissolved substance. Although alkalies and heat are applied and necessary for washing or scouring, that is, cleaning and ungreasing the feathers, strong alkaline and excessive heat operating together are as fatal for feathers as they are for any other animal fibre,—wool or silk,—and strong heat applied to dry feathers is apt to irretrievably ruin them. The soap with which feathers are to be treated, must, therefore, be as neutral as possible, and if recipes speak of "boiling" the feathers, it must be understood in the sense as in wool-dyeing, that is, to apply a heat near the boiling point, when the baths begin to throw up bubbles, but not actually boil.

That the water used for any purpose in ostrich feather dyeing, for washing, bleaching, dyeing or rinsing, must be perfectly clean, needs hardly to be mentioned.

UTENSILS.

The utensils required for feather dyeing are of a very simple character, few and inexpensive. For small establishments an ordinary stove, a common wash-boiler, to have constantly hot water on hand, an ordinary wash tub, a white china wash-basin for dyeing, a clean board for starching and a few bottles, together with a small tin pan or kettle and funnel, for making solutions and decoctions and filtering them, is all that is necessary besides the work table. More recently, flat, oval upper pans, tinned for special purposes, have been introduced as dye-vessels, which are neatly provided with a moveable perforated false-bottom, and are heated either upon a direct fire, or a gas jet, or by direct steam. In large establishments copper pans are generally used, for the better grades of ostrich feathers especially, and for ordinary goods wooden tubs, both heated by steam. Where wooden tubs are used, several of them are set apart for the color most in demand, such as black, brown, gray, mode, etc.

PREPARATION OF THE FEATHERS.

The bundles received from the dealer being opened, the feathers are sorted according to color and size, and those for white and light colors, to be bleached, are laid from those of dark colors, which are ordinarily not bleached, that is, the black or gray ones. When going to work the feathers are put on strings, that is, they are firmly tied singly, about an inch apart from one another, and about an inch above the end of the quill, 20 or 25 with one string, seldom more, as they would make the bundle too thick and unhandy. The feathers are then ready for the steep, which operation ought always and for any method be the first step of treatment before proceeding to washing, scouring and bleaching proper.

For this purpose a strong solution of soap is made in boiling water; when cooled down to about 150° F., it is well stirred, the feathers entered and left in the bath over night. The temperature may be kept up over night. It is necessary, however, to lay the feathers down in the steep so that the liquid can reach every part of them, and to keep them well immersed in the steep, for which purpose it is advisable to weigh them down by clean sticks of wood or some other means. Instead of soap, soda may be used for the steep; taking about one and one-half ounces of soda crystals to one gallon of water.

By the steep the impurities, dirt and grease, covering the feather are loosened, and thereby the following cleaning operations materially facilitated.

CLEANING AND BLEACHING OF FEATHERS.

The ostrich feathers, like all material taken from the covering of the animal body, wool, hair, etc., which are embellished by dyeing for the use of man or woman in dress, contain by nature a certain amount of fat, and in their raw condition are more or less covered with dust, dirt and a greasy exudation, which must be removed for dyeing; that is, they are scoured or washed and then bleached or whitened, because the feathers, like all other so-called white animal matter, have always a faint yellowish tint, sometimes yellowish spots which cannot be removed without injury to the material, but obliterated by bleaching, which, in the case of white feathers, is called bleaching or whitening. The bleaching of gray and black feathers and the stripping or decoloring of dyed feathers are different operations.

For scouring or washing, novel methods are recommended, which, however, differ from one another very little, and are, on the whole, represented by the following: Prepare a good handwarm bath (100-120° F.), in which dissolve two ounces Marseilles soap in per gallon of water and beat up to a good lather. Enter feathers and rub them well, string for string, by hand. They may even be taken upon a wash-board and rubbed with a brush, which does not hurt them, notwithstanding their delicate structure, because the soap steep has given them great elasticity and resistance to the manipulation. Continue the operations until the tank is exhausted and dirty; then give another fresh bath of the same composition and temperature; treat the feathers as in the first bath and rinse them perfectly clean from every particle of soap in two or three luke-warm (100°) waters; for, every trace of soap remaining upon the feathers will hinder the dye from running up and cause uneven colors, or, upon white feathers, yellow stains. Then prepare a cold bath with solution of bioxolate of potash (one-eighth to one-sixth ounce of salt to one gallon), enter feathers, pass them in the bath for 15-20 minutes, take up and rinse them in cold water three to four times to remove the salt. For feathers which have to remain white, the latter bath is composed of one and one-quarter ounces bioxolate of potash and one and

one-eighth ounces oxalic acid to one gallon of water, and the feathers laid down in it until perfectly white; when they are taken out and rinsed clean from acid in luke-warm water.

The feathers being rinsed clean from the oxolate of potash bath, if destined for white, are then whitened, or rather blued, for the purpose of covering the yellowish tint above mentioned. To this purpose a cold bath is prepared with only so much methyl violet or methylene blue, as to give the water a very faint tint. To ascertain whether this is the case, a white china plate is held about a foot below the surface of the bath, when its appearance will show the shade of blue that will be produced by the bath. The feathers are then entered and gently agitated in the bath until they have the desired tint.

DRYING OR STARCHING.

The feathers coming from the bioxolate of potash bath, after rinsing, or from the blue baths, are squeezed out by pulling them through the hand, and pressing them between the laps of a dry clean piece of white muslin, whereupon they are immediately passed through a bath of raw starch, that is, unboiled starch, consisting of about one-half pound of starch to a gallon of water. After passing them through the hand the feathers are then again pressed between the cloth; then the waves are lightly drawn by hand over the stems, and the feathers either beaten between the hands or upon a clean board over a stove until dry, or they are agitated by hand or by a suitable mechanical contrivance before an open fire or gas-jet, or hung in a warm room and frequently shaken until dry, that is, until all starch has dropped out; and finally the remaining starch is beaten out between the hands or upon the board by means of a soft brush. By this treatment the feathers are not only dried, but the flues opened besides. It needs not to be specially mentioned, that the feathers are dried, or finished, as it were, in the same manner after dyeing.

In case the flues are not sufficiently opened, although all the starch has been beaten out, dip the feathers into clean benzine and swing or agitate them until dry, which takes place in a few minutes, while the flues are opened in the most perfect manner. For white feathers the benzine may be blued, but in this case, they must be dried between muslin.

BLEACHING OR DECOLORING NATURALLY GRAY FEATHERS.

The feather dyer is often required to dye light colors upon naturally gray or even black feathers. As above remarked, the natural color would show even under dark colors dyed upon them to a greater or less extent, unless they are first decolorized, that is, their natural color destroyed or blackened. Much more necessary is, therefore, this operation for light colors to be dyed upon naturally colored ostrich feathers. The only known chemical agent affecting such a bleach to nearly white is peroxyd of hydrogen or oxygenated water. For bleaching ostrich feathers a bath is prepared of peroxyd of hydrogen to which so much liquid ammonia is added as to give the bath a sharp pungent odor. The feathers, which must be previously cleaned as above mentioned, and well rinsed, are entered and left immersed in the bath until they have assumed a nearly white cream color, whereupon the feathers are taken up and thoroughly rinsed or laid down in running water until every trace of ammonia has disappeared. It must be observed, however, that only acid aniline colors can be dyed upon such decolorized feathers and that in dyeing only a moderate heat must be applied.

Dr. P. Ebell, of Linden, near Hanover, one of the first and still largest manufacturers of peroxyd of hydrogen, writes on the subject of feather bleaching as follows: The assorted and picked feathers are cleaned from dirt and fat with soap and water by means of soft brushes, which operation is continued until the feathers (after drying) are readily wetted by water; when they are laid down for some time in pure water. The liquids are removed from the feathers by a centrifugal machine or a wringer (the latter is evidently meant for ordinary feathers, other than ostrich feathers). Mix in a clean wooden tub twenty litres peroxyd of hydrogen (Königswäter & Ebell) with four hundred and fifty grammes ammonia 20° B. (=0.9 sp. gra.) and heat to 34° C., by a leaden steam pipe at the bottom of the tub. Enter 5 kilo. cleaned feathers, which work by hand and turn every hour. In twenty four hours the bleach is completed. If pure white feathers are wanted, give a second bath, but for a shorter period. The bleached feathers are carefully removed from the bleaching bath and laid down, for an hour and a half, in a cold bath of one hundred litres water containing one hundred grammes sulphuric acid 66° B., and then completely lixiviated in pure, soft water.

While moist (after squeezing) they are then passed through a milk of unboiled starch which is lightly blued with aniline blue or violet, and slowly dried, in the air or in a warm room, under repeated shaking to prevent the flues from sticking together. After removing the starch by beating, the feathers are ready for curling, etc.

PEROXYD OF HYDROGEN.

This most valuable bleaching agent is a contraction of hydrogen and oxygen, of the formula HO_2 , sp. grv. 1.45 (chemically given), or 94.12 per cent. oxygen with 5.88 per cent. hydrogen. It consists in a limpid, syrupous liquid, of characteristic color, and when heated to 15°C. , is decomposed into water and oxygen, upon which property its great bleaching power is based. Experiments to reduce it to a solid form by refrigeration and pressure have thus far been unsuccessful. The commercial article is somewhat modified by the addition of water to prevent its ready decomposition under the influence of a warm temperature. For the same reason it is advisable to always keep it in a cool place.

LIGHT BLUE.

I. To dye this delicate color well, special care must be taken in cleaning the feathers, for which purpose only olive-oil soap of the best quality, with a little ammonia, ought to be employed. When they are perfectly clean and no more grease upon the stems, rinse them first in one or two lukewarm waters, then in cold water until the last trace of soap is removed. Then fill your basin or dyeing pan three-quarters full of cold water; put in, for a dozen feathers, one hundred and eighty grammes (about eight ounces) of raw starch in a sufficient quantity of good indigo extract to give the starch-bath the desired shade. Enter the feathers and work them gently until they are completely dyed, that is, for about fifteen or twenty minutes. Then take them out, squeeze out the starch by putting them between the fingers and thumb of your hand, and shake them before the stove, or in a well-warmed chamber until dry. While drying, beat them from time to time upon the board, or between the hands to remove the adhering starch.

II. Prepare a lukewarm bath acidulated with a few drops of sulphuric acid, so as to give a faint sour taste, to which add, according to shade, solution of methyl blue B. (Actien Gesellschaft fuer Anilin Fabrikation, Berlin). Enter the feathers and leave them in the bath until cold, or until uniformly dyed.

NOTE.—Some dyers use alkaline blue, which is not, however, recommendable, because alkaline baths, as above remarked, are injurious to the feathers and must be avoided as much as possible.

III. Prepare bath of lukewarm water, dissolve in it about one-half ounce tartaric acid per one quart, and add one ounce indigo carmine per quart of liquid; stir well, enter the feathers and agitate or lay down in the bath until the required shade is obtained. This color shows little fastness to light and air, which can be improved, however, by adding to the dye bath one-quarter ounce alum per quart. The shade being obtained, take up the feathers and pass, without rinsing, through raw starch milk, dry and beat as described.

Light blues, as is easy to understand, can only be dyed upon white feathers for the most delicate shades; nearly white, or developed gray feathers may be used for the shades approaching a light medium blue.

NAVY BLUE.

I. For this color naturally gray or semi-bleached feathers may be used. It requires a mordant, like wool. For this purpose prepare a bath of forty per cent. (of the weight of feathers) tannin at 167° F., enter the feathers and agitate them from time to time for three hours. Then take them up, drain and squeeze them out, enter a cold bath of pyrolignite of iron (black liquor) marking 5° B., and work them for half hour; take them out, drain and squeeze, and then expose them, well spread out upon the strings, for one hour to the action of the air. Then rinse and dye upon a fresh warm bath with a mixture of aniline blue and a little methyl violet, using about twenty per cent. of the weight of feathers. Add the dyestuff in the beginning only in small doses and slowly in order to prevent the production of a bronzy, undesirable lustre upon the stem, as is often the case in dyeing with aniline dyestuffs if they are added to the bath in too large doses.

II. Prepare a hot bath, to which add as much indigo carmine as to bring the color of the bath pretty near the shade to be produced. Enter the feathers

and agitate them in the bath for one hour. Then take up the feathers, add alum and a solution of cloth-blue S. to the bath, re-enter the feathers and work them while raising the temperature to boiling point, when the steam or gas is turned off, or the pan removed from the fire, and the feathers allowed to lie for fifteen or twenty minutes longer in the bath. They are then taken out, rinsed, starched and dried and beaten.

III. Have the feathers properly cleaned and well rinsed from the soap, respectively soda. Gray feathers may be used unbleached, but a purer color is obtained upon them when bleached. Prepare a hot bath, to which so much sulphuric acid is added, that it has a feeble sour taste; add the solution of two per cent. (of the weight of feathers), navy blue, one per cent. fast blue or black, and one-eighth per cent. acid fuchsine. Stir well, enter the feathers, manipulate while raising the temperature to boiling point, but not to actual boiling, continue at this temperature for one half hour; then stop off the steam, lay the feathers down in the bath until cool, lift and dry as usual.

GENDARME BLUE.

This color requires a pure bottom, that is, naturally white or bleached. After cleaning, respectively washing in warm soap, which must not even be omitted with bleached feathers, and thorough rinsing, prepare a bath and dye as for dyeing light blue with indigo carmine. Then add some aniline green and navy blue to the bath, re-enter the feathers which have been taken up before making the addition, work them well while raising the temperature to the boiling point; continue at this temperature for one-half hour longer, lift, rinse, starch and dry as usual.

PLUM OR PRUNE.

I. For this color, which has in itself a subdued tone of brown, or has the color of gray ostrich feathers, such naturally colored feathers may be used unbleached, but well cleaned and rinsed before dyeing. Prepare a lukewarm bath, to which add about one-half ounce tartaric acid to per quart of water and solution of methyl violet 6 B., according to shade, with a little aniline ponceau or fast brown for toning. While working the feathers, raise

the temperature and continue dyeing at nearly boiling for one-half hour; then take out, wash and dry.

Or,

II. Prepare a boiling hot bath with alum, sulphuric acid and tartar; to which add acid fuchsine; enter the feathers, and dye one-half hour to a blue red, which tone, by the addition of decoction of logwood, continue at nearly boiling heat for one-half hour longer, lift, rinse lightly, starch, beat and dry.

III. Take a hot bath, upon which violet has been dyed, and refresh it with some solution of methyl violet, 5 B., and a few drops of sulphuric acid, or prepare a hot bath with the same ingredients, and indigo carmine, according to shade; or, instead of indigo carmine, indigo substitute, fast blue B. A., and indigotine; preferably, however, use indigo carmine, which develops more slowly, and therefore is surer to give better results, while the aniline dyestuffs run up more rapidly, and are apt to dye unevenly, unless their solutions are added gradually and the feathers handled quickly and carefully.

LIGHT YELLOW.

I. Light yellow is comparatively very little in demand for ostrich feathers, and scarcely used for trimming hats of children and young misses as a set-off for other colors. To produce it, prepare a pretty hot bath with a little sulphuric acid, so as to give it a slightly acid taste, add very little quinoline yellow, lay down the feathers in the bath for one-half hour, turning and agitating them from time to time, lift, rinse and dry. For this color, as well as for light blues and roses, the feathers must be perfectly white. (For this dye the quinoline yellow manufactured by the Actien Gesellschaft fuer Anilin Fabrikation, Berlin, is specially suitable). As the purity of all light shades of delicate colors greatly depends upon the purity of the water, it is advisable to bring the bath, before preparing it, to boil with some bran and chloride of tin and skim it off well.

MEDIUM YELLOW.

Various shades of yellow can also be produced with the old natural dyestuffs, which are not, however, equal in brilliancy to the foregoing described colors. The feathers must be bleached for these as well as for any clear color, which would be materially impaired by an impure bottom; still developed grays may be employed. After scouring and thoroughly rinsing the feathers, prepare a cold bath of alum, about one ounce to one gallon of clear water, or of acetic acid; lay the feathers down until well opened, so that the liquid can uniformly act upon all parts, for one hour. Then take them out, squeeze and centrifugate them, and dye the shade upon a fresh warm bath with the required quantity of flavine, decoction of color or of fustic; lift, rinse and starch as usual.

Or, dissolve a sufficient quantity of turmeric in boiling water, filter and enter the feathers while the filtrate is still well hot. Agitate them for five minutes, then take them up, add to the bath a small quantity of tartaric acid, this to promote its dissolution; then re-enter the feathers, work them again for five minutes, lift, rinse in cold water, and dry.

If these colors are to have a light reddish or warmer tone, add, when nearly done, some anotto to the dye bath.

DARK YELLOW.

Bleached grays answer for this color as well as naturally white feathers. Scour and rinse them well. Prepare a bath, feebly acidulated with sulphuric acid, and add the filtered solution of two and one-half per cent. of the weight of feathers, dark yellow, (manufactured by the Leipziger Anilin Fabrik, formerly Bayer & Kegel). Enter the feathers in the cold and work them diligently until the color is well up, then raise the temperature slowly to 170° F., dye to shade, lift, rinse in clear cold water, starch and dry.

II. A new light yellow, which is fast to light and air, is obtained by products of Leonhardt & Co., at Manheim, viz: redarine and acme yellow. Add to a hot bath of 170-190° F., a quite small quantity of redarine and still less acme yellow; enter the feathers, manipulate for one-half hour, take out, rinse and dry them with starch, and beat well out. This color being extremely sensitive, the purification of the water for the bath is as necessary as the most scrupulous cleanliness of utensils and workshops.

GOLDEN YELLOW.

I. The feathers being scoured and rinsed clean of soap, prepare a bath of five per cent., of the weight of feathers, bisulphate of soda, add solution (filtered) of azo orange, according to shade. Enter the feathers at 120° F.; heat up slowly to 170° F., while working the feathers; lift when the shade is obtained, squeeze out, starch and dry.

II. Prepare a bath with three per cent. (of the weight of the feathers) Glauber salt and one per cent. sulphuric acid. Enter the feathers at 100-120° F., after adding to the bath the solution of one per cent. golden yellow S. (of Gust. Doerr, Frankfort-on-Main), work the feathers repeatedly during one-half hour, when they all have assumed a rich, nourished color; take up, rinse lightly; starch and beat them dry.

OLD GOLD.

Have the naturally white or decolorized gray feathers well washed in soap and rinsed clean from it. Prepare a hot bath at 170° F., to which add so much aniline cream as to color it dark reddish yellow. Enter the feathers and agitate them from five to ten minutes, according to the shade desired. Then take them up, add some sulphuric acid to the bath, re-enter the feathers, work for two minutes; then lift, rinse and dry. The bath can be preserved for further use.

GRAY.

So unpretending this color appears, so difficult is it to produce, and it requires a considerable amount of practice and good judgment to bring out a good color from the beginning, as very little too much or too little will spoil a color either in tone or in shade. A very good logwood gray, which with proper attention seldom fails to turn out satisfactory, is made as follows: Prepare a hot bath, to which add a small quantity of decoction of logwood; enter the feathers and work them in the bath for fifteen or twenty minutes, according to shade desired. Then take them up, add to the bath very little pyrolignite of iron, that is, only as much as to turn the color of the bath; re-enter the feathers, agitate them again for fifteen or twenty minutes

in the liquid; then lift, rinse and starch as usual. This color might be best described as dark ash gray. Instead of pyrolignite of iron, some solution of copperas may be used. It will be easily understood, that the more concentrated the decoction of logwood is, the darker turns out the color, and it is in that respect particularly that the dyer has to use good judgment in producing shades from silver gray to dark ash gray. This color, besides, presents the advantage, that by topping it with solutions of blue, brown, yellow or green coal-tar dyestuffs a great variety of mode colors can be produced.

PEARL GRAY.

After scouring and rinsing well, prepare a warm bath (100-120° F.) with five per cent., of the weight of feathers, bisulphate of soda, to which add solution of Victoria blue and of extract of archil, according to sample. Acid violet may be used, but requires a temperature near the boiling point, which ought to be avoided wherever possible in dyeing ostrich feathers. To be on the safe side, make the solutions of the dyestuffs of medium concentration, use only the clear of them, or better filter the same, and add it slowly and gradually first in small doses, finally by drops, for which purpose the use of a burette with squeeze-cock is recommendable.

SILVER GRAY.

Scour, respectively bleach, and rinse the feathers well clean, prepare a bath, work the solution of five per cent., of the weight of feathers, silver gray (Actein Gesellschaft fuer Anilin Fabrikation, Berlin), feebly acidulated with sulphuric acid; enter the feathers in the cold, work well to make the color dye up evenly; then raise the temperature slowly under diligent working, to 170° F., continue at this temperature for five to ten minutes, lift, rinse and dry.

BROWN.

The series of brown colors, partly produced by combinations of spectrum colors, partly of direct brown dyestuffs, presents a large range of modifications and shades, from a light rust brown or buff to nearly black,

blueish, yellowish, reddish, olive brown, etc., and is in this respect only inferior to the non-descript endless variety of modes. With the exception of the very lightest shades, which require perfectly white feathers, they can be dyed upon half-bleached, and the deeper shades upon unbleached gray feathers; the dyer, must, however, in the latter case, bear in mind, that the gray bottom color always influences to a certain degree, the tone of the color that is to be dyed upon it. Nevertheless, as to the proportions of the dyestuffs to be employed for a given tone or shade cannot be given, because the tinctorial value of artificial dyestuffs is very changeable and not even constant with the same makers. Experience, skill and trial dyes must, therefore, guide the dyer in composing the baths for browns as well as for modes, the majority of the latter being modifications of brown. In general the following may be observed:

I. After scouring the feathers and rinsing them perfectly clean of the scouring material, whether soap or soda, prepare a bath of 170 to 190° F., to which add fifteen per cent., of the weight of feathers, bisulphate of soda, indigo carmine, extract of archil and azo yellow.

According to the proportionally greater or smaller quantity of either dyestuff added to the dye bath either browns are obtained, or olives, Russia green, reseda, or a variety of modes. The trouble with all colors into whose compositions indigo carmine enters is, that this dyestuff requires a comparatively high temperature to run up, preferably a boiling bath, which, however, is decidedly objectionable with ostrich feathers. To avoid this difficulty, the new acid Victoria blue is used instead of indigo carmine, and fuchsine S. instead of extract of archil. Victoria blue dyes up readily at a moderate temperature.

II. The feathers being scoured and rinsed clean, prepare a boiling bath with so much sulphuric acid as to give a feebly sour taste, and add fast aniline brown, turmeric, and indigo carmine or cloth-blue S., according to the tone and shade desired. Prepare the bath so that it shows exactly this tone of color which is to be dyed, and bring it to boil in order to produce a perfect mixture of the three dyestuffs, or rather their filtered solutions. Then chill the bath to about 120° F., enter the feathers, while raising the temperature in about fifteen minutes to near the boiling point; then dye to shade, lift, rinse and dry.

It is advisable, in order to obtain a level dye, to add not the whole amount of dyestuff solution required at one time, but at least in two times; which rule altogether applies to all aniline dyestuffs, more or less, as they mostly run up very rapidly and are apt, therefore, to give uneven dyes.

If a yellowish tint is wanted, use a little azo yellow or azo orange; picric acid, which was formerly very freely used for this purpose, has been almost entirely abandoned.

LIGHT BROWN.

Clean and rinse them as usual, prepare a bath of 170-190° F., with redarine, a trace of orange O, and some acid green; enter feathers and work for one-half hour, then lift, rinse and dry. By varying the proportions of dyestuffs, a series of modes is obtainable. (Dyestuffs manufactured by Leonhardt & Co., of Mannheim).

RUST BROWN.

Prepare a slightly acidulated warm bath with three per cent., of the weight of feathers, fast aniline brown, one per cent. azo yellow, one percent. extract of indigo, and a little sulphuric acid; enter the well scoured and rinsed feathers at 120-140° F., work the feathers for one-half hour, while slowly raising the temperature to the boiling point; continue dyeing at that degree of heat, but not boiling, for five minutes; lift, rinse, starch and dry.

RED BROWN.

I. Scour and rinse well; prepare a warm bath, in which dissolve three per cent., of the weight of feathers, alum, add twenty-five per cent. extract of archil, one and one half per cent. azo yellow, and if required for shade, one-half per cent. indigo carmine; enter at 170° F., dye to shade while slowly raising the temperature to near the boiling point, continue at that temperature for ten to fifteen minutes longer; then lift, rinse and dry. Instead of indigo carmine, cloth-blue S. may be used, in which case enter at 120° and raise the temperature slowly, not above 190° F. Or,

II. Prepare a bath at 190° F., add five per cent. bisulphate of soda, when dissolved, add solution of extract of archil, fast yellow and indigo carmine as required for the shade, and dye at that temperature to sample. Instead of archil, any red or orange azo dyestuff may be used, preferably bordeaux.

COFFEE BROWN.

Have the feathers well cleaned and rinsed, bleaching being not required, prepare a bath with three per cent. alum (of the weight of feathers), at 170° F., add indigo carmine, bordeaux and azo yellow, according to sample, and dye to shade while slowly raising the temperature to near the boiling point, but bring not to boil, but continue until the indigo carmine is well up. A less fast color is obtained with archil, indigo carmine and picric acid. When finished dyeing, rinse, starch and dry as usual.

The dyestuffs for brown being nearly the same for all shades, while the depth and tone of the color is produced by differently proportioning the quantities of the different dyestuffs and the time of dyeing, it is advantageous to have the solutions of dyestuffs near by on hand; it is advisable, however, if good work is intended, to always filter before using solutions which have been standing for some time. This precaution is necessary, because from most solutions, if allowed to stand for a day or longer, some dyestuff which was not dissolved but only suspended in the liquid, separates out forming a more or less copious sediment which, if it passes into the dye bath, settles upon the feathers causing spots or streaks of a different shade than the rest of the feathers.

PUCE.

Scour and rinse the feathers well; grays can be used in their natural color without bleaching. Prepare a warm bath, in which dissolve eighty per cent., of the weight of feathers, tartaric acid and eighty per cent. Glauber salt; then add sixteen per cent. aniline fast brown; eight per cent. azo yellow, and sixteen per cent. induline or nigrosine, and bring the bath to a boil; after a few minutes of boiling, chill by the addition of cold water, enter the feathers and work them at hand-heat for fifteen or twenty minutes until the color has become level; then bring the bath again to near the boiling point; lay the

feathers down in the bath, shut off the steam, or withdraw from the fire, and let the bath cool down. When cold, that is in about one or two hours, take out the feathers, rinse and dry.

FAWN.

Prepare a warm bath with five per cent. bisulphate of soda, add solutions of azo orange, acid violet and some archil cautiously in several doses until the bath has the desired color. Enter the scoured and rinsed feathers and agitate for fifteen or twenty minutes, to produce a level dye; then raise the temperature slowly to 190-200° F., dye for a few minutes longer, lift, rinse and dry.

By varying the proportions of the dyestuffs, drab, wood brown, lead color, etc., can be obtained, and olives by increasing the quantity of acid violet and omitting the extract of archil.

CHESTNUT BROWN.

Scour and rinse the feathers well; natural grays may be used unbleached. Prepare a decoction of one and one quarter pound cudbear, and six ounces turmeric in two gallons of water, strain through a cloth and enter the feathers at hand heat (about 90-100° F.); work them for twenty or thirty minutes, or until they have attained a nourished garnet color. Then take them out, rinse, lay them down for five minutes in a cold solution of about six ounces copperas in one-half gallon of water, take them up and rinse in cold water. Then return to the first bath, operate for fifteen minutes at hand heat, enter again, after rinsing, the iron bath, and continue alternately dyeing upon the two baths until the required shade is obtained. Rinse every time on shifting from one bath to the other, in clean water, and finally rinse well, starch and dry.

HAVANNA.

I. For this color it is advisable to use naturally white or bleached feathers, scour or wash them clean in soap and warm water and remove the soap by thoroughly rinsing in two warm and one cold waters. Prepare a bath slightly

acidulated with sulphuric acid, to which add eighty per cent., of the weight of feathers, tartaric acid, eight per cent. azo yellow, six per cent. fast brown, and three and a quarter per cent. acid green. Enter the feathers at 100-120° F. and manipulate at that temperature for ten or fifteen minutes. Then raise the temperature to the boiling point (but do not boil), lay the feathers down in the bath for one-half to one hour, while the bath cools down, lift, starch and finish as usual.

II. Prepare a bath slightly acidulated with sulphuric acid, bring to nearly boiling, add a concentrated solution of orange S. and some acid green, enter the feathers and dye to shade; then pass them through a weak oil-bath, and dry them, placed straight between several laps of clean muslin.

III. Prepare a bath of the decoction of twelve and a half per cent., of the weight of feathers, alum and twenty-five per cent. turmeric; strain, enter the feathers at 170-190° F., and let them lie in the bath over night. On the following day dye, at 100° F., with decoction of fustet, tone with decoction of logwood or of brazil, according to sample, starch and dry.

MUSHROOM.

I. For this elegant color take naturally white or bleached gray feathers, scour and rinse them well. Prepare a hot bath with five per cent., of the weight of feathers, bisulphate of soda, to which add, as required, filtered solutions of fast yellow, indigo carmine and ponceau G. Enter the feathers at 170° F., work them for ten or fifteen minutes and raise the temperature slowly to near the boiling point. Add the dyestuffs in small quantities gradually, making the additions only when the dyestuff of the bath has been completely absorbed, and then by drops so as to be able to correct the color without waste of dyestuff. Bear in mind, that the indigo carmine dyes up slowly and requires a high temperature. An easier process is, therefore, the following:

II. After cleaning and rinsing well, prepare a bath at 170° F., with four per cent. bisulphate of soda, to which add gradually in small quantities, as required, some nigrosine, azo orange and a little mandarin or nigrosine, alkaline blue and fuchsine S., rinse, starch and dry.

LIGHT DRAB.

Scour and rinse the feathers as usual; bleached grays may be used. Prepare a bath with five per cent., of the weight of feathers, bisulphate of soda and the clear solutions of acid violet, azo orange and fuchsine S.; add the dyestuff in small portions and finally by drops, until the bath has the desired shade of color; then enter the feathers and dye at 170° F. to sample, squeeze or centrifugate, starch and dry.

BEIGE.

I. For this color take either naturally white or well-bleached gray feathers, scour or wash and rinse them clean. Prepare pretty thin solutions of aniline orange (chrysaniline) and violet, add very little of them at a time and finally by drops to the dyestuffs containing either five per cent. bisulphate of soda or a small dose of sulphuric acid; enter the feathers at 145° F. and dye to shade at the same temperature, which will require about twenty or thirty minutes; lift, rinse, squeeze and starch.

II. Have the feathers well cleaned, respectively, bleached, and rinsed. Prepare a hot bath (170-190° F.), with a little sulphuric acid, just enough to give it a slightly sour taste, add a few drops of solution of fast brown and a little more solution of acid green (both dyestuffs of the Farbwerke, formerly Meister, Lucius & Bruening, Hoechst-on-Main); take of them one or two drops, respectively two or three drops per gallon of water for a light shade and increase quantities proportionally for darker shades. Rinse after dying, starch and dry.

III. Take white feathers or grays very well bleached to nearly white, scour and rinse them well. Prepare a bath of warm water, 100-120° F. and some vinegar so as to give it a distinct sour taste; add to a basin full, or about one-half gallon of the bath a little solution of fast brown, one or two drops of indigo carmine, and a trace of turmeric. Lay the feathers down in the bath for fifteen or twenty minutes and agitate them repeatedly in the liquid to make them level.

For a Gray Beige, add a little nigrosine to the bath and proceed as above.

MODES.

For the modes it is impossible to give generally applicable directions, as these colors are of an indefinitely varying character, consisting in modifications of other compound or mixed colors which are affected by sometimes very trifling, unmeasureable additions of a toning dyestuff, and coloring effects are produced which cannot be described nor defined by names, but must be judged by the experienced eye of the dyer. Most of these colors are derived from grays or browns as above remarked, and the safest way for the dyer is, to begin dyeing with light shades of the prevalent characteristic color and give them the peculiar tone by the addition of other colors by drops. The proportions of dyestuffs thus ascertained for light shades, are then easy to increase for deeper shades. It needs not to be remarked, that for these colors the feathers must be bleached, especially for light and medium shades, and that, if unbleached, grays are to be dyed in dark shades, the effect of the natural color must be considered in composing the dye.

In general all modes are dyed upon a bath which is acidulated with bisulphate of soda, with azo orange, azo yellow, azo brown, acid violet, indigo carmine, solid blue or cloth blue, induline or nigrosine, archil or acid fuchsine. For brown modes, solution of Bismarck brown may be added at the beginning, in which case the other dyestuffs serve only for giving the peculiar tone.

For a yellowish green mode take orange O, azo yellow, and solid blue (fast blue); for darker shades add a little violet 6 B., or a few drops sulphate of indigo. If alizarine dyestuffs are to be employed, use tartaric acid as mordant, but for neutral dyestuffs add also a little alum to the dyebath.

For gray modes use the same dyestuffs as above, excepting the orange, instead of which a blue-red dyestuff is to be employed, such as azo rubine, bordeaux, fuchsine, etc., with the addition of a little acid green. The bath must be acidulated with a little sulphuric acid, or better with tartaric acid, or tartaric acid and alum, and after dyeing the feathers must be rinsed, starched and dried as usual.

For particularly fast modes add only tartaric acid to the dyebath and no alum, and a few drops of solution of thio-scarlet, thio-rubine, and thio-

brown; for grays add a little azo yellow and sadden with solid blue. Alum does not agree with the thio dyestuffs which are manufactured by Dahl & Co., Barmin, and are fast against soap and light. Feathers dyed with these dyestuffs which have become soiled, can be washed, therefore, with neutral soap without injury to the color, but must naturally be dressed anew.

RESEDA.

Scour the white, respectively bleached feathers and rinse well. Prepare a bath with five per cent., of the weight of feathers, bisulphate of soda, to which add gradually and carefully the filtered solution of acid violet, fast yellow and fuchsine S., making the additions from the beginning in small quantities only, until the desired tone and shade are obtained; then enter and work the feathers to sample at 173° F.

It is for this dye particularly important that the bisulphate of soda used be crystallized, that is, pure bisulphate free from surplus sulphuric acid, while the commercial article is often nothing but a mixture of Glauber salt (sulphate of soda) into sulphuric acid, answer for this dye.

A good reseda is also easily obtained by adding to the acidulated bath small quantities of decoction of logwood and turmeric, so as to give a feeble bath. Enter the scoured and bleached feathers, after rinsing, at 170° F., work them for about fifteen minutes, until level, and sadden with a little solution of blue stone. Rinse, starch and dry as usual.

ORDINARY GREEN.

For two and one-half pounds of feathers boil two and one-half pounds of fustic for one-half hour with three quarts of water, pour the decoction off and boil the chips again for one-half hour with three quarts of water, mix the two decoctions and strain. Add three ounces alum and one and one-half ounces tartar, enter the feathers well scoured and rinsed, and dye to shade at 170° F. Or, prepare the dyestuffs of decoction of fustic or turmeric, and indigo carmine, according to shade, enter at 170° F., work for one-half hour while slowly raising the temperature to near the boiling point, and dye to sample; lift, rinse, squeeze and starch as usual.

LIGHT GREEN.

I. Scour, respectively bleach, and rinse the feathers. Prepare a hot bath with the solution of forty per cent., of the weight of feathers, tannin, and treat the feathers in it for 1 hour at 170°F. Prepare a bath with a filtered solution of methyl green, according to shade, tone, if a yellowish green is wanted, with the clear solution of picric acid, and dye to sample at 150° F. Lift, squeeze and starch without rinsing.

II. A better color is obtained upon a lightly acidulated bath (with sulphuric acid) with acid green, malachite green, fast green, etc., that is, with the filtered solutions of these dyestuff's, added to the bath in quantities of from ten to twenty per cent. to suit the shade. Enter at 170° F.; dye for twenty or thirty minutes, lift, rinse, squeeze and dry with starch. If a yellowish tone is wanted, add the clear solution of picric acid, or of acid yellow.

MOSS GREEN.

Scour and rinse the feathers well; for dark shades unbleached grays may be used. Prepare a feebly acidulated bath with sulphuric acid, at a temperature near the boiling point; add turmeric freely and Guinea green G less. Enter the feathers and manipulate at the same temperature for fifteen or twenty minutes, according to the desired tone, re-enter and dye to shade.

By varying the proportions of the three dyestuffs, a great variety of green-brownish modes can be produced, which approach medium and dark bronzes the more the fast brown predominates in the composition of the color.

BOG GREEN.

This color is preferably dyed upon unbleached gray feathers. Scour and rinse them, prepare a decoction of green walnut husks or of sumac; lay the feathers down in it for two hours, working them from time to time; then add some decoction of logwood and dye to shade at 170° F., or indigo carmine and dye to shade while slowly raising the temperature to near the boiling point. Rinse, squeeze, starch and dry.

GRASS GREEN.

Scour, respectively bleach, and rinse the feathers. Prepare a boiling bath with turmeric and indigo carmine; chill, enter the feathers at 170° F., dye for one-half hour, raise the temperature slowly to near the boiling point and dye to shade, take up, rinse and pass through a handwarm bath of tartar; lift, squeeze, starch and dry.

RUSSIA GREEN.

I. Scour the feathers as usual and rinse well. Prepare a bath slightly acidulated with sulphuric acid, add two per cent., of the weight of feathers, acid green and one per cent. aniline navy blue dissolved in warm water and filtered acid, according to sample, some filtered decoction of turmeric or solution of fast yellow. Dye at 170° F. to shade, lift rinse and dry with starch.

II. Have the feathers well cleaned and rinsed. Prepare a bath twenty per cent., of the weight of feathers, new green, eight per cent. canarine, sixteen per cent. aniline blue black, sixty per cent. alum, and one-quarter litre sulphuric acid (for two and a half pounds of feathers). Bring the bath to a brisk boil, then chill with cold water, enter the feathers and work them for one hour; finally sadden and tone by adding some decoction of fustic and of logwood. Lift, rinse, squeeze, starch and dry.

III. Prepare a sharp hot bath with a little sulphuric acid; add Guinea green, according to shade, and tone by the addition of indigo carmine and turmeric; for very deep shades add also some nigrosine or fast blue-black, dissolved and filtered. Enter as hot as the feathers can be handled, work for one half hour; then raise the temperature slowly to near the boiling point and dye to shade.

The bath for deeper shades being not exhausted can be preserved for further use, refreshed by suitable additions of dyestuffs as required, but caution must be used as regards the subsequent additions of sulphuric acid, that not so much be added as to injure the feathers.

For Russia green, especially the darker shades of it, naturally gray and even black feathers can be used unbleached.

ROSE.

I. For this delicate color, as well as for the lightest shades of blue and pure yellow, absolutely white feathers must be used; scour them carefully and rinse them perfectly clean from soap or soda, and have the dyestuffs well dissolved and the solutions filtered. Prepare a handwarm bath with a little tartaric acid or acetic acid, to which some solution of eosine, rhodamine, azoeosine, safranine, coccine or ponceau 6 R. B. or ponceau R. R. Be particularly cautious in adding the dyestuff solutions gradually in small quantities, even by drops, to avoid over-dyeing, as by partly stripping of a too dark shade, a fine color can never be obtained, and the nature of the material demands that all unnecessary handling be avoided. After dyeing, rinse lightly, pass through starch and dry. The dyes with the ponceaus are faster than those with eosine or safranine.

II. Prepare a bath at hand heat with carthamine (extract of safflower), well dissolved and filtered, which add very gradually in small quantities, taking up the feathers each time before making a fresh addition, until the desired shade is nearly obtained, then add a little tartaric acid to the bath, re-enter the feathers and dye to shade; or dye first to shade upon the safflower bath, and then pass through a fresh, handwarm feeble bath of tartaric acid, which in this case can be used again for other colors, either as a fixing bath or in the composition of the dye bath.

RED.

Scour and rinse the feathers well; grays must be bleached as near to white as possible, and these ought only to be dyed dark shades of red. Prepare a bath with twenty per cent., of the weight of feathers, bisulphate of soda, and see, as in all cases, that it is well crystallized and dry. (Never use the article when it looks decayed, forms lumps or is moist). Add four to six per cent. azo red, according to the shade wanted, raise the temperature to 170° F., enter the feathers and work to shade; take out, starch and dry.

FAST ALIZARINE RED.

I. Scour and rinse the white, respectively bleached gray, feathers and prepare a bath of boiling water with eight per cent., of the weight of feathers, alum, four per cent. tartaric acid, two or three per cent. oxalic acid, and three per cent. alizarine red; let the bath boil for fifteen minutes, then let the temperature go down just below the boiling point. Lay down the feathers in the bath, which keep at near the boiling point for at least one hour before allowing it to go down to hand-heat; then continue for two or three hours longer, agitating the feathers from time to time; lift, rinse, starch and dry.

By using alizarine acid 2 A. bl. bl., a pure red, similar to Turkey red is obtained. Alizarine 1 W. S. gives scarlets. If the feathers are passed, before rinsing, through a strong soap bath, pretty blue tones are produced.

II. For a fuller red, striking towards bordeau, prepare a well concentrated boiling bath in the same manner with three per cent. bichromate of potash, one and one-half to two per cent. tartaric acid, one per cent. oxalic acid, and eight per cent. alizarine red 2 A. bl. bl. When all is dissolved, let the temperature go down below the boiling point, enter the feathers, and proceed as above.

SCARLET.

I. For this color naturally white feathers are preferably used, but well bleached grays may also be employed; scour and rinse well. Then fill your pan with boiling water, add a few handfuls of bran, let it well boil up, remove the bran from the bath and rub the feathers in the bath as in washing; then pass them three times through clean, cold water. While the feathers are draining, prepare another fresh bath of lukewarm water, to which add a little chloride of tin and, for one pound of feathers, about two pinches of starch and ninety grammes cochineal; then bring the bath to boil and let it gently boil for eight or ten minutes, shut off the steam or remove the pan from the fire, let it stand for a few minutes. Then lay the feathers down in the bath, taking care that they are well kept down in the liquid, work for twenty minutes diligently, then let them lodge in the bath for six to eight hours. As the combination of cochineal and the chloride is readily oxydized and changed to violet by the oxygen of the air, it is advisable to dye in a tinned pan with cover to shut out the air.

Then pass through three lukewarm waters, the last of which contains a little chloride of tin and about a pinch of cream of tartar.

II. Prepare a hot bath with twenty per cent. (of the weight of feathers) bisulphate of soda, well crystallized and dry, and four to six per cent. azo red, according to shade. Enter the feathers at 170° F., dye to sample in fifteen or twenty minutes, lift, starch and dry.

According to the brand of azo red which is used, either scarlet or ponceau is obtained. By mixing the various brands of azo red, a very fine ponceau is produced. If a very blue tone is desired, add to the bath some solution of coccine (azo red blue touch).

PONCEAU.

I. Scour and rinse the white, respectively bleached feathers well. Prepare a nearly boiling bath, acidulated with sulphuric acid, to which simply add ponceau R. R. extra. Enter the feathers, operate at boiling heat one-half hour, then lay down the feathers and let them lodge until level; lift, rinse, starch and dry.

II. Prepare a sharp handwarm bath with one per cent. tartaric acid (of the weight of feathers), or with one per cent. Glauber salt and one-fourth per cent. sulphuric acid, to which add the filtered solutions of ponceau R. B., ponceau 6 R. B., and eosine S. extra B. Enter the feathers and agitate for twenty to thirty minutes, or until the desired shade is obtained; lift, rinse, starch and dry.

III. Lay down the feathers for four hours in a cold bath in which some chloride of tin has been dissolved; then dye for one-half hour in a hand warm bath of cochineal, lift and dry.

IV. Prepare a bath with one and one-half per cent., of the weight of feathers, saccharic acid, one-quarter per cent. tin salt, and six to seven per cent. cochineal, bring the bath to boil for one minute; then chill. Enter the scoured and rinsed feathers at hand heat, dye for three-quarters of an hour, take up and expose them for two hours to the air, rinse, starch and dry.

BORDEAUX.

Scour and rinse the feathers well. Prepare a boiling hot bath slightly acidulated with sulphuric acid, to which add a liberal quantity of ponceau 6 R. B., a few drops solution of aniline blue, and some yellow dyestuff, such as turmeric, fast yellow, or quinoline yellow, and bring the bath to boil for a few minutes. Then chill to sharp hand-heat, enter the feathers and work until level, and sample; if still too light, add some more of all these dyestuffs. As the bath shows from the beginning the color it will produce, it can be corrected before entering the feathers.

GARNET (RED).

I. Scour and rinse the feathers well clean, grays ought to be bleached. Prepare the dyebath as for ponceau (I.), or use an old ponceau bath, and add to it some aniline cerise (cherry red) and very little extract of indigo, or solution of fast blue-black. Enter feathers as hot as possible to handle, work for fifteen to twenty minutes while raising the temperature to boiling heat; then stop heating, lay down the feathers, and let them lodge until level; lift, rinse and dry.

II. Prepare a boiling hot, not boiling, bath of anatto, according to shade, enter the feathers, work them well through, then lay them down in the bath for twelve hours. Take them up, rinse, pass through a moderately strong alum bath, rinse again, and dye at 170°F., with either decoction of red wood (brazil, camwood, etc.,) or fuchsine; lift and dry.

GARNET (BROWN).

For very deep shades naturally gray feathers may be used unbleached with proper consideration of the tone of the bottom color. Have the feathers well cleaned and rinsed, and add to a bath of two gallons of water, one and one-quarter pounds cudbear and five ounces turmeric. Bring the bath to boil, boil for five or ten minutes, let cool down to 100° F., enter the feathers and dye to shade. Lift, starch and dry.

Or, utilize a used ruby bath (following) and add to it five ounces turmeric.

RUBY.

For a good color the feathers must be white, naturally or bleached; scour and rinse them well. Add to two gallons of water one and one-half pounds good cudbear, stir well, enter the feathers and work them, while slowly heating, as long as the hands can stand it. Then lay them down until colored to shade, lift, rinse well, starch and dry.

SALMON.

I. Salmon or "flesh" may be dyed upon bleached naturally gray feathers, in which case the creamy tint of the feathers must be taken into consideration and can be utilized for certain broken tones of the color. Have the feathers well washed in soap or soda, and rinsed perfectly clean. For dyeing prepare a bath as for rose, preferably with ponceau B. R., or utilize an old bath for rose, according to its strength and the shade to be produced, and add in either case a suitable, small quantity of filtered decoction of turmeric. Proceed as stated for dyeing rose, with the difference only, that the acid may be added to the dyebath at once, if the bath is made fresh. Particularly fine shades are obtained with rhodamine and turmeric, in a bath slightly acidulated with acetic acid, upon bleached grays.

II. Prepare a bath as for rose, with some solution of eosine, a little quinoline yellow, according to tone, and a little acetic acid, just enough to give the bath a slightly sour taste. Enter the well cleaned, or bleached feathers after rinsing, at hand heat and agitate them until the bath is well exhausted, or a level color, according to sample, obtained, rinse lightly, starch and dry.

AMARANTH.

After scouring and rinsing, prepare a bath with one and one-half ounces alum per gallon of water, at 75-80° F., and lay the feathers down in it over night. On the next morning rinse them in cold water; then dye them at hand-heat to nearly boiling heat in a strained decoction of Brazil wood (or camwood, hypernic, etc.) until the required shade is obtained, and rinse in warm water to which some tartar has been added; starch and dry.

BRONZE.

For this color naturally gray feathers may be used if a deep shade is to be dyed; for light shades they ought to be bleached. Scour and rinse the feathers well; then prepare a bath with five per cent., of the weight of feathers, bisulphate of soda, to which add azo orange, acid violet and extract of archil. Dissolve the dyestuffs, each separately in water, filter, add the clear solutions gradually in small quantities until the shade is nearly reached, then, in order to correct, by drops, until the exact depth and tone are obtained. Enter the feathers and dye to shade at 170° F. Instead of acid violet indigo carmine may be used; in this case, however, as the dyestuff runs up slowly and difficultly, work at 170° F., for twenty to thirty minutes, then raise the temperature slowly to near the boiling point and continue at that temperature, without actual boiling, until the required color is obtained. Then rinse, squeeze, starch and dry.

Bronze is also produced like drab, that is, with azo orange, acid violet and fuchsine S., but with greater quantities of dyestuff. Bronze is also obtained with the recipe for any dark brown, by making the yellow in it predominant; particularly good bronzes are in this manner obtained from dark chestnut brown.

OLIVE.

I. Clean the feathers by laying them down for six hours, or over night, in a weak warm solution of soda crystals (1° B.) to which add so much ammonia as to give it a faint odor; take up when completely ungreased and rinse well in lukewarm and cold waters. Prepare the dyebath with five per cent., of the weight of feathers, bisulphate of soda, to which add filtered solutions of indigo carmine, archil and fast yellow as required for the sample. As the indigo carmine is slow to dye up and requires boiling heat or a temperature near it, dye first the feathers blue with indigo carmine, then let the bath cool down to 170° F., and add the solutions of archil and fast yellow in small successive quantities, so as to be able to give the accurate tone.

Instead of extract of archil, fast red or bordeaux may be employed.

To avoid any possible injury to the feathers by the high temperature necessary for indigo carmine to run up, in its stead a solution of alkaline blue or of acid Victoria blue. Take up and dry the feathers without rinsing.

II. Scour well and rinse the feathers, and prepare a bath with three per cent. alum (of the weight of feathers), to which add azo orange and some indigo carmine; enter at 170° F., dye for fifteen or twenty minutes, then raise the temperature slowly to near the boiling point and dye to shade. Lift and dry.

By beginning with small quantities of the dyestuffs and successively increasing them and varying their proportions, a series of fine shades from light old gold to the deepest olive, near black can be produced.

III. Prepare a boiling bath, in which dissolve one per cent., of the weight of feathers, alum, one per cent. Glauber salt, and add a little sulphuric acid; let cool down to 170° F., add some fast yellow, a little solution of archil and of sulphate of indigo, work for fifteen minutes while raising the temperature to the boiling point, and sadden with blue black, lift, rinse and dry with starch.

IV. Prepare a sharp handwarm bath with a little sulphuric acid, to which add the clear solution of a little quinoline yellow or turmeric, and acid green; enter the feathers and work for fifteen minutes, or until they have taken a sufficiently nourished yellow-green color; then take them up, add to the bath some solution of fast brown, as required by the sample and dye at 170° F. to shade; rinse, squeeze, starch and dry. The brown dyestuff must be added very carefully in small doses, best by drops, in order to obtain with certainty any of the great varieties of shades, from olive green to olive brown, as required.

VIOLET.

Naturally gray feathers may be used unbleached, but only for very deep shades as the bottom color acts dulling upon the dye, and brilliant colors can only be obtained upon a pure white bottom. Scour, respectively bleach well, and rinse clean. Prepare a hot bath to which add some filtered solution of methyl violet, according to tone, that is, more or less blue, enter the feathers and work until cool, then add gradually more dyestuff solution according to shade while raising the temperature to near the boiling point and continue at this temperature until the desired shade is nearly obtained.

If too blue, tone with a little solution of fuchsine S. Towards the end of the operation take up the feathers, add some alum to the bath, and when it is dissolved, shut off the steam, re-enter the feathers and work to shade for about ten minutes. Then lift, rinse and dry.

HELIOTROPE AND LILAC.

I. These colors being simply medium and light shades of violet, proceed as for the latter color, selecting for heliotrope the bluish brands of methyl violet, and for lilac the red touch mark. The dyebath is acidified with a little tartaric acid, so as to give it a feeble sourish taste and dyeing done at hand-heat until a level color is obtained with very little solution of the dyestuff, and more of it gradually added, while the temperature is raised to nearly boiling, as required for the shade to be produced. Or,

II. Prepare the dyebath simply of cold water acidulated with a little sulphuric acid, add a few drops of the filtered solution of methyl violet (4 B. for heliotrope), and dye to shade without heating. In both cases rinse after dyeing, pass through a bath of raw starch and dye as usual.

CREAM.

I. The lightest shade of this delicate color can be produced upon naturally gray ostrich feathers by simply bleaching them; this color, however, is extremely sensitive, probably because the action of peroxyd of hydrogen continues under the influence of the oxygen of the air. Bleached grays require, therefore, dying as well as naturally white feathers. The feathers being well scoured and rinsed, prepare in a white basin (preferable to the copper pans, because the coloring of the dyebath is easier and more correctly discerned over the white bottom) a bath of pretty hot water, to which add a pinch of tartaric acid, and a little decoction of turmeric or solution of fast aniline yellow or of azo yellow, but only enough to give the water a light tint; work the feathers in it for four to six minutes. Then sample and correct, if necessary, by adding more dyestuff solution. The shade being obtained, pass through cold water, starch and dry as usual.

II. Prepare in a white basin a handwarm bath with three or four drops of sulphuric acid and a few drops of the filtered solutions of picric acid, fast

aniline yellow, quinoline yellow, or mandarin yellow extra, but preferably turmeric which dyes up more evenly than the other dyestuffs. Enter the feathers and agitate them for fifteen or twenty minutes; then lay them down in the bath for one-half hour longer to insure a level dye; lift, draw through lukewarm water, starch and dry.

WHITE AND BLACK.

Science teaches that white is the source of light or the product of combination of all other colors, because the light of the sun, which is assumed to be white, when broken up by means of a prism, shows in its image reflected upon a white plain, the three primary, and three secondary colors with the uncounted number of intermediary products of combinations of fractions of the primary colors forming the transition from one to the other, which can be perceived by the eye but not exactly separated from one another, but may quantitatively determined to an approximate degree of accuracy. Black, on the other hand, is described as the absence of all light, and it is denied, therefore, by theory a place among the colors.

Practice asserts the direct contrary of the theory developed by science by way of conclusion. While philosophers assert that they have succeeded in producing white light by the combination of lights of the various colors, for which combination, however, they give no formula, no dyer with the greatest patience and with the most subtle proportioning of dyestuffs, giving pure reproductions of the primary colors as seen in the spectrum or image of the broken sunbeam, can ever be able to produce anything of a color approaching white. But every dyer knows how to produce white by bleaching, that is, by the destruction of all color. And this operation is comparatively simple and easy to perform, since the great achievements of modern chemistry have placed into the hand of the dyer the most energetic and effectious color destroying agents. The ostrich feather dyer of to-day is able to convert naturally gray and even black feathers into nearly pure white, which undertaking his father would have called the boast of a deranged mind and an absolute physical impossibility. And with the aid of a complementary color dyed upon the bleached feathers the tint remaining upon them is obliterated, or neutralized, which operation is generally called "white dyeing," although certainly white cannot be "dyed" with a blue or

violet dyestuff as little as blue or violet can be produced with a yellow dyestuff.

Black, on the other hand, although the name and rank of a color is denied it by the doctrines of theory is, for the dyer, most essentially a color requiring for its production the contribution of all colors, as can be shown by a simple experiment. If, for instance, within a circle, three equal circles, whose diameters are greater than half the diameter of the surrounding circle, are printed, one blue, one red and one yellow, so that the points of contact with the periphery of the outer circle are equidistant from one another, or form a regular triangle, their segments overlapping one another form four spherical triangles, one violet where red and blue cover one another, one orange where red and yellow come together, one green where yellow and blue are mixed, but the fourth is the centre, where parts of the three differently colored segments cover one another, is black, but toned by the color of the greatest intensity. In fact, black requires for its production more color, and is more difficult to dye than any color of the spectrum. It is not strange, therefore, that many more methods have been proposed and have been tried and adopted to dye black than for the production of any other color. Yet all these blacks are more or less tinted and are nothing more than the deepest shade, which can be produced with the aid of metallic salts, of blue, brown, gray, violet, green and even red. The only black upon ostrich feathers which may be justly called an absolute black is that produced after the method of Mr. Paul, as described in the front part of this book (page 53-56). It presents, besides, the advantage, that it can be completely done in eighteen minutes, while there are older methods which require not less than three days. Such, of course, will find no room here, but only the most reliable and expeditious, and safest of the older methods will be selected with due regard both to beauty of color and preservation of the precious material.

WHITE.

Naturally white ostrich feathers and bleached grays, like all material taken from the animal realm, retains even after scouring and bleaching a more or less noticeable yellowish tint, which becomes visible after some time even upon such as appear snow-white immediately after scouring. The cause is, probably, that the fat which is contained in feathers, as well as in wool and

hair, and a part of which remains after the cleaning process, is oxydized by the action of the air. To perfectly and completely extract this remaining small amount of fat which does not hinder the subsequent dyeing operations, is not advisable; for, it has been observed that in that case, the feathers become brittle, and for this reason, scouring with soap is preferable to scouring with soda. It has been observed that colored matter taken from the animal body in the course of time loses its brilliancy of color and becomes dull; white feathers turn yellowish, even if perfectly protected against dust. To prevent this alteration, the bleached feathers are "dyed white," or rather blued or tinted; that is, the brilliancy of the white is heightened and the faint yellowish tinge neutralized by the application of a very light, scarcely perceptible tint of a complementary color.

For this purpose, for instance, indigo carmine (greenish white), indigo carmine with a very small addition of ammoniacal cochineal (reddish white), induline or extract of indigo (bluish white), or methyl violet 6 B. (direct white), are employed, and a few drops of acid added to the bath, either sulphuric, acetic or oxalic acid. The acid, however, can be dispensed with, as it scarcely has anything to act upon, and as only a diminutive amount of it is employed, the blued feathers are not rinsed but immediately dried from the blue-bath. The additions of dyestuff to baths must be made so small that they do not affect a coloring of the feathers but only a faint tinting. Rather too little dyestuff may be added, which defect can at all times be corrected by adding a few drops more of the coloring solution, than too much. In the latter case it would become necessary to bleach the feathers again and go over the whole process of preparing the raw material for dyeing.

BLACK.

I. CHROME BLACK.—Black being the most difficult color to produce, as above remarked, the feathers require a specially careful preparatory treatment in order to remove everything that might interfere with the purity, uniformity and brilliancy of the color, or cause less dyed, dull spots and streaks. Naturally gray feathers, however, need not to be bleached or decolorized but only careful treatment and attention. The feathers are for twenty-four hours laid down in a solution of twice their weight of calcined

soda, ammonia soda being preferable for this purpose to Lablanc soda (old process soda), then taken up and carefully rinsed clean from the alkaline in warm water, or better, in two warm waters. In the case of particularly valuable feathers it is recommendable, before laying them down in the soda solution, to rub the stains of the feathers off with a piece of carbonate of ammonia or with a large soda crystal. After rinsing, the feathers are entered for one hour, at 170° F., in a bath containing forty per cent., of the weight of feathers, chromate of potash, forty per cent. copperas, and twenty per cent. tartar, and several times turned and agitated during the specified period while the entering temperature is maintained. Then the feathers are taken up, and the adhering liquid squeezed out by hand or by rolling them through a clothes wringer with rubber roller. In the meantime a logwood bath of medium concentration is prepared either with a fresh decoction or with extract of logwood and twenty per cent. Marsailles soap dissolved in it. The feathers are entered in this bath at hand-heat, diligently agitated for twenty or thirty minutes and, if necessary, while the temperature is raised to 200° F., laid down in the bath until the correct shade and a level dye are obtained. The feathers are then lifted, squeezed, very thoroughly rinsed in cold water, passed through starch and dried with frequent shaking, respectively beating upon the board or between the hands.

It occurs sometimes, that the stems of the feathers are imperfectly died and present light brown or gray places. This is attributable to insufficient scouring. In this case the defective portions of the stems must be scraped with a sharp penknife and dyed over. This operation, however, is difficult and requires much practice and a light hand, as too much scraping removes the horny glossy surface of the stem and, when dyed over, the only change effected is, that a dull black mark takes the place of the discolored or brown spot. Often, however, the defect can be remedied by touching the imperfect portions up with a feeble alcoholic solution of shellac, in which some nigrosine is dissolved. With properly scoured feathers this mishap does not occur.

Another trouble, however, which is not unfrequent with blacks, is that the feathers are over-dyed and become brownish black instead of black. But in this case the remedy is as simple as its occurrence is frequent; a quick passage through sulphuric acid diluted with water to 2° B. strips off the excessive dye and produces a good color. Besides, this operation gives the

feathers a brilliant lustre. Many dyers, therefore, methodically avail themselves of this effect of sulphuric acid and deliberately over dye their feathers (See IV below) and then apply the sulphuric acid passage for the purpose of imparting the feathers that peculiar lustre. A passage through a solution of sodium chloride, of 2° B. strength, has the same effect as a passage in sulphuric acid 2° B.

For this purpose lay the feathers down in the warm sodium chloride solution, until the black cotton strings with which the feathers have been tied together, as in the beginning described, begin to turn gray. Then take up the feathers, rinse them very thoroughly in cold water, drain, starch and dry.

Sodium chloride can be prepared in a simple way as follows: rub one-half pound fresh chloride of lime in a porcelain mortar with a little water into a smooth milk, which pour into a bucket, dilute with cold water, and add, under stirring, the solution of one pound Glauber salt; let settle and use the clear liquid. Instead of Glauber salt (sodium sulphate), soda crystals (sodium carbonate) may be used; the latter, however, is a little higher in price and renders the solution strongly alkaline.

II. IRON BLACK.—Lay down the feathers over night in a warm bath, in which one hundred per cent., of the weight of feathers, soda crystals have been dissolved. On the following day take them up, squeeze them out and lay them down for two hours in a proportionally strong solution of carbonate of ammonia, take them up and rinse well in warm water. Lay down for six hours upon a bath of nitrate of iron 10° B.; take up, rinse, and dye at 170° F. with the decoction of ten per cent. logwood in which five per cent. Marseilles soap has been dissolved. If a dead black is wanted, add some decoction of quercitron or turmeric to the bath. The desired depth being obtained, lift, rinse, starch and dry. In case the color is over-dyed, strip with sodium chloride (or sulphuric acid) 2° B., as above described, drain, squeeze and dry.

III. LOGWOOD BLACK.—Scour and rinse the feathers well. Prepare a bath with three per cent., of the weight of feathers, carbonate of lime, six per cent. blue stone, and five per cent. tartar. Enter the feathers at 170° F., maintain this temperature for one hour; then let it go down, but leave the feathers in the bath for six hours longer, agitating them frequently during that time. Take them up, drain and squeeze, or centrifugate, and enter a

handwarm bath containing some decoction of logwood, to which some decoction of fustic is added. Work for fifteen or twenty minutes, then raise the temperature to nearly boiling heat. Continue adding decoction of logwood, until a nourished black is obtained. This dye being hard to correct by the ordinary means, the additions of logwood decoction must be made with caution towards the end of the operation, in order to prevent over-dyeing. If a brownish touch is desired, add some more decoction of fustic when the black is nearly done. Then lift, rinse, starch and dry as usual under continual agitation, beating and shaking of the feathers. This chrome black is superior to iron black, because it imparts, to the ostrich feathers, more lustre.

IV. Whatever kind of feathers are to be dyed, white, grays or old blacks, wash them perfectly clean in two or three warm soap baths and remove the soap by rinsing in two or three warm and several cold waters. Colored feathers which are to be redyed blacks must be stripped of their color as much as possible by washing in hot soap to which some ammonia is added, whereupon this must be rinsed in several waters absolutely clear from soap and alkali; it is an erroneous notion to neutralize the last trace of alkali which may remain, by a passage through a feeble acid bath. The feathers thus prepared for dyeing, make a bath of two parts nitrate of iron to one part hot water at 170° F., enter the feathers, work them through a few times, and then lay them down in the bath for twelve hours (over night). Then lift and rinse the feathers in several (three or four) cold waters. Prepare a pretty strong decoction of logwood and fustic, for which take two parts of the former to one part of the latter; let the temperature go down to about 208-210° F., when enter the feathers and maintain that temperature for fifteen or twenty minutes. Then shut off the steam or remove the dye-vessel from the fire, as the case may be, and let the feathers cool in the bath. When cold take them out, prepare a fresh bath of logwood and fustic like the first, enter the feathers at 208-210° F., after fifteen or twenty minutes add about a teaspoonful of copperas for one gallon of water, and leave the feathers in the bath for six or eight hours longer; then lift and rinse in several cold waters. The feathers are at this stage black with a strong brown touch which is removed by a cleaning bath of Eau de Javelle (sodium chloride). The latter is prepared by rubbing one-quarter pound chloride of lime to a smooth milk with a little cold water (in a porcelain or a marble mortar) and adding this milk to the solution of one-half pound Glauber salt in three parts water.

After good stirring the mixture is then allowed to settle, when the clear solution is poured off and put up for use in well stoppered bottles. Of this liquid so much is added to a basin or pan full of warm water that it gives a slippery feel between the fingers, similar to that of a solution of soda.

In this bath the feathers are agitated for six or eight minutes, or until the liquid has assumed a yellowish color. Then the feathers are taken out, rinse in two or three warm waters, passed through raw starch, pressed out between several laps of a clean piece of muslin, and dried either by rubbing them in pulverized and sifted potato starch or by shaking them before an open fire or gas-flame.

The nitrate of iron bath can be preserved and used for the same purpose for eight or ten days, but the first logwood bath becomes useless and is let out. As above observed (I) sulphuric acid can also be employed for correcting the over-dyed feathers and reducing that brownish color to a pure lustrous black, but a much shorter passage is given: the feathers are entered by single strings, well opened, agitated in the sulphuric acid bath for a few seconds, and immediately rinsed. Where week work is done, it is advisable to have two men employed at this operation, one of whom passes the feathers in the acid bath and hands them over to the other man for rinsing.

CONTRASTS AND SHADINGS, OR OMBREES.

Fashion and fancy sometimes requires the dyer of ostrich feathers to dye upon one feather two, or even three contrasting colors, or different shades of the same color, that is, the tips of the feathers in another color or shade than that of the lower part of the feather. Generally in these combinations the tip is dyed the lighter color or shade, and the lower part considerably deeper or in a heavier color. Very popular combinations are: the tip light blue and the bottom brown, the tip rose and the lower part bordeaux, the tip light orange or dark yellow and the lower part garnet brown, tip rose with olive bottom part, or even three colors, such as the tip rose, the part below it medium olive green, and the lowest part deep violet. That such combinations are very handsome cannot be asserted; but fashion dictates, and fancy sometimes prefers oddity to beauty. More rational are at any rate the ombrées, or combinations of two or three shades of the same color upon one feather. The operation is the same for both styles; but contrasts are

generally dyed only upon single feathers, while ombrées, being in greater demand, are dyed by strings or even in greater lots. The feathers being scoured and rinsed as usual, are first dyed wholly in the lightest color or shade to be produced, according to recipe, say light blue for the tip, and dried. Then wrap the top, as far as it is to be light blue, in paper (some dyers use for this purpose oiled or waxed paper) and tie the paper firmly, but not so hard as to injure the feather, with a string, not so loosely as to allow the paper envelope to slip out of place during the manipulation. Then, holding the feathers by the top, dip them into the boiling hot bath for the other color, or shade, to be dyed, but only so deep that the paper just touches the surface of the dye-liquid. This method is the safest for learners or new beginners. For more experienced workers it is unnecessary to use the paper wrapping; they simply first dye the light bottom shade, dry or not, according to the characters of the two colors (for shadings, half-dry feathers, that is, drained and squeezed out, are rather preferable), and loosely hold them in the bath for the second color, or deeper shade. They have it thereby in their power to effect a more gradual transition from one color or shade to the other. As the color becomes deeper, the longer the feather is immersed in the bath, it is plain that the dyer can easily produce upon one feather a complete graduated scale of shades. Each time, after a shade has been dyed to the required depth, the feathers are rinsed in cold water and some more dyestuff solution added to the bath. These additions require good judgment, because too much dyestuff added would cause an abrupt, dull contrast instead of a desirable gradual shaking off, or transition from one shade to the other. There ought only a little more dyestuff be added each time, than has been absorbed from the bath by dyeing the preceding shade. If paper wrappings are used, they must naturally be untied for rinsing and replaced by longer pieces before entering the bath for the following darker shade. After rinsing the feathers must always be well squeezed out. If two colors are to be dyed, for instance light blue tip with brown lower part, dye first the whole feather light blue, rinse, dry, tie up the tip in paper, and dye the lower half brown.

It needs not to be mentioned that for dyeing two or three contrasting colors upon one feather only such dyes must be chosen as can serve for bottoming and topping one another without materially altering the character of the topping color.

EDGINGS OR BORDERS.

For this style of feather dyeing, use feathers of good quality, with wide and well developed vanes. They are dyed in two colors and shades only, presenting one color, mostly of a light shade, or a white "black" on both sides along the stems, while the outer edges for the vanes, or ends of the fibres, are dyed in a different color or darker shade. They make a particularly handsome effect when curled over the stem, setting off the edges in a fine contrast against the black showing through the curls.

To produce edgings an oval pan, as described in the beginning, or other dye-vessel of greater length than the feathers, and three or four inches deep must be used. The well scoured, respectively bleached, and rinsed feathers are first dyed the color for the middle part, as usual on strings. After rinsing and drying they are taken off from the strings and "edged" singly. For this purpose prepare the dye-bath for the edging color, heat to the proper temperature, take the tip and quill respectively between the fore-finger and the thumb of both hands, dip the feathers edgewise, that is, with the ends of the fibres on one side of the stem, or the edge of the vane only, into the dye-bath as deep as the edging is to be wide, and move the feather in this position horizontally forward and backward in the bath until the shade is obtained. Then place the feather between several laps of clean dry muslin, squeeze it out by passing the hand over it, and dye the other edge in the same manner as the first. Finally rinse, starch and dry the feather as usual.

In this connection a chemical reaction is worthy of mention, which was discovered about two years ago by an accident, and may be advantageously employed for the production of edgings upon ostrich feathers, if further developed by experiments. In a large feather-dyeing establishment, in Berlin, a sheet of paper which had been wetted with ammonia, and had become dry, had been left on a work-table, when one of the employees, who was handling a lot of feathers freshly dyed with methyl violet, inadvertently put one of the feathers, which was still moist, upon the impregnated paper. After a while, when the feather was picked up, it was found that the violet, all round the edge of the feather, had turned brilliant green, producing a very pleasing effect. It is rational to suppose that with mixed colors, in whose composition methyl violet largely enters, similar effects can be

produced by the action of ammonia; and probably the same is the case with other aniline dyestuffs.

GILDING AND SILVERING.

Gilded and silvered ostrich feathers are but seldom in demand, and then only for grand evening dresses or stage effects, and for short seasons, which generally return far between. Their production is by no means a dyeing process, but rather an operation of surface ornamentation, still the dyer is sometimes requested to perform it. While goose feathers and other feathers of small value are wholly gilded or silvered, the gilding of ostrich feathers consists chiefly in a sprinkling with metallic dots, or sometimes in an edging, or is only applied to the tip of the feathers, which, from the nature of the operation, are treated singly. Such ornamented feathers, white as well as dyed, being only used for short periods, a permanent fixation of the gold or silver upon them is not required, but rather undesirable, as they will soon be redyed for other uses.

For gilding, respectively silvering, a sufficiently adhesive solution of possibly colorless gum arabic is prepared and distributed by hand, and by means of a fine hair-brush, in smaller or larger dots, as required, over the upper side of the feather or along the edges, and before the gum solution becomes dry, sprinkled over with finely divided gold-leaf or silver-leaf. The feather is then turned over, given a few light taps with the hand to remove the loose dust of metal, and vigorously shaken, partly to prevent the fibres from sticking together, partly to remove the remaining loosely adhering particles of metal. The operation must be performed as rapidly as possible to prevent the gum solution from drying before the metallic dust is shaken off. The smaller the gum-dots are made, the quicker must the work be done, but the less is the danger of the fibres being pasted together, and the more elegant the appearance of the feather. The dots or spangles are made of different shapes, in little circles or squares, and sometimes arranged so as to form angular designs, according to taste and skill of the operator.

Another very pretty, scarcely more permanent, but more frequently applied ornamentation of ostrich feathers, is the following.

FROSTING.

For this purpose the feathers are first dyed in a light or medium shade of any color, the effect of frosting feathers of a dark color being rather unfavorable. The feathers are then, after drying, covered on the upper side with a solution of clear gum arabic, as for gilding, but more closely, or may even be entirely brushed over with the gum solution, and are then, before the gum dries, sprinkled over with finely ground white glass, or mica, the latter giving the appearance of frosted silver. The glass powder or mica powder is then quickly and vigorously shaken off, to open the fibres and flues as much as possible, while drying. Finally, to complete the opening of the fibres, the feathers are steamed at the under side, and shaken in the air until open and dry.

Great care is required in curling gilded or frosted feathers, that the metal or glass powder is not rubbed off in passing the fibres of the vane over the curling-knife. This operation being extremely difficult and dangerous, the use of a curling-iron, like that used by hair-dressers, is preferable to that of the knife. The iron is moderately heated, so as not to singe the feathers; then, beginning at the lower end of the feather, a part of the fibres on one side of the stem are taken by their ends between the shanks of the iron, the latter closed and the fibres wound downwards around it, the iron being carried on the under side of the feather towards the stem. Then first one side of the vane is successively curled from the quill up to the tip, when the same operation is repeated upon the other half of the feather. If, in this manner, the feather should be curled too strongly, the fibres are taken between the shanks of the warm curling-iron at the stem and simply drawn through the iron.

Numerous ostrich feather dyers and dressers use the curling-iron altogether, instead of the knife; the only difficulty for the beginner is to get the proper heat, which, however, is soon learned.

Very pretty effects are also obtained by dyeing the feathers a light shade of color, drying, gumming and sprinkling them with either powdered black glass or jet.

RENOVATING FEATHERS.

White ostrich feathers which, by long exposure to the show-window, or by lying in store for a protracted time, have lost their whiteness and turned

yellow, and dyed feathers which, from the same causes, have become dirty, pale and discolored, can be restored to their former beauty by washing, respectively redyeing, as follows:

I. A washing process, which is ordinarily only applied to white feathers which have become yellow, is as follows: Prepare a bath of two gallons of water at 145°F., to which add half a gallon of liquid ammonia (spirits of sal ammoniac, ammonia water); enter the feathers, work them once well through with the hands, and lay them down in the bath over night. On the following day take them up, wash them once through a soap-bath at 145°F., pass them again through the first ammoniacal bath, and rinse well and let them drain. Then prepare a bath of cold water, to which add so much of a clear solution of methyl violet 6 B., that a white china plate held about a foot below the surface of the water, appears with a faint bluish tint, or such a blue tone as is desired; and add to the bath so much sulphurous acid, that it gives the liquid a well defined odor. When the sulphurous acid mixes with the tinted liquid, the violet color of the latter disappears and changes to a greenish tint, which, however, turns again to blue upon the feathers when they are afterwards exposed to the action of the air. The feathers are then passed, singly, if possible, through the blue-bath, well drained, centrifugated or whizzed, starched and dried as usual.

Colored feathers which have lost their freshness, and are to be redyed, are simply washed clean with soap and rinsed, or they are stripped of their color, as much as possible, with soap and oxalic acid, or bleached with peroxyd of hydrogen, as described in the beginning; whereupon they are dyed and treated like bleached new feathers, always taking into consideration, however, what of the old color may remain upon the feathers, may be utilizable as a bottom for the new color, or even as a component of it, for instance, in the case of many modes and several browns.

II. Another method of renovating ostrich feathers presents the advantages that it is executed without the application of heat, that it is a simple cleaning process which attacks no color, and that it leaves the curling of the feathers intact, which is unavoidably taken out of them by washing with warm water and soap, or any other alkaline detergent substance. It is, therefore, only applied to feathers which have lost their purity of color by exposure, and whose curling is to be preserved, or is worthy of preserving. It is, in part, the same process which is known as "dry washing" among scourers and

dyers of garments, and can be applied to feathers of any color and shade, white and even black, without exception.

For this operation fill a basin or small wooden hand-tub with benzene, add a handful or two of potato flour (sifted potato starch), enter the feathers and rub them well through with the starch until clean; then squeeze them out by hand and press between muslin, finally whiz or shake them in the air until dry.

This process is partly chemical, in so far as the benzene loosens the dust and other impurities which have settled upon the feathers, partly mechanical, as the numerous fine particles of the potato starch, which do not dissolve in benzene as soap does in water, rub these impurities off from the feather. By the combined action of the benzene and starch, and the friction applied, the feathers are not only cleaned, but the flues completely opened, so that the feather thus treated looks perfectly like new.

A remarkable feature of this process is that the starch carries nearly all the impurities down with itself to the bottom of the wash-basin, and becomes soiled, while the benzene takes up every little of them, and can, therefore, after settling, be poured off from the starch sediment, and can be used several times before it needs to be purified or eventually becomes unfit for use.

In using benzene, which is a highly combustible substance, the utmost precaution must be observed that no open flame or fire be in the work-room, neither open lamps nor a fire in the stove burning. Even doors leading to adjoining rooms, where lights or fires are burning, ought to be kept closed while working with benzene, because the benzene vapors, which may be carried to the flame by a draft of air, would inevitably ignite and cause an explosion and fire. Occurrences of this kind have been not unfrequently observed.

Feathers which have been cleaned by this process, as well as new feathers, may be dyed by the following process.

DYEING IN THE COLD WAY.

This process is a real dyeing process, as well as a renovating process, both, however, to a limited extent, inasmuch as it can be applied only to white

feathers or to such as are dyed with light and medium shades of certain colors which are to be freshened up; but it does not answer for dark colors. It is, however, extremely simple and easy to execute; besides, almost instantaneous, and therefore of great utility where rapid work is required, because it leaves the feathers perfectly in shape, like the benzene washing process, and does not affect the curling of the feather, if there is such. Old feathers which were already dyed cream, rose, salmon, light blue, light gray, light green, sea green, golden yellow, heliotrope or beige, can be redyed in the same colors, but must previously be washed with benzene; new white feathers do not require such washing.

For this method of dyeing, aniline dyestuffs soluble in alcohol are used, viz.: for

Cream,	Curcumine or Aniline Orange,
Rose,	Eosine or Ponceau,
Salmon,	Curcumine and Eosine,
Light Blue,	Water Blue and Methylene Blue,
Gray,	Nigrosine,
Sea-green,	Malachite Green,
Golden	
Yellow,	Orange and Fast Brown,
Heliotrope,	Methyl violet 6 B.,
Beige,	Methylene Blue, Curcumine and Fast
	Brown,
mixed according to tone and shade.	

Operate as follows: Fill a white basin with a sufficient quantity of alcohol to completely wet the feathers in it; add, according to shade, a smaller or greater quantity of the clear alcoholic solution of the required dyestuff, or mixture of dyestuffs, pass the feathers singly, without previously wetting them, three or four times through the alcohol bath; then press them out between clean muslin, put a few handfuls of sifted potato starch upon a clean sheet of paper, and rub the feathers with it until thoroughly dry; finally, shake out the starch.

RECAPITULATION OF GENERAL RULES.

At all times have the feathers, which are to be dyed, scoured well, that is, washed clean from all externally adhering impurities, fat, etc.; naturally colored feathers bleached for all light and medium shades to be dyed upon them, and rinsed perfectly clean from the scouring or bleaching bath, first in two or three warm waters and then in cold water.

On taking the feathers from any bath, always squeeze the liquid out first by drawing the feathers through the hand closed upon them, then by placing them straight between several laps of clean dry muslin and repeatedly passing the hand with quite a smart pressure over it. Never transfer the feathers, in any case, from one bath to another in a wet, but in a moist condition, or nearly dry.

Never allow the feathers to become dry in the course of operations. If it is necessary to interrupt work, or to put feathers to one side for further treatment, dry them properly by first passing them through a bath of raw starch, in order to have the flues at all times as well opened as possible.

In no case let the temperature of a bath, in which feathers are treated, rise to actual boiling, although for some dyestuffs a temperature near the boiling point is required to make them dye up, to become level or to fix them.

In every instance, where an acid or acid salt is employed, either in a separate mordanting or fixing bath, or as a component of the dyebath, rinse well before drying.

When sulphuric acid is used in the composition of a bath, add only so much of it as to give the water a very slight, scarcely perceptible acid taste.

Although some artificial dyestuffs dye up without an addition of acid to the dye bath (basic dyestuffs), the addition of sulphuric acid, in a very small quantity, to the dye-bath is advantageous, rendering the colors brighter and also faster.

When bisulphate of soda is employed, it is not necessary to also add sulphuric acid to the dye-bath; if it is added, however, it must only be in a very small quantity; careful rinsing in several warm and cold waters after dyeing is required.

When alum alone is used without any other addition as mordant, sulphuric acid may be added, but only in the proportion of one tenth or, at the most, up to one fifth of the weight of alum, and careful rinsing in several warm

and cold waters is the more indispensably required the more acid has been employed.

All solutions of dyestuffs, as well as of chemicals, ought to be carefully filtered, and decoctions of woods, etc., strained before adding them to the bath; never add dyestuffs, drugs or chemicals in substance to any bath, in order to prevent solid particles from settling upon the feathers.

Never add all the dyestuff probably required or prescribed by a recipe to the dye-bath at one time, but in several small quantities, each time after taking up the feathers, stir the bath after making the addition, re-enter the feathers and watch the progress of the dyeing carefully; when approaching the desired shade, add the dyestuff very cautiously, by drops if necessary, particularly with mixed colors, such as modes.

Sample in proper time, and take not a whole feather for it, but pull off two or three fibres from the lower part of a feather, dry them quickly by squeezing between dry muslin, match, correct the bath and finish dyeing.

While drying keep the feathers as much as possible in constant motion, shake and beat them.

Do not interrupt operations, if it can be avoided, but do the work rapidly and continuously, without pausing.

Keep every utensil scrupulously clean.

THE END.

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**DYES, DYESTUFFS, AND CHEMICALS AS
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Textile Machinery, Carding, Spinning, Weaving,

**DESIGNING AND IMPROVED PROCESSES IN TEXTILE
MANUFACTURING.**

ESTABLISHED, JANUARY, 1879.

Published on the 15th of each month, at

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Transcriber's Notes:

Corrected obvious typos and inconsistencies in punctuation:

[p34a](#). BIEGE -> BEIGE (also in Index and Index to samples).

[p54](#). half a century age -> half a century ago.

terra-cotta (Index) -> terra cotta (text).

triluel -> trileul.

[p65](#). luke water -> luke-warm water.

[p104](#). certian -> certain.

[p122](#). turmeric is boiling -> turmeric in boiling.

[p124](#). analine -> aniline.

[p151](#). Lay down the feathers for four hours is a cold. bath -> Lay down the feathers for four hours in a cold bath.

[p162](#). most energetic and and effectious -> most energetic and effectious.

[p173](#). and hand them over -> and hands them over.

[p187](#). Operate of follows -> Operate as follows.

[p188](#). to put feathers one side -> to put feathers to one side.

[p189](#). composion -> composition

In the Sample Index, the page for BLUE, MEDIUM should be 34a not 24a.

Some inconsistencies and oddities in spelling and hyphenation have been left as printed:

hand heat and hand-heat.

centrifugated (in text) and soluble (in printed advertisement).

redyed, re-dyed.

peroxyd and peroxide.

luke warm, lukewarm and luke-warm.

[p114](#) The correct formula for hydrogen peroxide is H₂O₂

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OSTRICH FEATHER DYER ***

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